

Two Modes, Not Many Types: The Structure of Football Player Roles in Season-Aggregate Data

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Abstract

Studies of football player roles routinely report a set of discovered archetypes, but a partitioning algorithm returns a partition whether or not the data contain groups, so the usual evidence cannot tell a real taxonomy from a cut through a continuum. We test existence directly on 13,822 player-seasons from 5 seasons of the big five European leagues, described by 33 statistics from a pre-withdrawal archive, calibrating cluster quality against three clusterless nulls: a matched Gaussian, a copula that keeps every real marginal and correlation, and the gap statistic’s uniform reference. A two-mode division is real: the observed silhouette exceeds every simulation under every null in every scope, by at least 8.01 standard deviations against the hardest. Finer structure is not: beyond two modes the observed quality converges on the null and density based clustering labels every point noise within each position group. A bootstrap cannot see the difference, since clusterless data resample at an adjusted Rand index of at least 0.911. The modes are not positions relabelled: 92.9 percent of defenders fall one side and 99.8 percent of forwards the other while midfielders divide almost evenly, and a classifier at 93.4 percent accuracy routes all but 1.31 percent of its errors through midfield. The division transfers across leagues at a mean adjusted Rand index of 0.944. Against real results over 490 team-seasons, the two-mode composition predicts points beyond possession out of sample, raising the coefficient of determination from 0.566 to 0.664, while the six archetypes add nothing beyond the two modes. Football role space has two poles and a populated middle, and finer taxonomies are artefacts of the method used to find them.

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Contents

1	Introduction	3
2	Related work	4
3	Data	6
4	Methods	8
5	Results	15
6	Discussion	41
7	Conclusion	45
A	Feature dictionaries	52
B	Minutes threshold sensitivity	55
C	UMAP hyperparameter grid	56
D	Cluster selection criteria in full	57
E	Cluster membership	58
F	Supplementary figures	58
G	Individual archetype radars	63
H	Further analyses on the reduced sample	69

1 Introduction

Public football data now describes every player in a major league by dozens of per-90 measures, and a standard genre of analysis has grown around it: cluster the players, name the clusters, and present the result as a taxonomy of roles. Published examples report four archetypes within a single position group [El Foukahi et al., 2026], and in basketball nine [Kalman and Bosch, 2020] or thirteen [Alagappan, 2012] across the whole sport, with labels such as deep-lying progressor or advanced playmaker. The genre rests on an inference that is rarely examined. A partitioning algorithm always returns a partition. Run k -means on a single multivariate Gaussian, which by construction contains no groups at all, and it will hand back tidy clusters with a respectable silhouette. That a method produced clusters is therefore not evidence that clusters exist.

This paper asks how much group structure football role data actually contains, and answers it against an explicit null. We assemble 13,822 player-seasons covering 5 seasons from 2018 to 2022 across the 5 major European leagues, with 33 per-90 features spanning shooting, chance creation, ball progression, touches by pitch zone, take-ons, defensive actions and passing accuracy by distance. On that sample we run dimensionality reduction, clustering under a selection rule fixed before any fitting, and, crucially, a calibration against simulated data built to contain no clusters whatsoever.

Six findings are stated in advance.

First, the structure is real, and it survives the hardest null available. Against three clusterless references, a Gaussian matched on covariance, a copula that keeps every real marginal and correlation, and the gap statistic’s uniform box, each simulated 25 times, the observed silhouette at two clusters exceeds every simulation under every null in every scope. Against the hardest of the three the margin runs from 8.01 standard deviations for forwards to 13.87 for the outfield pool. Hartigan’s dip statistic on the leading component clears the hardest null by 13.67 on the pool. Football role data is not a featureless cloud. A bootstrap, by contrast, cannot show this: clusterless data return a resampling adjusted Rand index of at least 0.911, against an observed 0.962, so the stability check the taxonomy genre leans on is close to uninformative about whether groups exist.

Second, the structure is two modes and nothing finer. The selection rule returns two clusters globally and within every position group, on the full 33 feature set and on 9,263 eligible players. Beyond two clusters the observed silhouette converges on the clusterless null, so the additional clusters that a taxonomy would report are indistinguishable from what structureless data produces. Density-based clustering, which is permitted to conclude that nothing is clustered, assigns 0.865 of the outfield pool to noise and every position group entirely to noise.

Third, the two modes divide players by attacking against defensive involvement, and they do not recover the positions the data already records. Agreement with listed position group is 0.293 on the adjusted Rand index, far above chance and far below recovery, and the shape of the disagreement is the point: 92.9 percent of defenders fall on one side and 99.8 percent of forwards on the other, while midfielders divide almost evenly between them. A supervised model reaches 93.4 percent accuracy on the same features and yet routes all but 1.31 percent of its errors through midfield, which is the same geometry recovered by machinery that shares nothing with the clustering. This is not a claim about one league or one season: refitted independently, every one of 5 competitions and 5 seasons returns the same partition, and centroids fitted in one league assign another league’s players at a mean adjusted Rand index of 0.944.

Fourth, two modes is not one dimension. The leading principal component carries 39.9 percent of the variance, which is far more than any other single direction, but 14 components are needed to reach ninety percent of it. The space has one dominant axis along which the density is bimodal, and a long tail of smaller axes carrying variation that is real, interpretable and continuous. A two-mode description is a statement about where players are concentrated, not a claim that role reduces to a

single number.

Fifth, the result is robust to the feature set. A separate live sample, collected after the largest free source of football statistics withdrew its advanced metrics in January 2026, retains only about half the features. The same analysis on that sample returns the same answer, which is reported in Section 5.14 as a robustness check, not as a limitation.

Sixth, the modes predict what happened. Over 490 team-seasons with real results, a club’s minutes-weighted composition on the two-mode axis predicts points per match beyond team possession, out of sample under season, league and club holdouts, raising the coefficient of determination from 0.566 to 0.664. The six archetypes add nothing beyond the two modes. Real results say what the null calibration said.

The practical implication is not that role analysis is impossible. It is that the correct output of role analysis on this kind of data is a position along an axis rather than membership of a type, and that a published taxonomy of a dozen archetypes is reporting structure its data cannot support.

The paper proceeds as follows. Section 2 places the work in the cluster validity literature it draws on and the football literature it responds to. Section 3 describes the two samples and what each can support. Section 4 gives every procedure, and states the calibration formally. Section 5 reports the results on the primary sample first and the reduced sample second. Section 6 says what the two modes mean, why finer taxonomies keep being published, and what the results do not show, and Section 7 closes.

2 Related work

Prior work on football roles has generally required data that is not public. Bialkowski et al. [Bialkowski et al., 2014] recovered formations and positional roles from tracking coordinates. Decroos and Davis [Decroos and Davis, 2020] learned player vectors from event streams, Pappalardo et al. [Pappalardo et al., 2019] built a performance rating from event data across several leagues, and Aalbers and Van Haaren [Aalbers and Van Haaren, 2018] distinguished playing styles from match events. All depend on proprietary feeds. Where public season aggregates are used instead, the standard output is a taxonomy. El Foukahi et al. [2026] apply k -means to twelve indicators from the same source used here and report four midfielder archetypes, and Seri et al. [2024] fit partial membership models to a season of Serie A counts. Two systematic reviews catalogue the genre, Plakias et al. [2023] for players and Gu et al. [2025] for teams; the first synthesises twenty-six styles from twelve studies and finds no shared framework behind them. The pattern is not specific to football. Alagappan [2012] reported thirteen positions in basketball from a topological summary of box score rates, and Kalman and Bosch [2020] nine from a Gaussian mixture. In none of this work is it asked whether the data contain groups at all. Fernández et al. [2024], reviewing 278 sports science articles that used clustering, found that 55 percent did not state the criterion by which the number of clusters was chosen and 27 percent did not state the clustering method, which puts the omission in context: the question of existence has not been answered badly so much as not asked.

The closest methodological relative of the present paper is the work of Akhanli and Hennig. Their aggregated validity index [Akhanli and Hennig, 2020] combines several criteria, each calibrated by comparing its value on a candidate partition against its distribution over random partitions of the same data, and applied to a season of player statistics from eight European leagues [Akhanli and Hennig, 2023] it returned a coarse partition into five groups, ordered from centre backs through full backs, midfielders and attacking midfielders to forwards, together with a fine partition into about one hundred and fifty small clusters intended for similarity search. The coarse partition runs along the same territorial axis reported here. The difference is in what the calibration is against. A

random partition of the observed points asks how good a clustering is relative to other clusterings of the same data, which is the right question for choosing among methods and numbers of clusters. Clusterless data matched to the observed sample asks whether any clustering of these points is better than a clustering of points that have no structure, which is the question of existence. The two are complementary, and the first does not answer the second.

The methodological position taken here has a longer history than the football literature, and the paper should be precise about what is and is not new. The problem of choosing the number of clusters has been studied systematically since Milligan and Cooper [1985] compared thirty stopping rules on data with known structure, and the silhouette [Rousseeuw, 1987], the gap statistic [Tibshirani et al., 2001] and the mixture information criterion all descend from that programme. The gap statistic is the direct ancestor of the calibration used here: it compares the observed within-cluster dispersion against the same quantity on reference data drawn uniformly over the principal box, which is one of the three nulls in this paper. What this paper adds to it is not a new statistic but a harder reference. A uniform box is easy to beat, a Gaussian matched on covariance is harder, and a copula that keeps every real marginal is harder still, and the paper reports the observed separation against all three so that a reader can see which reference the claim survives.

Stability-based validation is the second tradition the paper engages, and it engages it critically. Ben-Hur et al. [2002] proposed choosing the number of clusters by the stability of partitions under subsampling, and the bootstrap adjusted Rand index reported by most role taxonomies is a descendant of that idea. von Luxburg [2010] reviewed the theory and reached a conclusion that the applied literature has largely not absorbed: for algorithms such as k -means, stability is governed by the geometry of the objective rather than by the presence of clusters, so a partition can be perfectly stable on data that contain no groups at all. Section 5.1 measures exactly that on this data and finds it. The point is not that stability is useless, since an unstable partition is certainly not a taxonomy, but that a stable one is not certainly a taxonomy either, and the genre has been treating the second as if it followed from the first.

Clusterability as a question distinct from clustering has its own literature. Ackerman and Ben-David [2009] surveyed formal notions of it and found them pairwise inconsistent, so that the same data can be well clustered under one definition and not under another. Adolfsson et al. [2019] compared the practical tests and recommended those built on Hartigan’s dip statistic [Hartigan and Hartigan, 1985] applied to a low dimensional projection, which is one of the instruments used here. Hennig [2015] argued that there is no true clustering independent of the purpose the clustering is meant to serve, and that the honest question is whether a partition has the properties the analyst needs rather than whether it is correct. This paper takes that position: the property it asks for is that a partition be distinguishable from one imposed on a continuum, and it reports which partitions have it.

A different tradition describes athletes as mixtures rather than as members. Archetypal analysis [Eugster, 2012] represents each player as a convex combination of a few extreme profiles, and partial membership models [Seri et al., 2024] let a player belong to several clusters at once. Both are natural descriptions of a continuum, and both presuppose one, since neither asks whether the data would have supported discrete groups instead. The present paper supplies the test that decides between the two descriptions, and its answer, two modes with a populated middle, is one that a mixture representation expresses well and a taxonomy does not.

The role taxonomy genre has proceeded as though the cluster validity literature did not exist. It reports partitions without asking whether the data are clusterable, and validates them with criteria that, as Section 5.1 shows, a clusterless continuum satisfies equally well. The contribution here is not a new clusterability test but the application of existing ones to a field that has skipped them, and the finding that they return a real but far coarser structure than the field has been reporting. To

our knowledge no published study in any sport has calibrated cluster quality against data simulated to contain no clusters, measured bootstrap stability on such data, or applied a clusterability test to player statistics, and the searches conducted for this paper found none.

3 Data

3.1 Primary sample

The analysis uses season-aggregate player statistics for the 5 major European leagues, covering 5 seasons from 2018 to 2022, where a season is labelled by the calendar year in which it ends. The sample contains 13,822 player-seasons, of which 9,263 are outfield players meeting the minutes threshold and 1,035 are goalkeepers.

These tables originate from FBref [Sports Reference LLC, 2026], which published advanced statistics derived from a commercial event-data provider. They are obtained here from a public mirror maintained by the worldfootballR project [Zivkovic, 2026] rather than from the site directly, and that choice needs stating plainly. On 20 January 2026 the site’s operator removed every statistic derived from that provider, retroactively and across completed seasons, after the licence was terminated. The mirror predates the removal. It is therefore a secondary source in the strict sense, and a reader should weigh it accordingly: it was scraped by a third party, we did not observe the scrape, and we cannot re-derive it from the primary source because the primary source no longer holds the data. What we can do, and do, is audit it. Row counts, per-column coverage and per-season coverage are checked on load, and the checks are part of the pipeline, not a one-off inspection.

One season is excluded on those grounds. FBref restructured its ball-carrying columns for the season ending 2023 and the mirror did not follow the renames, so every carrying column is empty for that season while neighbouring columns are intact. An overall missingness rate hides this, because the affected columns are complete in the five retained seasons; a per-season rate exposes it immediately. The pipeline raises instead of imputing whenever a feature exceeds half missing within any retained season, so the exclusion is enforced by a check and not by a comment.

Retrieval is part of the pipeline. The module `src/archive_fetch.py` downloads each upstream file from the stated repository, converts it, and records a SHA-256 digest per file in a manifest alongside row and column counts, so that a run can be identified after the fact and an upstream change is detectable, not silent. The mirror rewrites its files in place on each scrape and publishes no tags, so the digests are the only available pin.

3.2 Feature set

The outfield feature set contains 33 features and is listed in full, with the FBref table each column was read from, in Table 6 of Appendix A. It spans six families: shot volume and quality including non-penalty expected goals [Eggels et al., 2016, Rathke, 2017]; chance creation including expected assisted goals, key passes and shot and goal creating actions; ball progression through progressive passes, carries and receptions and entries into the final third and penalty area; touches broken down across the five pitch zones; take-ons attempted and their success rate; and defensive work through tackles, interceptions, blocks, clearances, ball recoveries and aerial duel success, together with passing accuracy at short, medium and long range.

Counting statistics are divided by nineties played. Quantities published as rates are used as published and not divided a second time. Minutes and appearances are retained as context and never enter the feature set, because squad usage describes a player’s opportunity rather than the role he performs within it.

Several features are undefined for some players by construction, not by data loss: a player who attempted no shots has no shot accuracy, and a player who attempted no take-ons has no take-on success rate. After the minutes filter these affect at most eight percent of players in any single feature, and the affected values are median-imputed within position group and flagged.

3.3 Adjusting defensive counts for possession

Defensive actions can only be recorded while the opponent has the ball, so their raw rates confound what a player does with how often his team is out of possession, which in turn tracks team strength. The correction and the reasoning behind its exact form are set out in Section 4.1. The point for the data description is that the five opponent-ball counts, tackles, interceptions, blocks, clearances and fouls committed, enter the feature set in adjusted form.

3.4 A second, deliberately impoverished sample

A separate sample is collected directly from the live site for the seasons ending 2025 and 2026, after the withdrawal, through the soccerdata library [Robberechts, 2026], with expected goals taken from Understat [Understat, 2026] as an independent source. It is not used for the main analysis. Its purpose is a robustness check: it contains roughly half the features of the primary sample, having lost the site’s own expected goals, all passing and possession volume, shot and goal creating actions and the entire advanced goalkeeping table, and it therefore allows the central result to be re-derived on a much poorer description of the same game.

That sample carries a methodological warning that applies beyond this study. The affected tables are still served with complete column headers and one row per player, and only the values are gone, so a feature set assembled from the site’s published schema resolves correctly and yields an empty matrix. Table 1 reports the audit. Feature selection against observed values instead of documentation is now a necessary step when working with this source. The withdrawal was also applied unevenly: at season level the columns were emptied, while at match level they were removed from the schema outright, so match reports remain usable for what survived, which Section H.8.4 exploits.

Table 1: Populated columns in each FBref player table. The tables are still served with complete headers, so the empty counts are the number of columns that resolve correctly and contain no values.

Table	2024-25		2025-26	
	Columns	Empty	Columns	Empty
standard	17	0	17	0
shooting	11	0	11	0
passing	21	19	21	19
passing types	16	14	16	14
goal shot creation	17	16	17	16
defense	17	14	17	14
possession	21	20	21	20
playing time	17	0	17	0
misc	13	2	13	2
keeper	19	0	19	0
keeper adv	26	23	26	23

3.5 Known limitations

Season aggregates discard context. A tackle count does not say where on the pitch the tackles happened, and an expected goals total does not distinguish six poor chances from two good ones. The possession correction described above reaches only the opponent-ball statistics, so attacking output remains related to a team’s own share of the ball, and Section 5.12 sets out why correcting that would remove signal rather than bias. Per-90 rates computed over partial seasons remain noisy, which motivates the shrinkage analysis in Section H.8.3. The sample is recent European club football, so nothing here should be read as a claim about the game in other eras or at other levels. Finally, the clusters are descriptive and carry no causal content: a cluster label records how a player’s season looked, not why.

4 Methods

Every procedure below is seeded from a single constant, and every hyperparameter is declared in one configuration file that the analysis code reads. Nothing is tuned after seeing a result. All computation uses scikit-learn [Pedregosa et al., 2011] unless stated otherwise.

4.1 Possession adjustment

Defensive actions can only occur while the opponent has the ball. A player in a side that concedes possession therefore accumulates interceptions, tackles and fouls partly through exposure rather than through role or ability, and since possession correlates with team strength this confound would enter the feature set as a proxy for the quality of a player’s team. Team possession share survived the withdrawal even though the touch counts it is derived from did not, which is what makes a correction possible on both samples.

Possession adjustment of defensive counts was introduced by Knutson [2014], whose original form scaled tackles and interceptions by a sigmoid in team possession. The form in common use since, and the one tested here, multiplies a defensive rate by the ratio of a reference opponent possession to the actual one, which is equivalent to assuming that defensive volume is directly proportional to time spent out of possession. That is an empirical claim, not a definition, and it can be tested. Regressing

the logarithm of team defensive volume per ninety on the logarithm of opponent possession, over 490 team-seasons on the primary sample and 40 on the reduced one, gives the elasticity, which is the percentage change in defensive volume produced by a one percent change in opponent possession. Direct proportionality corresponds to an elasticity of one.

Every fitted elasticity, on either sample, falls below one. Defensive volume does rise with opponent possession, so the confound is real, but far less than proportionately. A side that spends more time without the ball does not simply make proportionally more tackles: it also defends deeper, concedes territory without contesting it, and its defensive actions saturate. The conventional exponent therefore overcorrects, and Section 5.11 reports by how much on each sample and shows that under an exponent of one the correlation between defensive rates and team possession does not fall toward zero but changes sign and grows, so the adjustment injects more team dependence than it removes.

We therefore adjust with the measured elasticity instead of the assumed one. Each defensive count is multiplied by the exposure ratio raised to the power of its own fitted elasticity, which leaves a player at a club with league-average possession unchanged and corrects the others by the amount the data says exposure actually matters. The exponent is estimated from team totals rather than player rates, because the quantity being corrected is a team-level exposure effect and player rates carry position and role variation that would swamp it. It is pooled across seasons, and on the primary sample across leagues, because a single season of one competition gives a noisy slope, and because the exponent is a nuisance parameter that uses no player-level or cluster-level information, so a shared value cannot manufacture the cross-season and cross-league agreement that Section 5.6 measures. Attacking statistics are deliberately left unadjusted, since they scale with a team’s own possession rather than the opponent’s and correcting both directions with the same instrument would double count.

The elasticities are estimates and are reported with their uncertainty. Team-seasons are resampled with replacement two thousand times and every elasticity is refitted on each resample, giving a percentile interval and a bootstrap probability that the true elasticity is at least one, which is the value the conventional correction assumes. Each elasticity is also refitted within every league and every season separately, so the paper can say whether a pooled value transports rather than hedging that it might not, and the constant elasticity form is checked against a model with a quadratic term in log possession.

4.2 Dimensionality reduction

Principal component analysis is fitted to the standardised feature set, both globally and separately within each position group. The global embedding is dominated by the axis that separates defenders from forwards, which is the least interesting distinction available because it is already recorded in the listed position, so the role profiling that follows uses the position stratified fits.

Uniform manifold approximation and projection [McInnes et al., 2018] is fitted over a grid of twelve configurations, with the number of neighbours in $\{10, 15, 30, 50\}$ crossed with a minimum distance in $\{0.0, 0.1, 0.5\}$. Configurations are ranked by trustworthiness, which measures the degree to which points that are close in the embedding were also close in the original space, and the full grid is reported in Appendix C. Selection by an explicit criterion matters here because a manifold method will produce a visually appealing picture at almost any setting.

t-distributed stochastic neighbour embedding [van der Maaten and Hinton, 2008] is computed at a perplexity of 30 for visual contrast only. It is never used as an input to clustering, because its objective preserves local neighbourhoods at the expense of the global geometry that a distance based clustering algorithm depends upon.

The three embeddings are compared on trustworthiness, on continuity, which is trustworthiness

computed with the roles of the two spaces exchanged, and on a nearest neighbour position purity score at $k = 10$ that reports the share of a player’s ten nearest neighbours that share his listed position group.

4.3 Clustering

Clusters are estimated with k -means as the primary method, with a Gaussian mixture model whose Bayesian information criterion is reported across the same range of component counts, and with HDBSCAN [Campello et al., 2013], for which we report the fraction of players assigned to noise. Each is fitted both to the standardised features and to the principal component space retaining ninety percent of variance, and each reported result states which space produced it. Clustering is performed globally and within each of the three outfield position groups.

The number of clusters is selected by a rule fixed in advance of any fitting, which matters because four criteria applied after the fact can be made to justify almost any choice. Over k from 2 to 12 we compute the silhouette coefficient [Rousseeuw, 1987], the Davies-Bouldin index, the Calinski-Harabasz index, and the gap statistic [Tibshirani et al., 2001]. The gap statistic has no implementation in the library we use, so it is implemented here against the original description, comparing the within-cluster dispersion to its expectation under fifty reference datasets drawn uniformly over the bounding box of the data in the fitted space. The rule selects the smallest k whose silhouette lies within 0.02 of the best silhouette observed and which ranks in the top three on at least two of the silhouette, Calinski-Harabasz and gap criteria, with Davies-Bouldin breaking ties. Which criteria agreed, and which did not, is reported alongside the choice.

4.4 Testing whether clusters exist at all

Selecting a number of clusters presupposes that clusters are there to be counted. That presupposition is the weak point of the whole genre, because a partitioning algorithm returns a partition whatever it is given, and the internal criteria used to choose k are computed on the partition rather than against any alternative. A silhouette of 0.273 sounds like evidence until one asks what silhouette structureless data of the same shape would have produced.

We therefore calibrate against an explicit null. For each scope we draw 25 synthetic datasets from a multivariate normal with the empirical mean and covariance of the observed feature matrix, matching the real data in sample size, dimension and correlation structure while containing, by construction, no group structure at all. The identical k -selection procedure is run on each. The observed silhouette profile is then read against the distribution of simulated ones. Where the observed value falls inside that distribution, the partition at that k is indistinguishable from one imposed on a continuum, and no claim is made from it.

A single Gaussian is the easiest reference to beat, because real per-ninety rates are skewed and bounded below and skew alone raises the silhouette a partition can reach. Two further nulls therefore run through the identical procedure. The first is a Gaussian copula that preserves every feature’s empirical marginal distribution and the observed rank correlation while destroying higher-order joint structure: each column is transformed to normal scores, a multivariate normal is drawn on the correlation of those scores, and each simulated column is mapped back through the observed column’s empirical quantile function. It carries the real skew and the real correlations. The second is the reference distribution of the gap statistic [Tibshirani et al., 2001], uniform within the bounding box of the data after rotation to its principal axes, which assumes no distributional shape at all. For every scope the paper reports the observed silhouette against all three, and treats the hardest of the three as the calibration that counts.

The same simulations answer a question the paper would otherwise be asserting. A bootstrap stability check is the second standard piece of evidence for a published taxonomy, and it is claimed here that a partition imposed on clusterless data passes it. Rather than assert that, the bootstrap adjusted Rand index is computed on the simulated datasets by exactly the procedure applied to the real data, so the observed stability can be read against what stability structureless data produces.

Three further instruments guard against the null calibration itself being misleading. Hartigan’s dip test [Hartigan and Hartigan, 1985] is applied to the leading principal components and asks directly whether a distribution is unimodal, which is a claim about the data, not about any clustering of it. At the sample sizes here the test rejects unimodality for departures too small to matter, so the dip statistic itself is reported as the effect size and is placed against the distribution of dip statistics computed on the simulated nulls, in the same way as the silhouette. A Gaussian mixture model’s Bayesian information criterion is computed across the same range of component counts, because unlike the silhouette it carries no built-in preference for few components. HDBSCAN [Campello et al., 2013] is fitted at several minimum cluster sizes, because it is permitted to answer that there are no clusters and does so by assigning points to noise rather than by forcing them into groups.

Agreement among instruments that fail in different ways is the standard of evidence used throughout the results. A continuum and a set of discrete types are different claims about the same data, and the paper takes care not to report the second when only the first is supported.

4.5 The calibration, stated formally

Let $X \in \mathbb{R}^{n \times d}$ be the standardised feature matrix for a scope, with n player-seasons and $d = 33$ features, and let $\pi_k(X)$ be the partition returned by k -means with k clusters, run with a fixed seed and multiple restarts. Let $s_k(X)$ be the mean silhouette of $\pi_k(X)$, computed on a fixed subsample of at most four thousand rows when n exceeds that, since the silhouette is quadratic in n . The number of clusters reported, $k^* = 2$, is not chosen by inspecting $s_k(X)$; it is returned by the selection rule declared in advance in Section 4.3, and the calibration then asks a single question about it.

Three generators produce clusterless data matched to X . The Gaussian generator draws n rows from $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ with the empirical mean and covariance. The copula generator transforms each column of X to normal scores by rank, estimates the correlation $\hat{R}$ of those scores, draws n rows from $\mathcal{N}(0, \hat{R})$, and maps each simulated column back through the empirical quantile function of the corresponding observed column, so that every marginal distribution of X is reproduced exactly and only the joint structure beyond rank correlation is destroyed. The uniform generator rotates X to its principal axes, draws n rows uniformly within the resulting bounding box, and rotates back, which is the reference distribution of Tibshirani et al. [2001]. For each generator g and each of $B = 25$ draws $X^{(g,b)}$, the identical pipeline computes $s_{k^*}(X^{(g,b)})$, giving a null sample of size B per generator.

The reported statistic is the standard score

$$z_g = \frac{s_{k^*}(X) - \bar{s}_g}{\hat{\sigma}_g},$$

where $\bar{s}_g$ and $\hat{\sigma}_g$ are the mean and standard deviation of the null sample under generator g . With B draws the null standard deviation is itself an estimate, so z_g is reported as a description of the margin, not as a test statistic with a known distribution. The claim the paper makes is the distribution-free one: that $s_{k^*}(X)$ exceeds every one of the B null values under every generator, which under the null of exchangeability between the observed and simulated statistics has Monte Carlo probability at most $1/(B+1)$ per generator and scope. The paper’s headline is a conjunction over three generators and four scopes, not a disjunction, so it is conservative with respect to multiplicity: it can only be weakened by adding a generator or a scope, never strengthened.

The same generators supply two further calibrations. The dip statistic of Hartigan and Hartigan [1985] on the leading principal component of each $X^{(g,b)}$ gives a null sample against which the observed dip is placed in the same way, which matters because at these n the dip test’s p-value rejects unimodality for departures too small to be of interest. The bootstrap adjusted Rand index of π_{k^*} is computed on a subset of the simulated datasets by exactly the resampling procedure applied to X , so that the stability the genre reports for real data can be read against the stability a clusterless continuum produces.

One assumption should be stated because the analysis makes it silently otherwise. The rows of X are player-seasons, and the same player appears in up to 5 of them, so rows are not independent. The generators match the marginal and pairwise structure of X but not this nesting, and the bootstrap resamples rows as if exchangeable. The replication of Section 5.6, which refits every season and every league independently, is the analysis that does not share this assumption, and it is the reason the paper does not rest the two-mode claim on any single pooled fit.

4.6 Stability and validation against listed positions

A partition can be produced from any dataset, so the question that matters is whether it survives resampling. We draw two hundred bootstrap resamples, refit the clustering on each, and compute the adjusted Rand index [Hubert and Arabie, 1985] between each bootstrap partition and the reference partition on the players common to both. We also accumulate a consensus matrix recording, for every pair of players, the fraction of resamples in which the two were assigned together. Clusters whose members co-assign below 0.6 on average are reported as unstable, and they are described as such in the results rather than being renamed or merged to make the account tidier.

Replication is tested separately and more stringently than resampling, because a resample reuses the same players. Each season and each league is standardised inside itself and clustered from scratch, borrowing no centre, no scale and no starting point from the pooled fit, and each independent partition is compared against the pooled one. Cluster labels are arbitrary per fit, so labels are aligned to the pooled partition by matching on the contingency table before any label-dependent quantity is computed, and an assertion fails the run if an aligned agreement drops to chance. Three comparisons are reported and they answer different questions: an independent fit scored against the pooled partition on the same rows, one season’s labels scored against another’s on the players present in both, which mixes replication with real change in those players, and one scope’s centroids imported into another scope’s players and scored against that scope’s own fit, which isolates replication. We also correlate each scope’s centroid-difference vector with the pooled one, to check that the scopes recover the same axis and not merely a similarly sized partition.

The relationship between the discovered clusters and the listed positions is quantified by normalised mutual information and by the adjusted Rand index, computed against both the position group and the full position string. Players whose cluster assignment disagrees with their listed group, and players whose Gaussian mixture posterior is split across components, are extracted as candidate hybrids.

4.7 Robustness to the feature set

A two-mode result on 33 features could in principle be carried by one block of them. The features are grouped into six families, shooting, chance creation, ball progression, pitch territory, defending and passing accuracy, with the grouping stated in the analysis code so that it can be audited and asserted to cover every feature exactly once. Each family is then removed in turn and the pre-declared selection rule and the Gaussian null calibration are rerun on what remains, in every scope. The

converse is run as well, each family alone, which shows which families carry the two-mode structure by themselves. For every ablation the paper records the number of clusters the rule selects, the standard score of the observed silhouette against the null, and the adjusted Rand index between the ablated partition and the full one on the same players, so that a surviving k of two can be distinguished from the same two modes surviving.

4.8 Validation against results

The structure the paper reports is tested against what happened. The archive’s team tables carry goals and expected goals but not results, so wins, draws and losses are recovered by summing the goalkeeper table, which credits every match to the goalkeeper who played it, and the recovery is audited: each league-season is checked for the right number of clubs, every club’s credited matches are checked against the schedule with the exceptions listed, goals for are taken from goals scored while each goalkeeper was on the pitch and accepted only where the same rows reproduce the goalkeeper table’s goals against to within one, and two published season records are asserted. Points per match, goal difference per match and league position follow.

Each team-season is then summarised by its minutes-weighted composition on the two-mode partition and on the six archetypes, using the eligible players, and the composition is correlated with results and used to predict them. Prediction is out of sample under three holdouts, one season at a time, one league at a time and one club at a time, the last being the strict one since no held-out team-season then shares a club with the training data, by ridge regression with the penalty chosen by cross-validation inside the training folds, comparing possession alone, the composition alone and both. The increment of the last over the first is the quantity of interest, because composition and dominance are confounded and only the increment says whether the modes carry information about results that dominance does not. The goalkeeper partitions are tested the same way, against the prediction that a partition sorting keepers by their teams will track points and a partition describing goalkeeping will not.

4.9 Archetype profiling

For every retained cluster we compute the centroid profile as a standardised deviation from the mean of its position group, take the eight features with the largest absolute deviation as the distinguishing features, and list the ten players closest to the centroid and three players near the boundary. Archetype names are assigned only after inspecting these profiles and the players they contain, and each name is accompanied by a justification that cites the distinguishing features and the example players. The names are hypotheses about what the statistical profile represents, not observations, and they carry no authority beyond the profile that produced them.

Outliers are identified within position group with an isolation forest [Liu et al., 2008], reported alongside a Mahalanobis distance so that a distance-based and a partition-based notion of unusualness can be compared.

4.10 Supervised validation

Unsupervised structure is easy to over-interpret, so we ask a supervised model what separates the listed positions. Logistic regression and gradient boosting, the latter with LightGBM [Ke et al., 2017], are trained to predict the position group from the feature set under five-fold stratified cross validation, and we report accuracy, macro F1 and the confusion matrix, every number computed from out-of-fold predictions. A majority-class baseline runs through the identical folds, so the reported lift is over the trivial model rather than over zero. Two properties of the confusion matrix are read as

evidence about geometry rather than about accuracy: whether the two extreme classes are confused with each other, and whether the errors concentrate in the class that lies between them. An ordered space with two poles produces few errors of the first kind and many of the second, while three genuine categories would not.

The purpose is interpretability, not accuracy: SHAP values [Lundberg and Lee, 2017] on the boosted model identify which statistics actually distinguish each position in this feature set. The model is fitted on every row and the attributions are computed on a stratified sub-sample, which is stated with the result. We also record, for each feature and class, the rank correlation between the feature’s value and the attribution it makes, which separates a class identified by what its members do from one identified by what they do not. Players the classifier places in the wrong group are cross referenced against the candidate hybrids from the clustering, since agreement between two methods that fail in the same place is more informative than either alone.

A secondary ridge regression predicts non-penalty expected goals per ninety from the non-shooting features only, so that the fitted value represents the shot volume a player’s role would ordinarily generate and the residual represents the part not explained by role. We report the cross validated coefficient of determination and treat the residual ranking with heavy caution, because a residual from a model with modest explanatory power is mostly noise.

4.11 Goalkeepers

Goalkeepers share almost no features with outfield players, so they are analysed with a separate pipeline on the 13 canonical goalkeeper features, under the same eligibility threshold, the same standardisation and the same selection rule. Saves above expectation are computed as post-shot expected goals [Vatvani, 2022] minus goals allowed, which conditions on the difficulty of the shots faced rather than only on their number.

One methodological choice needs stating in advance because the result turns on it. The goalkeeper feature block mixes quantities describing what a goalkeeper does with quantities describing what his defence permits him to face, and a partition fitted on both will separate teams rather than goalkeepers. We therefore fit twice, once on the full block and once on technique alone with the exposure variables excluded, and report for each partition how much variance membership explains in situational quantities against technical ones. The comparison is declared here, not chosen after seeing which fit looked better.

On the reduced sample the same analysis is not possible, because post-shot expected goals, sweeper actions, distribution and cross handling are all absent there. Only shot stopping and workload survive, so that version builds two composites instead of five and substitutes the residual from a regression of saves on shots on target faced for post-shot expected goals. That substitute adjusts for the volume of work a goalkeeper receives but not for its difficulty, and it is reported as the weaker instrument it is.

4.12 Team signatures

Each club is summarised in two ways: as a minutes-weighted centroid of its outfield players in the shared principal component space, and as the share of its outfield minutes played by each archetype. Clubs are then clustered hierarchically on archetype composition and compared against final league position.

4.13 Additional analyses

Three further analyses address specific threats to the interpretation.

Player similarity is computed as cosine similarity within position group, and the nearest neighbours of a set of well known players are reported so that a reader with football knowledge can judge whether the geometry corresponds to anything real.

Cross-season replication refits the entire pipeline on the replication season and computes the adjusted Rand index between the two seasons for players present in both, which distinguishes archetypes that describe stable playing roles from artefacts of a single season.

Per-90 rates computed over partial seasons are noisy, and that noise is largest where the minutes are fewest. We therefore apply empirical Bayes shrinkage toward the position group mean, with the degree of shrinkage determined by minutes played [Efron and Morris, 1977], refit the clustering on the shrunken features, and report how many assignments change. A partition that dissolves under shrinkage was substantially a description of sampling noise.

4.14 Reproducibility

The analysis runs from a single command against pinned dependencies and a fixed seed. Every figure and every number in this paper is generated by code in the accompanying repository and read into the document from generated definitions at build time, so no value in the text was transcribed by hand. The code, the retrieval step that fetches the archive from its stated upstream with a recorded digest per file, the cluster assignments for both samples, and the sources of this document are at <https://github.com/ZacharyTChung/pl-player-profiling>.

5 Results

5.1 Is there structure at all?

The first question is not how many clusters there are but whether there are any, and it is answered against the three nulls of Section 4.4, beginning with the softest: 25 synthetic datasets matched to the observed feature matrix in size, dimension and covariance but drawn from a single Gaussian and therefore containing no groups whatsoever.

The answer is that real structure exists, and that it is confined to a single division. Figure 1 shows the comparison. At two clusters the observed silhouette is 0.273 for the outfield pool against a null mean of 0.207, a standard score of 18.0, and the observed value exceeds every one of the 25 simulations. The same holds within each position group: 0.261 against 0.191 for defenders, 0.249 against 0.201 for midfielders and 0.247 against 0.166 for forwards, with standard scores from 13.2 to 24.2 and no simulation reaching the observed value in any scope.

A Gaussian is the softest reference available, and the numbers above are the ones a sceptic should discount. Two harder nulls run through the same procedure. The copula null keeps every feature’s real marginal distribution, skew and bounds included, and the real rank correlations, so it removes the objection that skew alone produced the separation. The uniform reference of the gap statistic keeps only the data’s principal bounding box. The copula is the hardest of the three in every scope, and the observed partition clears it everywhere: by 13.87 standard deviations on the outfield pool, 8.60 for defenders, 10.13 for midfielders and 8.01 for forwards, against 46.18, 56.35, 35.60 and 23.92 for the uniform. The observed silhouette at two clusters exceeds every simulation under every null in every scope. The smallest margin anywhere, over all four scopes and all three nulls, is 8.01 standard deviations, for forwards against the copula. Those are the paper’s numbers; the Gaussian ones are reported because they are what the genre would report.

Beyond two clusters the two curves converge. By around six clusters the observed silhouette is inside the simulated band in every position group, which is the central negative result of this paper

stated precisely: the third, fourth and subsequent clusters that a taxonomy would name are not distinguishable from what a partitioning algorithm extracts from structureless data.

The second standard piece of evidence for a published taxonomy is a bootstrap, and the same simulations show what it is worth. The bootstrap adjusted Rand index of the two-cluster partition, computed on 5 clusterless datasets from each null in each scope by exactly the procedure applied to the real data, runs from 0.930 to 0.985 across scopes and nulls, and no single clusterless dataset anywhere falls below 0.911. The observed data run from 0.962 to 0.990. The gap between the real data and data built to contain no groups is as small as 0.003 and never larger than 0.033, and a stability screen at 0.8, which is a demanding one by the genre’s standards, is passed by every clusterless dataset in every scope. A resampling stability check does not distinguish real structure from a partitioned continuum. It measures whether an algorithm is deterministic on similar inputs, which k-means is, and it is reported in this paper only because readers will expect it.

Three further instruments agree, with one exception that is reported as such. Hartigan’s dip test rejects unimodality on the leading principal component at $p < 0.0001$ for the outfield pool, $p < 0.0001$ for defenders and $p < 0.0001$ for midfielders, but at these sample sizes a p-value rejects for departures too small to matter, so the dip statistic is the quantity to read and it is read against the nulls. The observed dip on the pool is 0.0176, which sits 13.67 null standard deviations above the hardest of the three references, the uniform box, and 51.17 above the copula. Defenders clear the hardest null by 8.84 and midfielders by 5.66. The dominant axis is two-moded, not merely wide. Forwards are the exception in both readings: the test returns $p = 0.9936$, and the observed dip of 0.0053 sits -2.69 standard deviations against the uniform reference, which is to say below it. A forward’s leading axis is less bimodal than a flat box. The forward split is real by the silhouette calibration, clearing the hardest null by 8.01, and its leading axis is not bimodal by the dip, and both halves of that are reported because together they say what the forward split is: a cut through a continuum of involvement that the null cannot produce, not two kinds of forward. Density-based clustering assigns 0.865 of the outfield pool to noise and every position group entirely to noise, which is what a density method does when handed a continuum with a density gradient instead of separated groups. The Gaussian mixture criterion declines monotonically to the top of the candidate range in every scope, which is the signature of a model absorbing extra components into a smooth density rather than identifying real ones, and it is reported for completeness, not treated as selecting a number.

Taken together the instruments support one claim and refuse a stronger one. Football role data contains a real division into two modes. It does not contain a taxonomy.

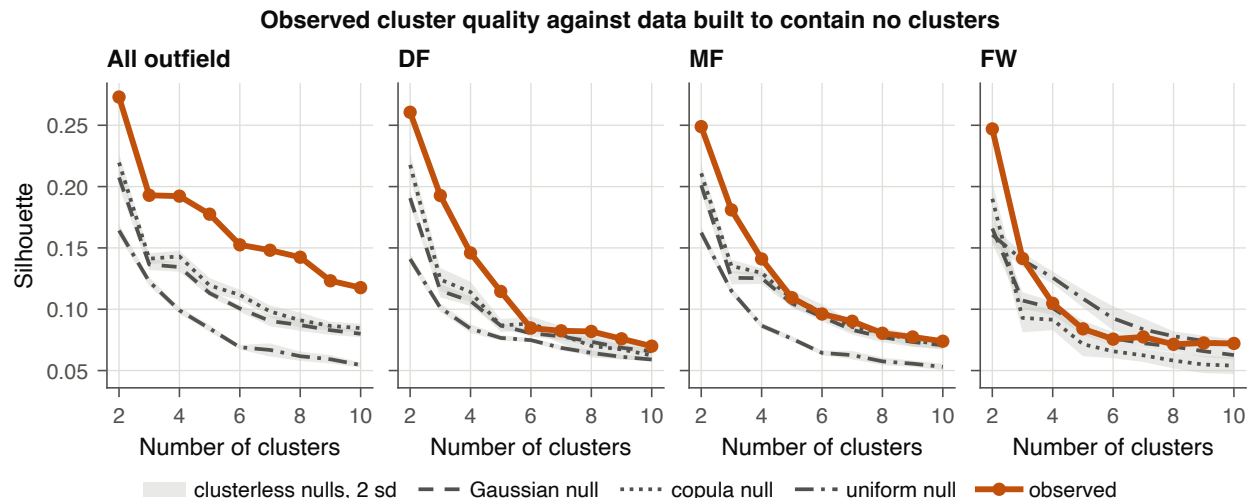


Figure 1: Observed cluster quality against three kinds of data built to contain no clusters. Each null is simulated 25 times and drawn as its mean with a band of two standard deviations: the Gaussian matched on covariance, the copula that keeps every real marginal and correlation, and the uniform reference of the gap statistic. The observed curve separates from all three at two clusters and rejoins the bands as the number of clusters grows.

5.2 How many dimensions the space has

A two-mode description invites a natural objection: if the data reduce to two groups, perhaps they reduce to one number, and the whole exercise is an elaborate way of rediscovering an attacking-to-defending scale that any observer could have drawn by hand. Principal components on the full 33 archive features answer that directly, and the answer is no.

The leading component carries 39.9 percent of the variance, which is far more than any other single direction and confirms that one axis dominates. It is nowhere near the whole space. The first three components together reach only 64.2 percent, 5 components exceed the Kaiser criterion, and 14 are required to reach ninety percent of the variance. The role space therefore has one dominant axis and a long tail of smaller ones, which is a different object from a line. Figure 2 shows the scree and the loadings.

The content of the leading components is readable and matches the football. The first separates players by where on the pitch they touch the ball, loading positively on progressive receptions and touches in the attacking third and the opposition box, and negatively on touches in the defensive third and the defending team's own box. The second is a progression axis that is close to orthogonal to the first, loading on progressive passes, progressive carries and passes into the final third while loading negatively on non-penalty expected goals and touches in the opposition box. A deep midfielder who moves the ball forward and a striker who receives it in the box sit at opposite ends of the second component while sharing similar territory on neither.

That pair explains why two modes coexist with a multidimensional space. The first component is bimodal and drives the partition. The components behind it carry real and interpretable variation that is continuous, not clustered, which is the variation a finer taxonomy would have to divide and, as Section 5.1 showed, cannot.

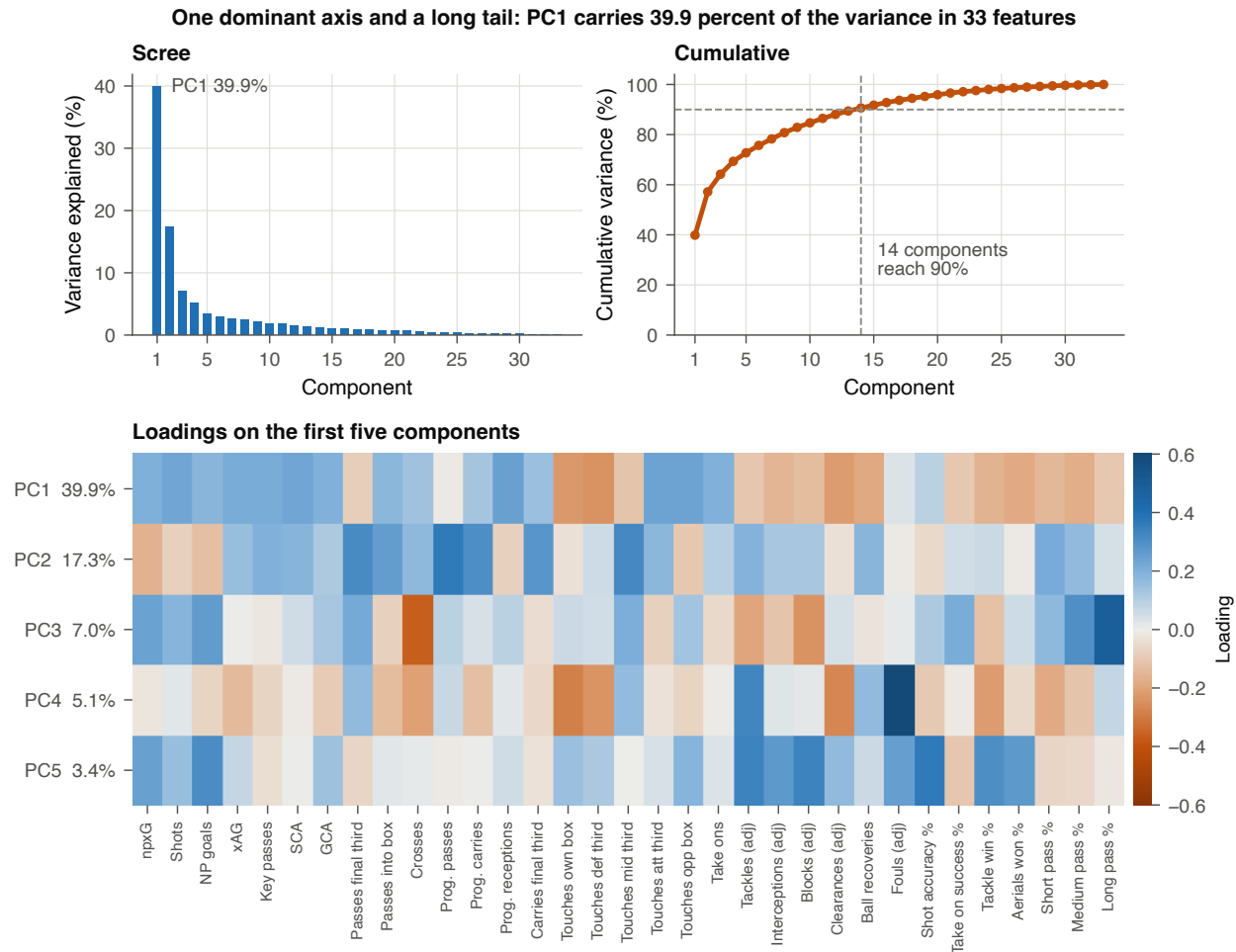


Figure 2: Principal components of the 33 archive features over 9,263 player-seasons. The scree panel shows one dominant component and a long tail rather than a sharp elbow, and the loading panels show that the leading component orders players by pitch territory while the second orders them by ball progression.

5.3 What the two modes are

Two modes within each of three position groups give six descriptions, and with 33 features they can be characterised far more sharply than a reduced feature set allows. Names were assigned only after inspecting centroid profiles and member players, and the justifications quoted here are generated from the computed centroids, not composed for the paper. Figure 4 shows the deviation profiles and Figure 5 the radar comparisons.

Among defenders the division is between 2,094 Deep central defenders and 1,699 Wide attacking defenders. The first take three times as many touches in their own defensive third as the second, more than twice as many in their own box, twice the possession-adjusted clearances and win considerably more aerial duels. The second take five times as many touches in the attacking third, attempt ten times as many crosses, receive far more progressive passes and attempt four times as many take-ons. This is the centre back against wide defender distinction, and the notable point is that the source's own position label does not encode it: the great majority of both groups are listed simply as defenders.

Among midfielders the division separates 1,939 Deep ball winning midfielders from 1,827 Advanced attacking midfielders. The first recover the ball more often, tackle more, and take far more touches in their own third. The instructive detail is that they also make more passes into the final third and more progressive passes than the second group. They are not less involved in progressing the ball, they progress it differently: by passing forward from deep rather than by receiving high. A description that treated attacking involvement as one axis would miss that, and it is the clearest case in the paper of the richer feature set buying something the reduced one could not.

Among forwards the division separates 1,241 Finishing centre forwards from 463 Creating and carrying forwards, and it is not a division by scoring. Non-penalty expected goals are slightly higher in the first group. What separates them is carrying and creation: the second group record nearly three times the progressive carries, almost three times the passes into the box, twice the take-ons and substantially more shot creating actions, while shooting slightly less. Forward is not one role, but the axis that splits it is involvement in build-up, not goal threat.

One structural property constrains how far these six can be read as independent. Features are standardised within position group before a two-way split, so the two centroids in a group are exactly anti-parallel and the larger cluster necessarily carries the smaller deviations. Each pair is therefore the two ends of one axis, and the six archetypes describe three axes, not six independently shaped profiles.

5.4 The modes are not the positions

The obvious deflationary reading of a two-mode result is that the algorithm has rediscovered the difference between defending and attacking, which nobody needed clustering to learn. The contingency between the pooled partition and listed position group rules that reading out, though not in the direction the deflationary account expects.

Agreement is well above chance and well below recovery. The adjusted Rand index against listed position group is 0.293 and the normalised mutual information is 0.338, with cluster purity at 58.8 percent against a three-class problem. Scoring against the finer listed position instead of the group is lower still, at 0.262. A partition that simply relabelled position would return values near one, and a partition unrelated to position would return values near zero. These sit in between, which means the two modes divide the same players along a different line.

Where that line falls is the informative part. The defensive mode contains 3,522 defenders, 1,846 midfielders and 3 forwards. The attacking mode contains 1,701 forwards, 1,920 midfielders and 271 defenders. The poles are close to pure. 92.9 percent of defenders fall on one side and 99.8 percent of forwards on the other. Midfielders divide 1,920 to 1,846, which puts 51.0 percent of them on the attacking side and is as close to an even split as the sample permits, and it is that even split that drags the agreement indices down.

The partition is therefore not a relabelling of position but a cut through it. Defenders and forwards sit at the ends of the same axis and are assigned almost deterministically; midfielders are spread along its middle and are assigned by which end they lean toward. Section 5.9 shows the same geometry from the supervised side, where the classifier confuses midfielders with both poles and almost never confuses the poles with each other. Two methods with no shared machinery agree that the space is an ordered continuum whose density is concentrated at two ends, and that listed position is a coarse and partly arbitrary partition of it. Figure 3 shows the contingency and the agreement indices.

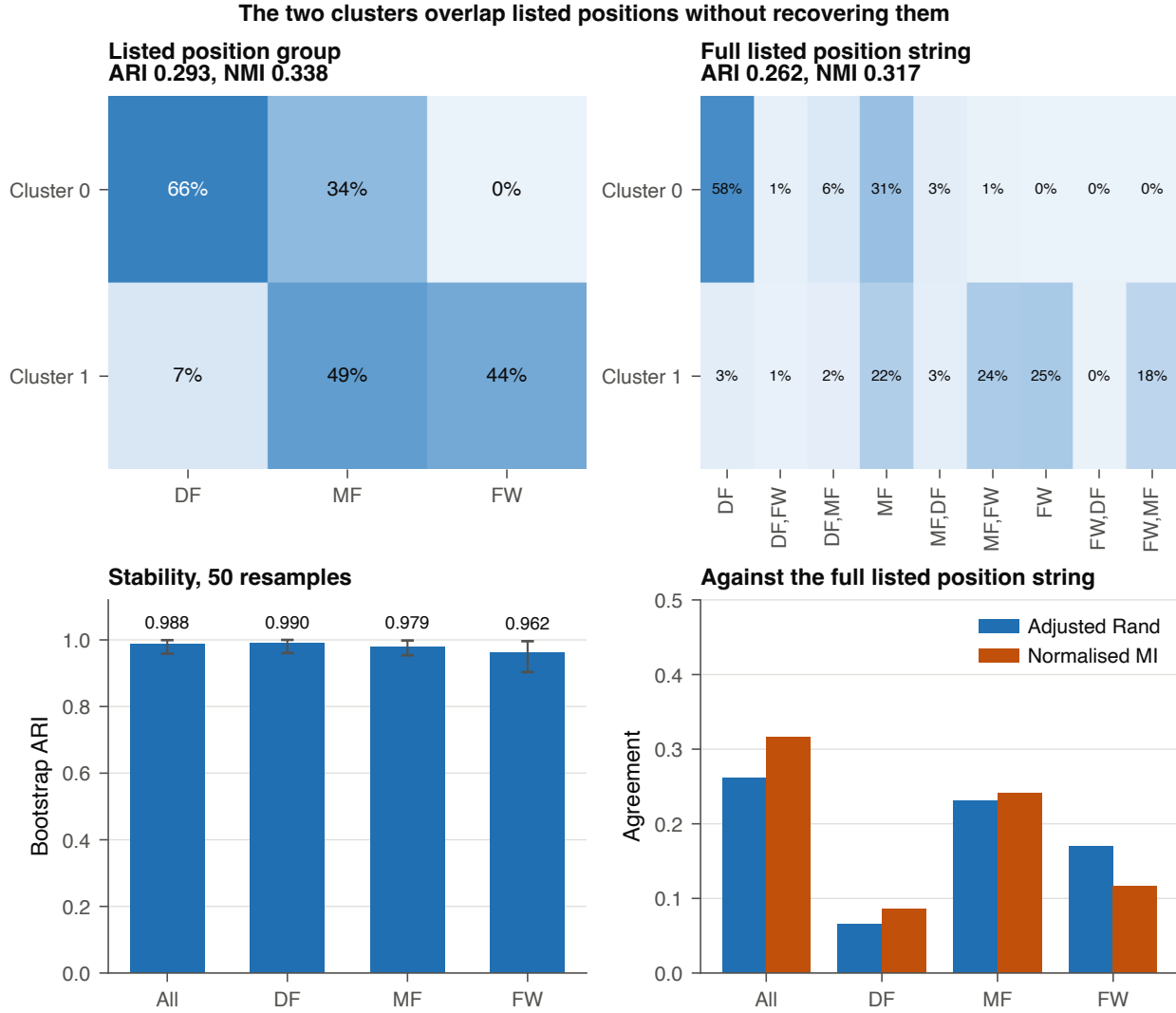


Figure 3: The pooled two-mode partition against listed position over 9,263 player seasons. The heat maps are normalised by row, so each gives the position composition of one mode rather than the split of one position group: the defensive mode contains almost no forwards, the attacking mode almost no defenders, and midfielders are the largest or second largest component of both. The text reads the same table by column, which is where the 92.9 and 99.8 percent figures come from.

5.5 Do the modes describe stable roles?

A partition that changed every season would describe a season, not a role. Tracking 3,479 players across the sample, of whom 2,327 appear in two or more seasons, membership changes in 7.8 percent of consecutive season pairs against a within-group chance baseline near half. The adjusted Rand index between consecutive seasons runs from 0.814 to 0.844 with a mean of 0.834.

Refitting the partition independently within each league and comparing against the pooled one gives a mean adjusted Rand index of 0.883, and refitting within each season gives 0.942. Bootstrap resampling returns 0.989 for defenders, 0.975 for midfielders and 0.958 for forwards. The two modes are a property of how players play rather than of which season or competition they were measured in.

The three groups are not equally well supported and the paper does not claim they are. The defender split is the strongest: the largest centroid separations, the lowest churn, the highest agreement across leagues, and a dip test that rejects unimodality on the defender axis. The midfield split is well supported with a more populated boundary. The forward split is the weakest, and should be read with more caution than the other two: its dip test does not reject unimodality, the division is uneven, the finishing group has no positive signature of its own, and one league's refit falls well below the others. Silhouette does separate it from the clusterless null, so there is real structure, but it reads more like a cut through a continuum of forward involvement than like two modes.

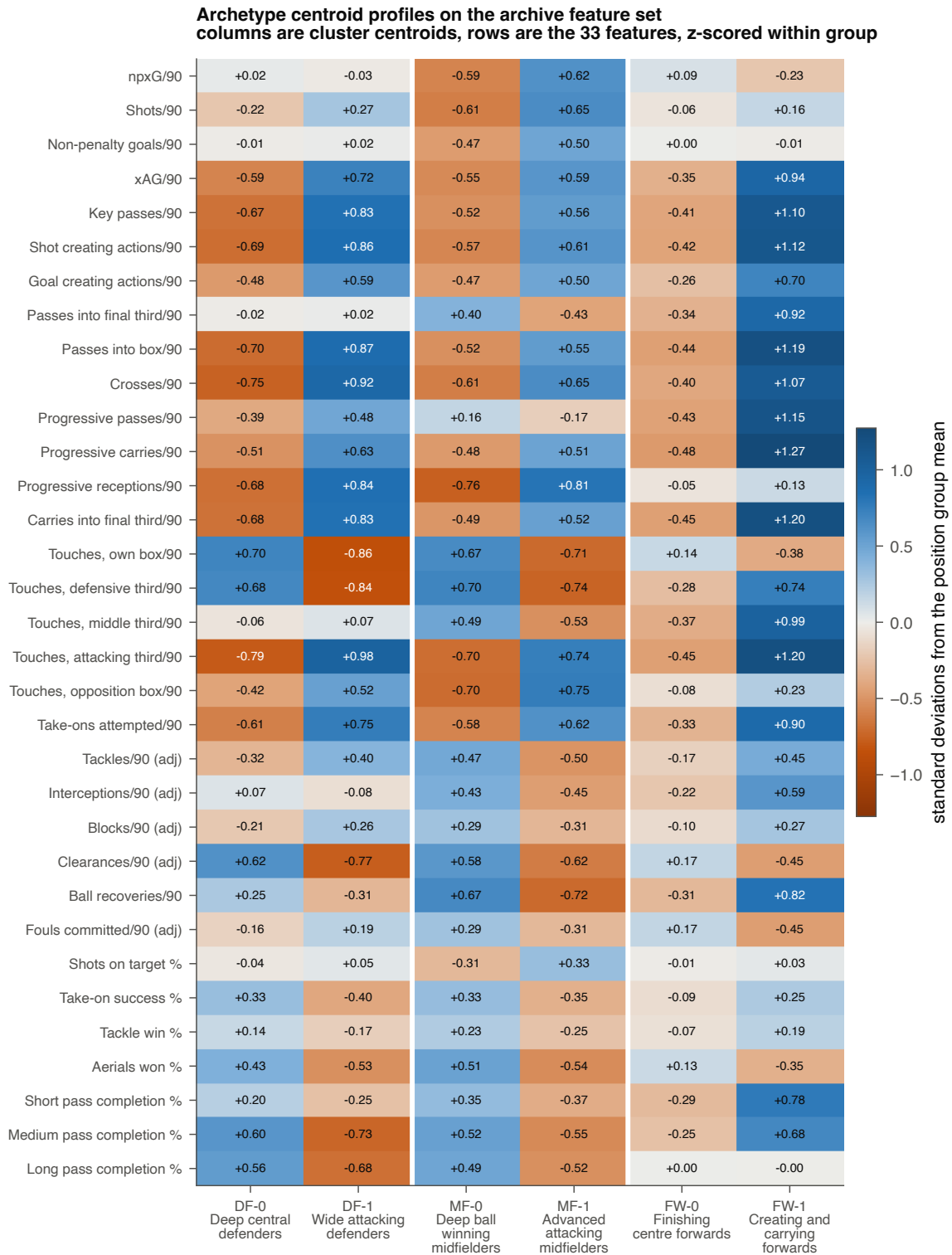
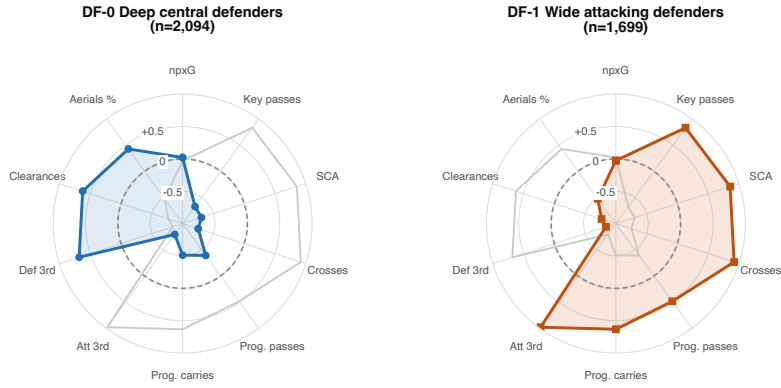
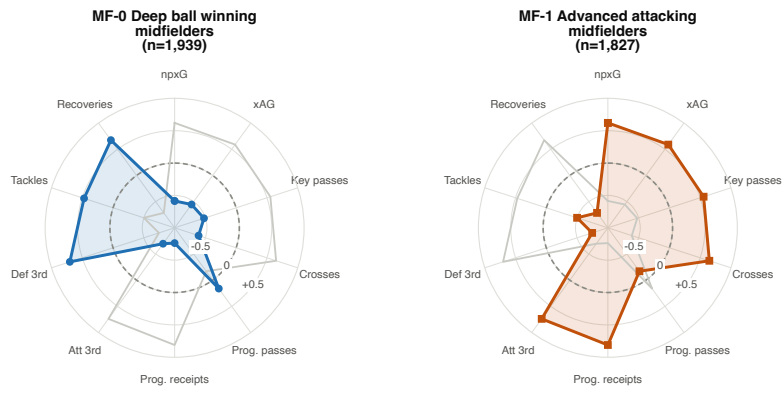


Figure 4: Centroid deviation of each archetype from its position group mean across the full feature set. Within a group the two profiles are mirror images by construction.

DF archetypes on the fixed radar axes, the other archetype shown in grey



MF archetypes on the fixed radar axes, the other archetype shown in grey



FW archetypes on the fixed radar axes, the other archetype shown in grey

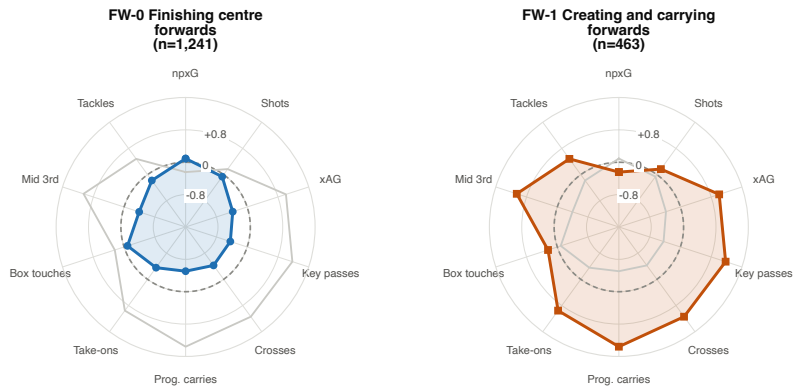


Figure 5: Archetype radars on fixed axes within each position group.

5.6 The same solution in every season and every league

A partition that appears once is a description of one dataset. A partition that appears independently in five seasons and five competitions is a description of the sport. The archive supports the stronger test, so each season and each league is standardised inside itself and clustered from scratch, borrowing no centre, no scale and no starting point from the pooled fit.

Every scope recovers the pooled partition. The weakest season agrees with the pool at an adjusted Rand index of 0.916 and the weakest league at 0.932, on 9,263 player-seasons in total. The separating axis is the same object each time and not merely a similarly sized one: the correlation between each scope’s own discriminant direction and the pooled direction never falls below 0.9983. Bundesliga, the league with the lowest agreement, is also the smallest sample here, and its axis correlation is indistinguishable from the rest.

Importing one scope’s centroids into another scope’s players separates replication of the solution from change in the players, and it is the cleaner test of the two. Across the twenty ordered league pairs the transferred centroids reproduce the receiving league’s own fit at a mean adjusted Rand index of 0.944, never falling below 0.893. A model of role fitted on Ligue 1 assigns Bundesliga players almost exactly as a model fitted on Bundesliga does. The two modes are not a quirk of one league’s tactical fashion.

The season-against-season comparison on players present in both seasons is lower, at a mean of 0.693 with a range from 0.570 to 0.753, and the pattern in Figure 6 explains why. Agreement decays monotonically with the gap between seasons, so adjacent seasons agree most and the seasons furthest apart agree least, while the transfer index over the same pairs stays flat. That decay is players changing role over four years, not a weakness of the partition, and the player-level view confirms it: across the same pairs 91.6 percent of players present in both seasons are assigned to the same mode in both, falling no lower than 87.8 percent on the pair furthest apart in time. Roughly one player in twelve moves between modes over a four-year span, which is a plausible rate of genuine role change and not the churn an unstable partition would produce.

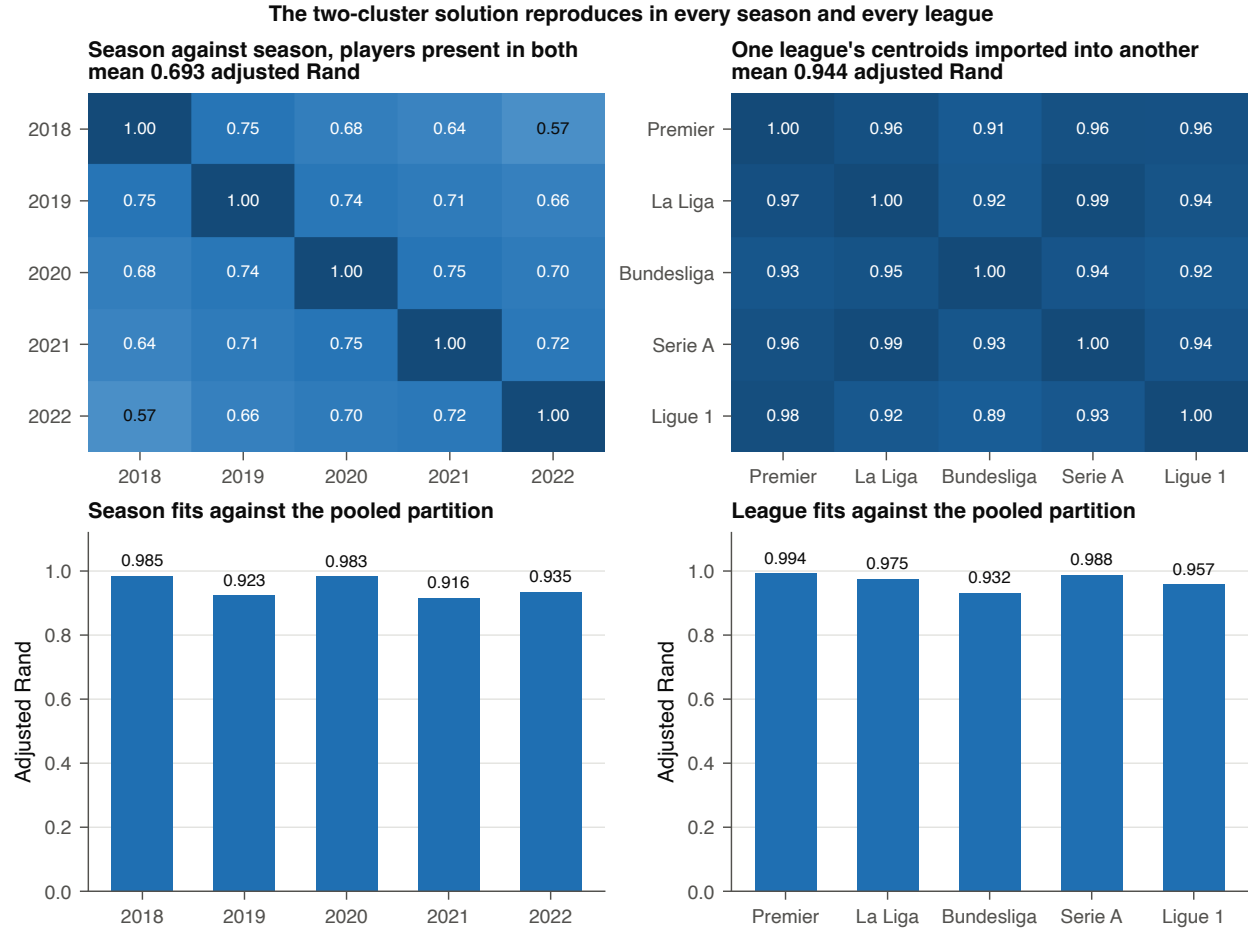


Figure 6: Independent refits of the two-mode solution. Top left, agreement between seasons on the players common to both, which declines as the gap widens because players change. Top right, one league's centroids imported into another league's players, which does not decline. Bottom, each season and each league scored against the pooled partition.

5.7 Do the modes predict results?

Everything above is a claim about the structure of the data. The archive also carries what happened: wins, draws and losses for every one of the 490 team-seasons, so the two modes can be tested against real results and not only against listed positions. Each club season is summarised by the minutes-weighted share of its outfield minutes played by the attacking mode, by the minutes-weighted mean position of its players along the axis between the two centroids, and by its share in each of the six archetypes, with eligible players covering a median 96.9 percent of each club's outfield minutes and never less than 89.5. Results are recovered from the match-by-match keeper record, which credits every match to the goalkeeper who played it, and audited against the schedule: 4 team-seasons close short of it, all for documented reasons, the largest being the Ligue 1 season abandoned in March 2020.

The share alone correlates with points per match at a Spearman rank correlation of 0.430 (0.348 to 0.502), and with goal difference per match at 0.442. The continuous axis position is stronger, at 0.655 (0.594 to 0.708) against points, which is what a continuum should show: where a club's players sit along the axis carries more than which side of the cut they fall on. Within leagues the

share correlation runs from 0.19 in La Liga to 0.61 in Premier League, and every season lies inside that range. Table 2 lists the ten team-seasons at the extremes of the share so that a reader can check them against memory, and they are the sides one would name: the champions of Germany, Spain and England at the top, and a Sheffield United season that finished last near the bottom.

A correlation between attacking composition and points is partly circular, because good teams have good attackers, so the test that matters is whether the modes carry anything beyond team possession, the simplest summary of dominance and one already in the paper’s exposure adjustment. Ridge regression of points per match on possession alone, on the composition alone and on both, with the penalty chosen inside the training folds, gives out of sample coefficients of determination of 0.566, 0.545 and 0.664 when one season is held out at a time, and 0.557, 0.534 and 0.656 when one league is. The strictest holdout leaves one club out at a time, so that no held-out team-season shares a club with the training data and a model cannot score by remembering which clubs are good; there the three figures are 0.558, 0.546 and 0.661. The modes add 0.099 (0.067 to 0.131) to what possession explains under the first holdout and 0.098 (0.066 to 0.132) under the second and 0.102 (0.071 to 0.135) under the club holdout, and for goal difference 0.114, 0.116 and 0.120. The composition on its own predicts slightly worse than possession on its own, which should be said plainly: the modes are not a substitute for dominance, they are information about how it is produced that dominance does not contain.

Where the information lives is the sharpest result in this section. Possession plus the two two-mode measures alone reach 0.660, an increment of 0.094 over possession that is the whole of the gain. Adding the six archetype shares on top adds 0.005 (-0.006 to 0.015), an interval that contains zero, and 0.002 (-0.009 to 0.012) under the club holdout; the archetypes without the two-mode measures reach only 0.579. Real results, then, say what the null calibration said: two modes carry information and a finer taxonomy carries none that the two modes did not already carry.

The goalkeeper claim can be tested the same way. The share of a club’s goalkeeper minutes falling in the full-feature partition’s second cluster correlates with points per match at 0.713 (0.667 to 0.753), which is what a partition that sorts keepers by the team in front of them must do. The technique partition’s share correlates at 0.087 (0.002 to 0.174), an interval that barely clears zero, so the line keeper against sweeper claimer division is unrelated to how good the team is, which is the property a description of goalkeeping rather than of goalkeepers’ teams should have. Shot stopping itself, post-shot expected goals minus goals allowed per ninety, sits between the two at 0.337. Figure 7 shows the share against points and the out of sample comparison.

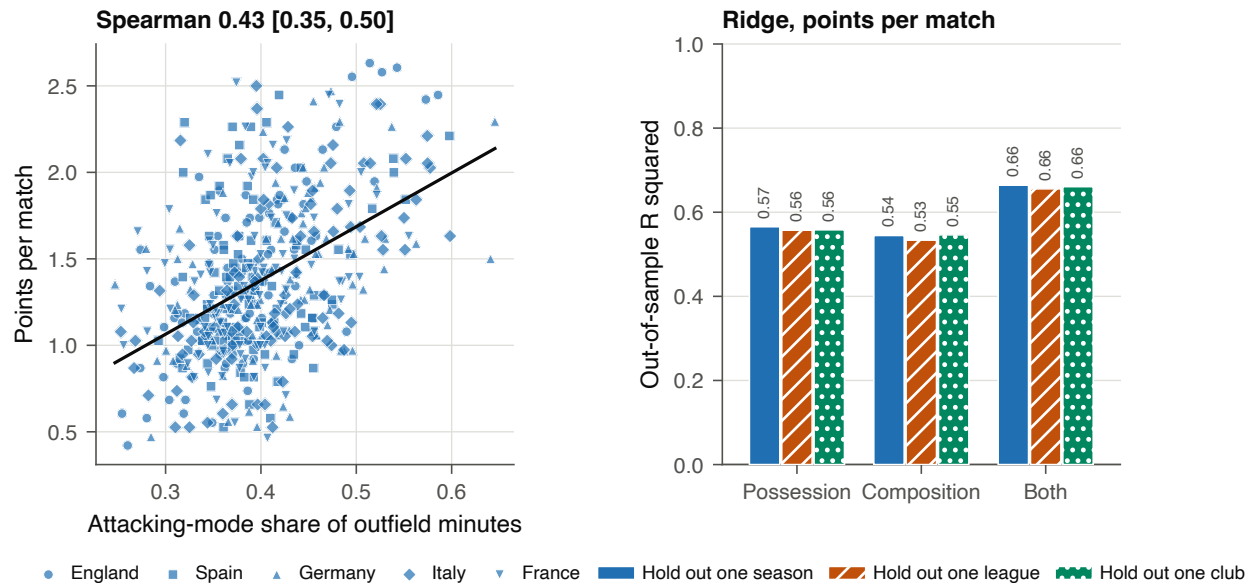


Figure 7: The two modes against real results over 490 team-seasons. Left, the minutes-weighted share of outfield minutes in the attacking mode against points per match, one marker shape per league. Right, out of sample coefficient of determination for points per match from possession alone, the composition alone and both, under leave-one-season-out, leave-one-league-out and leave-one-club-out holdouts.

Table 2: The five team-seasons with the highest and the five with the lowest minutes-weighted share of outfield minutes in the attacking mode, with their real points per match and final league position.

Club	League	Season	Share	Points per match	Position
<i>Highest attacking-mode share</i>					
Bayern Munich	Bundesliga	2019	0.645	2.29	1
Hoffenheim	Bundesliga	2019	0.641	1.50	9
Fiorentina	Serie A	2022	0.598	1.63	7
Real Madrid	La Liga	2021	0.598	2.21	2
Manchester City	Premier League	2022	0.586	2.45	1
<i>Lowest attacking-mode share</i>					
Mainz 05	Bundesliga	2022	0.247	1.35	8
Parma	Serie A	2019	0.253	1.08	16
Reims	Ligue 1	2022	0.254	1.21	12
Sheffield Utd	Premier League	2021	0.254	0.60	20
Troyes	Ligue 1	2022	0.256	1.00	15

5.8 Which features carry the modes

A result on 33 features could be carried by one block of them, and a reviewer who suspected that the modes were the expected goals block wearing a costume, or the defensive counts, would be right to ask. The features fall into 6 families, shooting, chance creation, ball progression, pitch territory, defending and passing accuracy, and each was removed in turn and each was tried alone, with the pre-declared selection rule and the Gaussian null calibration rerun every time in every scope. Table 3 and Figure 8 give every cell.

Table 3: Standard score of the observed two-cluster silhouette against the clusterless Gaussian null, by scope, under every feature-family ablation. A dagger marks a cell where the pre-declared rule chose a number of clusters other than two.

Feature set	All	DF	MF	FW
<i>Full feature set</i>				
all 33 features	18.0	24.2	15.0	13.2
<i>Leave one family out</i>				
without shooting	17.7	21.3	12.3	18.0
without creation	21.2	21.1	14.5	15.1
without progression	19.0	19.7	13.2	19.8
without territory	21.6	22.6	16.0	15.6
without defending	19.7	29.1	16.0	17.2
without passing	19.1	17.5	10.2	20.1
<i>One family alone</i>				
shooting only	29.3	19.8	16.1	2.9
creation only	12.8	22.6	11.7	18.4
progression only	8.3	15.6	9.0	12.9
territory only	11.4	26.6	16.2	20.2
defending only	19.0	6.9	10.7	8.5
passing only	6.6	11.4	5.3	1.4

Removing any one family removes nothing that matters. The rule returns two clusters in every leave-one-out in every scope, and of the 52 feature sets tested across the four scopes, 51 exceed every one of the 25 clusterless simulations. The weakest case is MF without passing, where the observed silhouette still sits 10.15 null standard deviations above the clusterless mean. The survivors are the same two modes, not two different ones: the adjusted Rand index between each ablated partition and the full one, on the same players, runs from 0.703 to 0.990, with the least similar case being FW without creation at 0.703. Forwards are the group where that family matters most, which is consistent with the forward split being a division by involvement in build-up, not by scoring.

The converse is more instructive than the headline. Judged by whether a single family produces a two-cluster solution that beats the null, most families carry structure alone: creation, progression, territory, defending do so in every scope, and every family does so in at least one. That criterion is generous, and the adjusted Rand index against the full partition shows why. Shooting alone in defenders separates from the null at 19.79 standard deviations and agrees with the paper’s partition at 0.000: it finds two groups, and they are the finishers against everyone else, not the deep against the wide. Passing accuracy alone reaches the null threshold in three scopes while agreeing with the full partition at no better than a third. Only territory reproduces the paper’s own modes by itself in every scope, at an adjusted Rand index from 0.732 to 0.856, and defending does so in the pool at 0.667. The two modes are where a player touches the ball, and every other family either sharpens that division or, alone, cuts a different one.

The two failures are forwards. Shooting alone in forwards clears the null by only 2.92 standard deviations, and passing accuracy alone reaches 1.42 and does not exceed every simulation. A forward’s shot volume and pass completion do not, on their own, divide forwards into anything the null cannot produce, which is a further reason to read the forward split as a cut through a continuum of involvement, not as two kinds of forward.

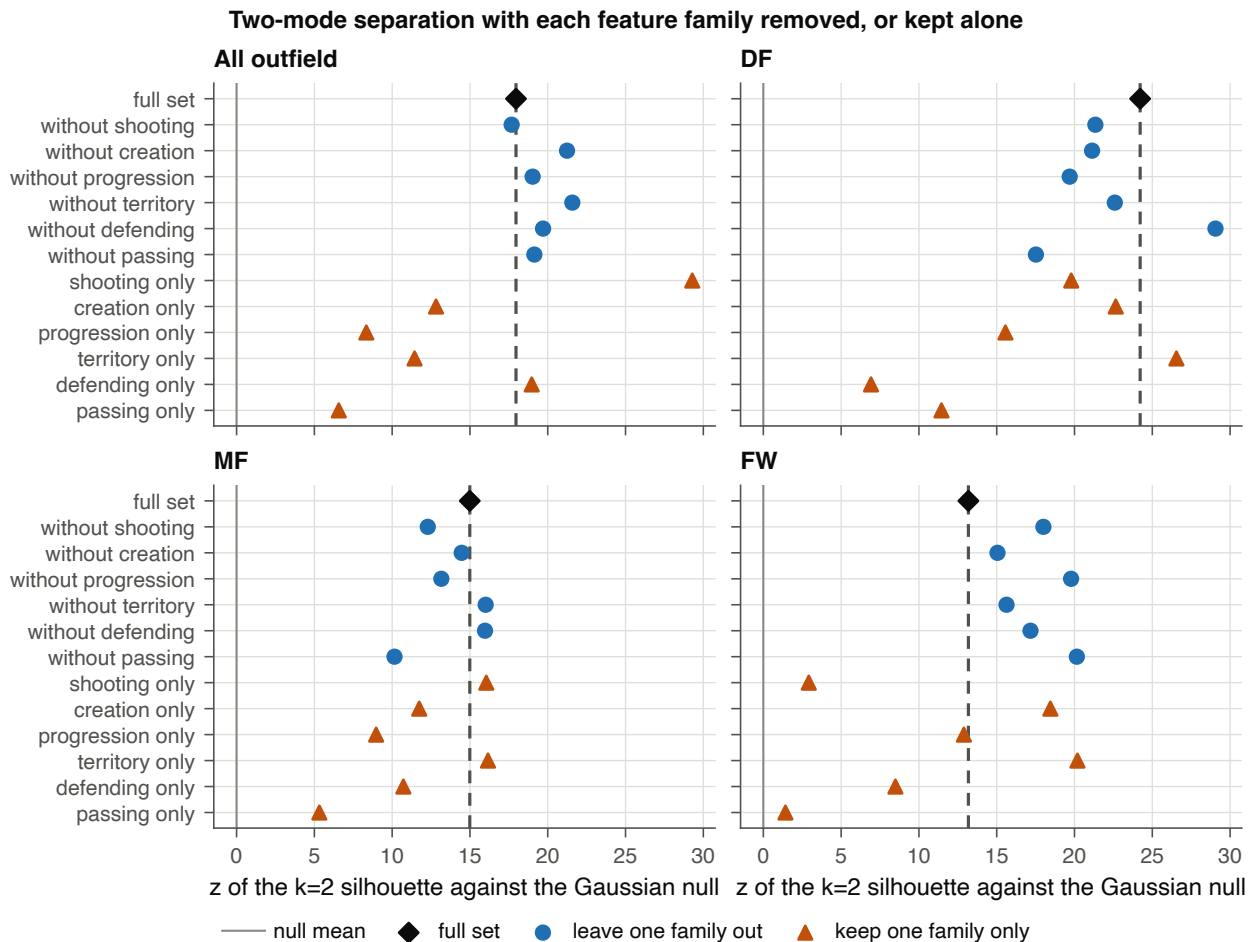


Figure 8: Standard score of the observed two-cluster silhouette against the clusterless null under every feature-family ablation, one panel per scope. Circles remove one family, triangles keep one family alone, and the vertical line is the full feature set. The adjusted Rand index against the full partition, which distinguishes the same two modes from a different pair, is in Table 3 and the text.

5.9 Position seen through the two modes

The unsupervised result says the space has two poles. A supervised model gives an independent way to check that claim, because if the space really is a continuum with two ends then the two ends should be easy to tell apart and the middle should be hard, and the errors should fall in a specific place rather than spread evenly.

Listed position group is highly predictable from the 33 features. Gradient boosting reaches 93.4 percent accuracy and a macro F1 of 0.921 out of fold against a majority-class baseline of 40.9 percent, and multinomial logistic regression is barely behind at 93.1 percent and 0.918. The near equality of a linear model and a boosted tree ensemble is itself informative, because it means almost all of the recoverable signal is linearly separable and there is little interaction structure left for the trees to find. A taxonomy of many discrete types would be expected to leave more.

The errors are where the argument is. Defenders and forwards are almost never mistaken for one another: of 9,263 player-seasons the model assigns 0 defenders to the forward class and 8 forwards to the defender class. Every other confusion runs through midfield, which is involved in 603 errors

against the 8 that link the two poles directly, so only 1.31 percent of all misclassifications skip the middle. Midfielders are recovered at 93.3 percent and forwards at 83.9 percent, against 97.8 percent for defenders, with per-class F1 of 0.921, 0.865 and 0.977 respectively. This is the confusion structure of an ordered line whose ends are far apart and whose middle overlaps both, and it is not the confusion structure of three distinct categories, which would scatter errors more evenly across the off-diagonal. Figure 9 shows both confusion matrices.

The feature attributions agree. The three strongest features overall are Touches def third, Clearances (adj) and Ball recoveries, all of which are territorial, not technical, and touches in the defensive third alone carries more attribution than any other feature for all three classes. Position, as the data record it, is mostly a statement about where a player stands. Figure 10 shows the attribution of every feature for every explained player. Figure 11 splits the attribution by class and shows that the three are separated by different things, defenders mostly by defensive volume and crossing and forwards mostly by where they receive the ball.

The direction of each attribution adds a detail that magnitude alone hides. Ranking the correlation between a feature’s value and the attribution it makes to a class separates a class identified by what its members do from one identified by what they do not, and the three classes differ on exactly that. 4 of the five strongest features for defenders are positively signed, so defenders are recognised by the presence of what they do. Forwards and midfielders are recognised as much by the absence of defensive volume as by the presence of attacking output, with only 2 and 2 of their five positive respectively. A class defined largely by negations is a class occupying one end of a scale rather than a category with a signature of its own.

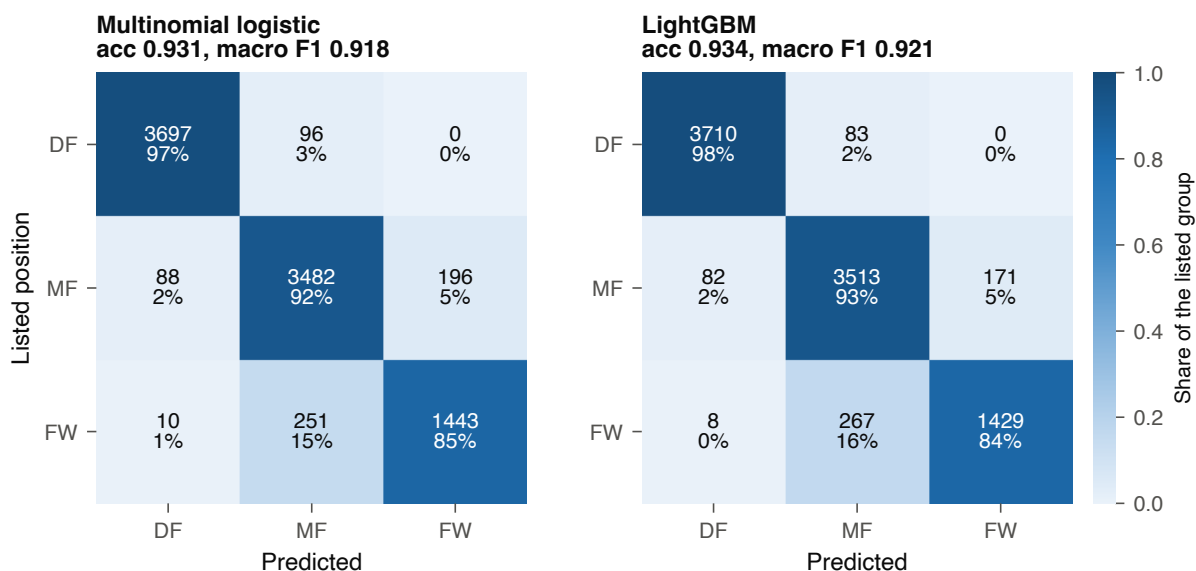


Figure 9: Out of fold confusion for both classifiers on 9,263 player-seasons. The corners linking defenders and forwards are close to empty and every substantial error passes through midfield, which is what an ordered space with two poles produces.

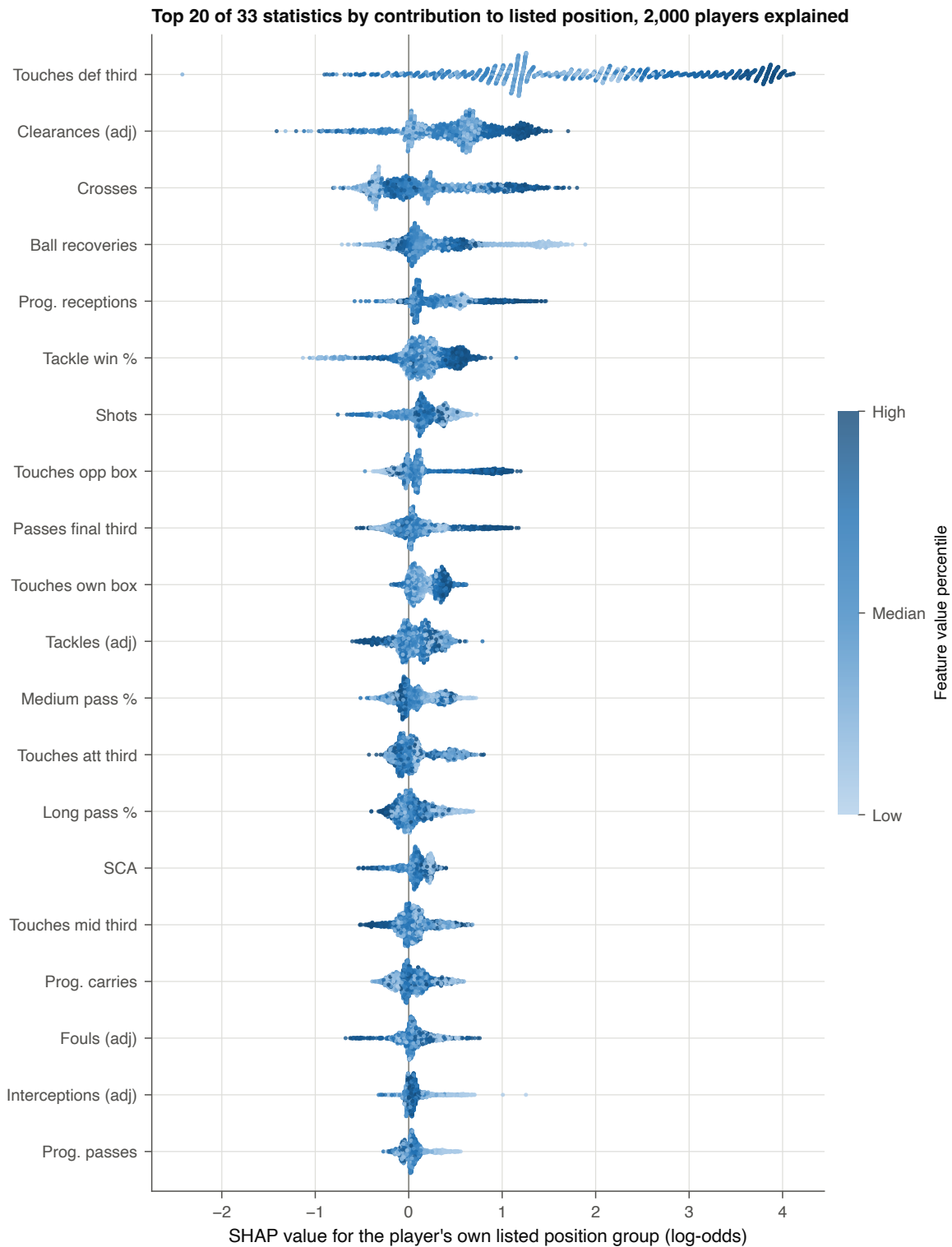


Figure 10: SHAP values for the gradient boosted model, showing for each player the contribution of each feature toward the position group the player is actually listed at. Territorial features dominate technical ones throughout.

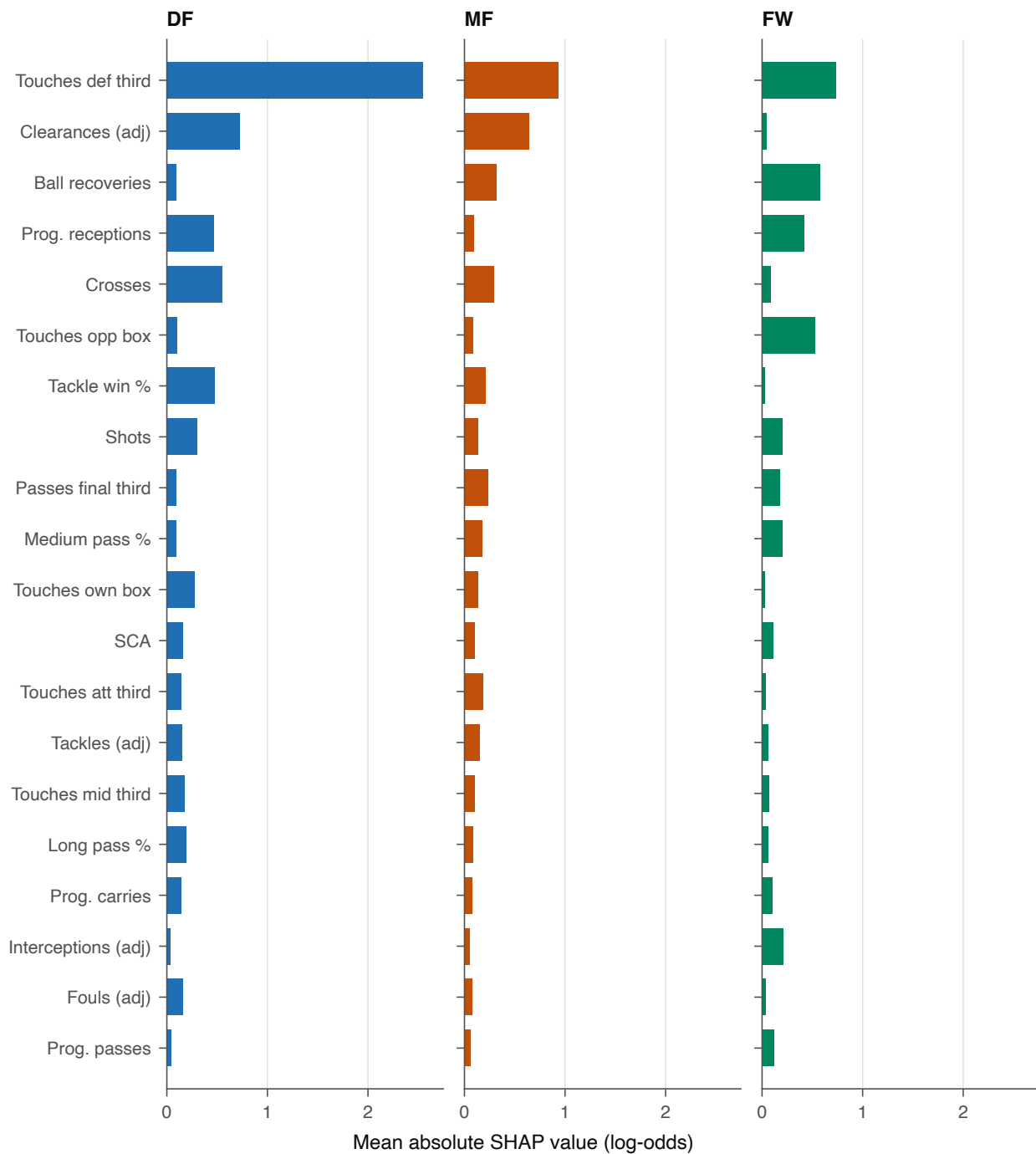


Figure 11: Mean absolute attribution per feature, split by position group. Touches in the defensive third dominate for all three, and the profile of what matters differs sharply between them: defenders are separated largely by defensive volume and crossing, forwards by where they receive the ball. The panel shows magnitude only; the direction each feature pushes is reported in the text.

5.10 Players the description fits worst

A two-mode description of a continuum will fit some players badly, and which players those are is a test of whether the description is measuring football or noise. Isolation forests and Mahalanobis distance under a shrinkage covariance were fitted separately within each position group, and the two detectors agree on 80.5 percent of the players they flag.

The 15 strongest outliers are 12 distinct players, headed by Neymar. Two features of the list stand out. The first is that 11 of the 15 carry a compound listed position such as midfielder and forward together, against a much smaller share in the pool, so the players the partition handles worst are the ones the listing system also declines to place. The second is that the detector is not simply finding stars: it also finds attacking full backs whose output profile resembles a winger's, which is the same boundary the defender split describes at Section 5.3 and a genuine tactical category, not an artefact.

One caveat belongs with the list, not after it. 9 of the 15 played under nine hundred minutes in the season concerned, so their per ninety rates carry more sampling noise than a full season's would, and extreme rates are easier to reach on a small denominator. The eligibility threshold removes the worst of this and does not remove all of it, and an outlier list is the place where the residual matters most. The list should therefore be read as evidence about the kind of player the two-mode description struggles with, and not as a ranking in which the fifteenth entry means what the first does. Figure 12 shows both detectors and the fifteen.

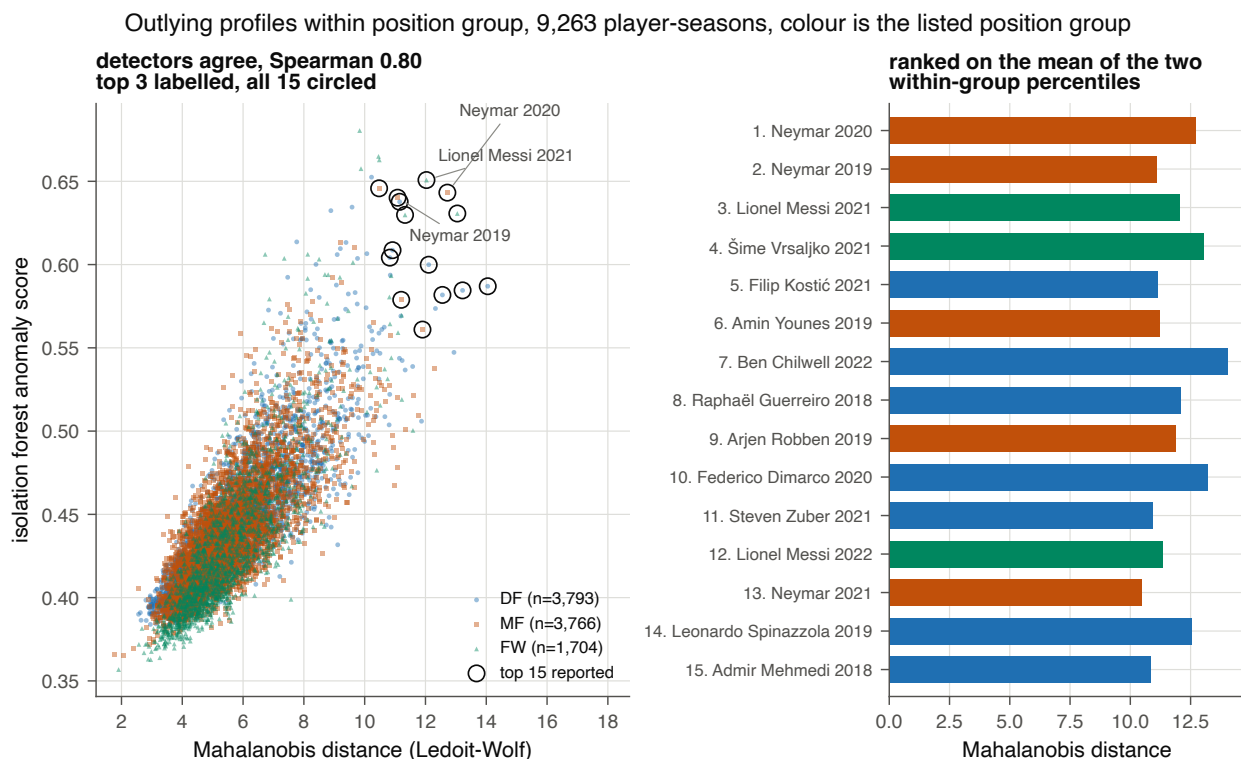


Figure 12: Isolation forest score against Mahalanobis distance within position group. The two detectors are computed independently and rank the same players highly, and the flagged points are dominated by players carrying compound listed positions.

5.11 Correcting defensive counts for possession

Before any structure is estimated, the defensive counts have to be separated from the opportunity to record them. Team defensive volume does rise as opponent possession rises, so the confound is real, and the conventional correction, in the proportional form that has succeeded the original of Knutson [2014], assumes the relationship is proportional: a side with twice the ball is taken to concede half the defensive events.

It is much weaker than that for four of the five statistics, and it is not distinguishable from proportional for the fifth. Fitted on the primary sample over 490 team-seasons, with 2,000 bootstrap resamples of those team-seasons giving each estimate an interval, the elasticities are 0.172 for tackles (0.109 to 0.235), 0.353 for interceptions (0.171 to 0.540), 0.439 for blocks (0.379 to 0.499) and 0.443 for fouls committed (0.359 to 0.526). All four intervals exclude one, and no bootstrap resample of any of them reached it. Clearances are different: the estimate is 0.958, the interval runs from 0.823 to 1.091, 28.2 percent of resamples land at or above one, and the bootstrap probability that the true elasticity is at least one is 0.2975. For clearances the conventional exponent cannot be rejected, and the paper does not claim otherwise.

That pattern fits the football. A clearance is the most purely reactive action in the set, made when the ball arrives rather than sought, so it is the statistic for which the exposure story should hold best, and it is the one where proportionality survives. A tackle is the most volitional, and it is furthest from proportional. The spread from 0.172 to 0.958 is itself the point: a single exponent applied to every defensive statistic is wrong for four of them, and the one it is right for is the one a reader would have guessed.

The reduced sample supplies the check the primary sample cannot, because it retains the team-level possession series needed to measure what is left after correcting. Across its 40 team-seasons, team possession ranges from 40.6 to 61.3 percent, and its fitted elasticities agree with the primary sample in falling well below one, at 0.525 for interceptions, 0.432 for tackles won and 0.283 for fouls committed. Table 4 shows what each choice of exponent leaves behind. The three raw correlations with team possession are -0.122, -0.111 and -0.102, all negative, since a player whose team keeps the ball records fewer defensive actions. Applying the conventional exponent overshoots every one of them: the correlations become 0.070, 0.105 and 0.129, reversing sign in all three cases and, for fouls committed, ending further from zero than where it started. The correction passes through the confound and out the other side. Adjusting with the measured elasticity instead leaves -0.022, -0.018 and -0.037, which are three to four times closer to zero than the conventional exponent achieves and are the smallest residual dependence available from either choice.

Two qualifications belong with this. The residual-dependence check runs on one league, whose possession shares are relatively compressed, so the exact residuals should not be transported to a competition with a wider spread without refitting. The elasticities themselves are fitted across 5 leagues and 5 seasons and are less exposed to that objection. Refitting within each league tests that directly, and the answer is mixed. For tackles, interceptions, fouls and clearances every league's interval overlaps the pooled one. Blocks do not transport: the league estimates run from 0.263 in La Liga to 0.615 in Premier League, and the intervals for both of those leagues miss the pooled interval entirely, so a pooled exponent for blocks is the wrong exponent in at least two of the five competitions. Clearances are also the statistic where leagues disagree most in absolute terms, with estimates from 0.768 to 1.217 and 3 of the five league point estimates at or above one, which is a second reason not to claim sub-proportionality for them.

The constant elasticity form is itself an assumption, and for 3 of the five statistics it fails. Adding a quadratic term in log opponent possession improves the fit for tackles, blocks and fouls, with a coefficient whose interval excludes zero in each case, and in each the response flattens toward zero for

Table 4: Possession adjustment. The elasticity is fitted from team totals. The final three columns give the correlation between each per-90 rate and team possession: unadjusted, adjusted with the fitted elasticity, and adjusted with the conventional exponent of one. A value near zero is the goal.

Statistic	Elasticity	Correlation with team possession		
		Unadjusted	Adjusted	Exponent one
Interceptions	0.525	-0.122	-0.022	0.070
Tackles won	0.432	-0.111	-0.018	0.105
Fouls committed	0.283	-0.102	-0.037	0.129

the sides that see least of the ball. Fouls committed bend most: the implied elasticity is 0.90 at 37.6 percent opponent possession and -0.07 at 58.9 percent, with the adjusted coefficient of determination rising by 0.040. The exponents reported above are therefore averages over the possession range rather than constants, and the correction they define is coarser than it looks. Clearances and interceptions show no evidence of curvature. Figure 13 gathers the pooled estimates, their intervals and the league refits.

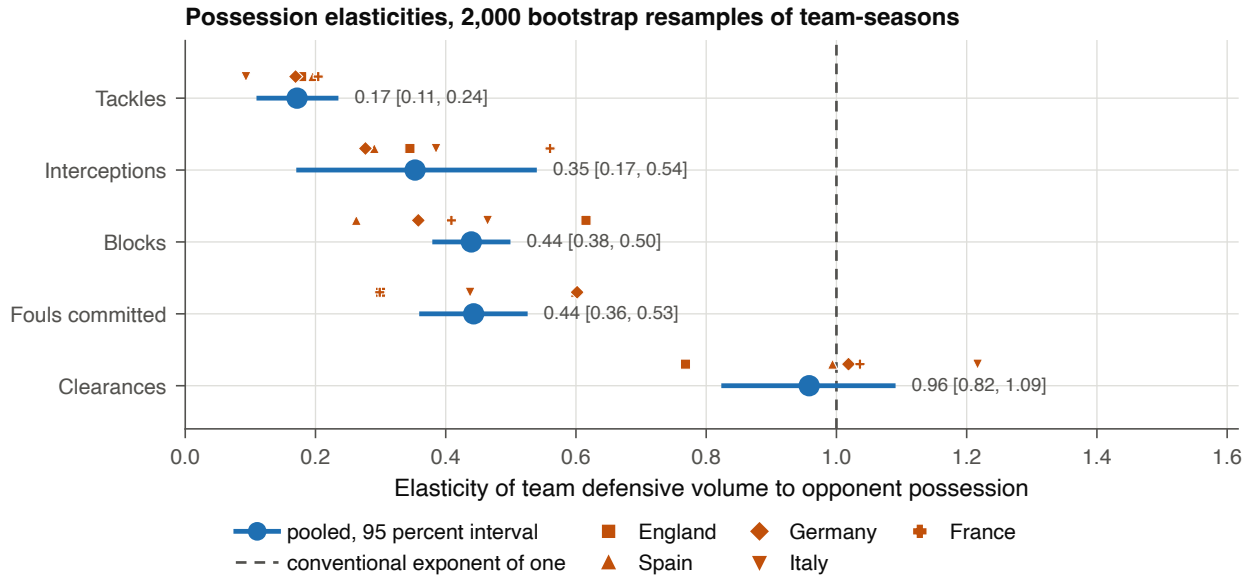


Figure 13: The five defensive possession elasticities with bootstrap intervals from 2,000 resamples of 490 team-seasons, against the conventional value of one, with each league's own refit shown as a smaller marker. Four intervals exclude one; the interval for clearances does not, and three league refits for clearances sit above one.

One further qualification matters more than any of these. Adjusting the defensive counts does not make the feature set as a whole team-neutral, because the attacking statistics still scale with a team's own share of the ball. What follows should be read as having removed one identifiable confound, not all of them.

5.12 Why the correction is one sided

Correcting only the defensive statistics invites an obvious objection. Attacking output also depends on possession, so the feature set is still team dependent and the correction looks like half a correction. The objection deserves a test, not an assertion, and the test produces the sharpest methodological result in the paper.

This test runs on the reduced sample, which is the only one of the two carrying an independent expected-goals source with involvement measures, and its conclusion is about the shape of the two relationships rather than about their exact magnitudes.

The two sides of the ball do not behave alike. Fitted the same way, against a team's own possession rather than the opponent's, the attacking elasticities are 1.09 for shots, 1.48 for non-penalty expected goals, 1.63 for non-penalty goals, 1.72 for assists, 2.12 for xGChain and 2.44 for xGBuildup. Every one of them except crossing, at 0.42, exceeds one, while four of the five defensive elasticities fall well below it and the fifth, clearances, cannot be distinguished from it. Figure 14 shows the two families either side of the proportional line.

That asymmetry is not a curiosity, it is the whole argument. For defending, opponent possession is an exposure denominator in the plain sense that a tackle cannot occur while your own side holds the ball, and the sub-proportional response says exposure is what it measures. For attacking, an elasticity above one says the opposite: ten percent more of the ball buys appreciably more than ten percent more output. Returns to possession are increasing. That is a statement about football, not about measurement, so dividing it out would remove signal along with any bias. The two Understat involvement measures make the point most starkly, since both exceed two: a side that keeps the ball does not merely have more possessions, its players are involved in more of each.

The counterfactual was nonetheless computed. Applying the fitted exponents to the attacking block as well moves the correlation between club centroids and final league position from -0.861 to -0.610, so a symmetric correction does remove a real part of the team relationship rather than nothing at all. It does not remove all of it, the partition moves at an adjusted Rand index of 0.872 against the one reported here, and the silhouette drifts from 0.285 to 0.281. The choice is therefore a judgement, not a fact, and it is recorded as one: we treat possession driven productivity as something a role profile should reflect, not something to divide away, and a reader who takes the opposite view has the numbers to see what that would cost.

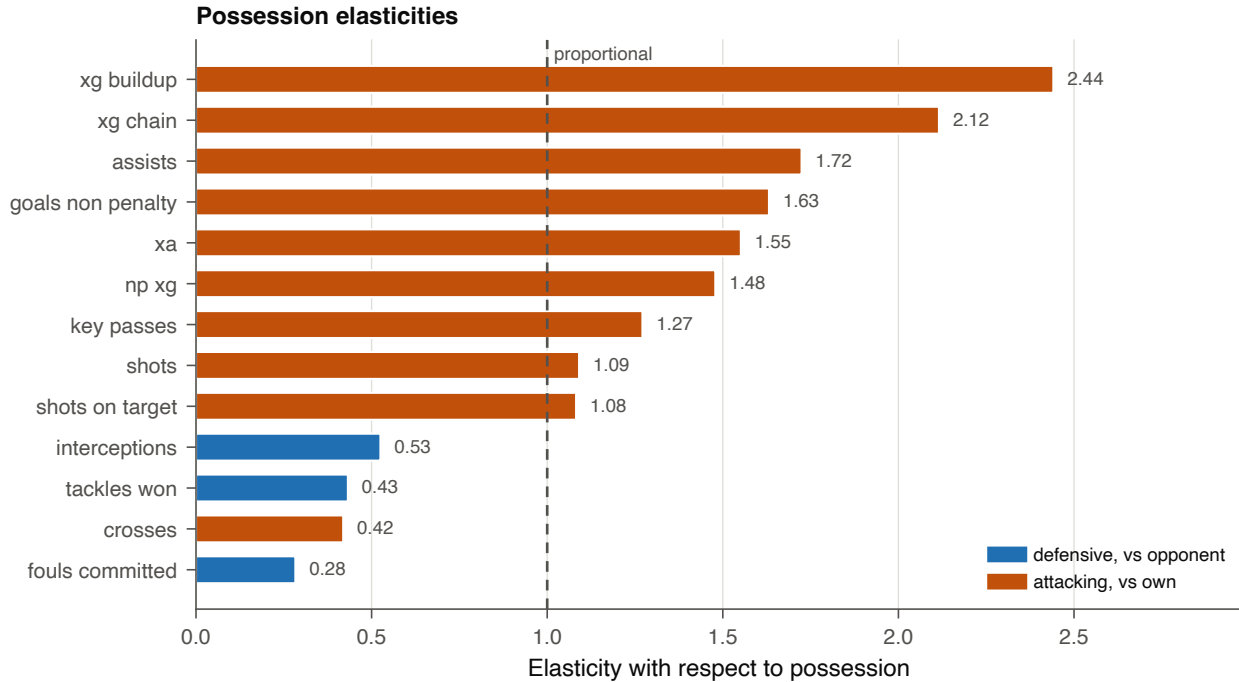


Figure 14: Possession elasticities for both families of statistic, fitted from team totals. Defensive volume responds to opponent possession less than proportionately, while attacking output responds to a team’s own possession more than proportionately. Crossing is the one attacking statistic that behaves like a defensive one.

5.13 Goalkeepers

Goalkeepers are described by a different vocabulary from outfield players, and the primary sample supplies all of it: 1,039 keeper-seasons with post-shot expected goals, sweeper actions and their average distance, cross stopping, launch rate and average pass length, at a worst per-column coverage of 0.962. Post-shot expected goals minus goals allowed is the proper measure of saves above expectation, because it conditions on the difficulty of the shots faced rather than only their number.

The first attempt at a partition walks into a trap, and the trap is itself a finding. Clustering on all available features returns 2 clusters at a silhouette of 0.207, and the partition looks interpretable until it is asked what it is made of. Membership explains a mean of 0.386 of the variance in situational quantities, the volume and quality of shooting a defence permits, against 0.109 in quantities describing what the goalkeeper does. The partition is largely a ranking of the teams in front of the goalkeepers. Named as archetypes it would be a league table wearing a tactical vocabulary.

Excluding the exposure variables and clustering on technique alone reverses that completely. The same rule again returns 2 clusters, at a silhouette of 0.189 and a bootstrap adjusted Rand index of 0.570, and now membership explains 0.135 of the variance in technique against 0.019 in situation, a ratio of about seven. This partition describes goalkeepers.

What it describes is not the expected taxonomy. The familiar trio of shot stopper, sweeper keeper and distributor is not what the data produces. The division is between Line keepers, 411 keeper-seasons who stay on their line, and Sweeper claimers, 282 who come off it, separating on cross stopping and sweeping together. Distribution style barely separates the clusters at all, and neither does shot stopping ability. Both are real and measurable, and both are continua lying across the off-line division rather than dimensions of it. A goalkeeper who sweeps is not thereby a better or

Shot stopping adjusted for shot quality, which the post-withdrawal data cannot measure

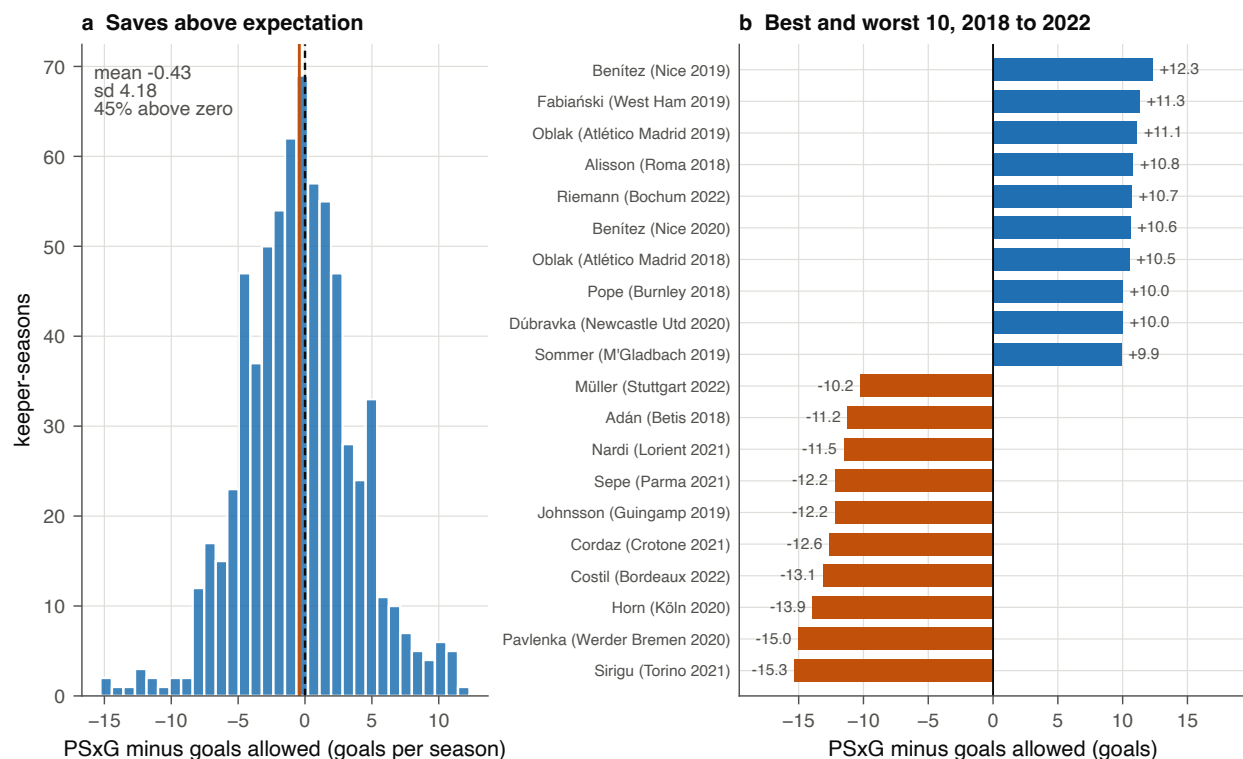
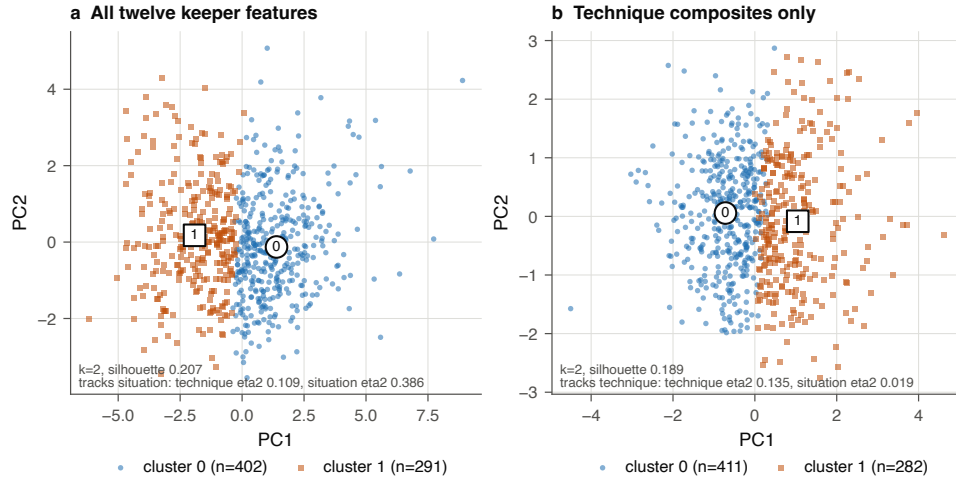


Figure 15: Goalkeepers ranked by post-shot expected goals minus goals allowed, which conditions on the difficulty of shots faced rather than only their volume.

worse shot stopper, and how far he kicks the ball is close to independent of both. One check belongs here because the paper applies it to every outfield defensive count. Sweeper actions are counts, and could be exposure driven in the way tackles are. They are not: fitted exactly as the outfield elasticities were, over 490 team-seasons, sweeper actions per ninety rise rather than fall with a team's own possession, with an elasticity to opponent possession of -0.631 (-0.899 to -0.358), because sides that keep the ball defend a high line and ask their goalkeeper to cover behind it. Sweeping is a tactical count that tracks line height rather than an exposure count, and adjusting it for possession would remove the signal the partition is built on, so it is left unadjusted.

That is a substantive finding, not a taxonomy, and it echoes the outfield result: the data supports one binary distinction plus continuous variation, not a set of types. Figures 15 and 16 show the shot stopping ranking, the partition and the two profiles.

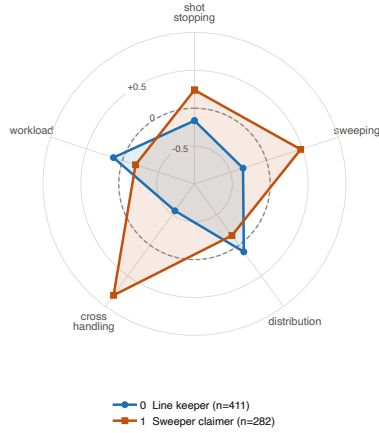
Removing the exposure block turns a team strength partition into a technique one



(a) The technique partition.

Goalkeeper cluster centroids on the five composite axes

standard deviations from the eligible pool mean, dashed ring is that mean
distribution is signed so that higher means shorter passing



(b) Composite profiles.

Figure 16: Goalkeeper structure on the primary sample. The clusters separate on off-line activity; distribution and shot stopping vary continuously across both.

Analyses on the reduced sample

The two subsections that follow use the reduced sample of Section 3.4, the Premier League for the seasons ending 2025 and 2026, which retains roughly half the features. The first asks whether the main result survives that description, since a finding that holds on both is worth more than one that holds on either alone. The second reports a property of the feature set that does not depend on the clustering at all. The remaining reduced-sample analyses, which support the argument without carrying it, are in Appendix H.

5.14 Robustness: the same result on half the features

The primary sample describes each player with 33 features. A natural worry is that the two-mode result depends on that description, and in particular on the expected goals and possession blocks that dominate it. The second sample of Section 3.4 provides an unusually clean test, because the withdrawal removed a specific and large part of the feature set without touching the population: the same competition, the same kind of players, roughly half the features.

Repeating the analysis there returns the same answer. The declared rule selects two clusters globally and within every position group, agreement with listed positions remains weak at 0.226, and bootstrap stability of the global partition is 0.914. Density-based clustering again assigns every position group entirely to noise. Removing expected goals, all passing and possession volume, shot and goal creating actions and the entire advanced goalkeeping table changes what the two modes are made of without changing that there are two of them.

That is informative in both directions. It supports the central claim, since a result that survives halving its own feature set is not an artefact of any particular measurement. It also bounds what richer data buys: the modern feature block sharpens the description of the two modes considerably, as Section 5.3 shows, but it does not reveal additional modes that the poorer description was hiding.

The remaining reduced-sample analyses are reported in Appendix H. Several of them, in particular the within-season role drift of Section H.8.4, depend on match-level data that exists only for that window, which is why they are retained at all.

5.15 Statistics that change meaning by position

The correlation analysis produced the clearest substantive finding available from the reduced feature set. Correlations were computed within each position group and compared across groups, and the ten pairs whose correlation varies most between groups are reported in Table 5 and shown in Figure 17. The largest spread between position groups reaches 0.87, against a median spread across all pairs of 0.38.

Every one of the ten most divergent pairs changes sign between position groups. The relationship between shots on target and crosses is positive for midfielders and negative for forwards, which is to say that a midfielder who crosses often is generally a more active attacker while a forward who crosses often is generally not the player getting on the end of chances. The relationship between key passes and tackles won is positive for defenders and negative for midfielders, distinguishing the full back whose defensive work and creative output rise together from the midfield division of labour between a destroyer and a creator.

The practical implication is that a single pooled correlation matrix, of the kind that appears routinely in public football analysis, averages over these sign reversals and reports a number that describes no actual population of players. It also supplies the central justification for the position stratified analysis that follows, since a global embedding fitted across all outfield players is fitted across relationships that point in opposite directions.

Table 5: The ten feature pairs whose Spearman correlation differs most across position groups. Every pair changes sign between at least two groups.

Feature A	Feature B	DF	MF	FW	Spread
Shots on target	Crosses	-0.01	0.43	-0.44	0.87
Crosses	Shot accuracy %	-0.22	0.26	-0.57	0.82
Crosses	Interceptions (adj)	0.21	-0.50	0.30	0.81
npxG	Crosses	-0.15	0.36	-0.43	0.79
Key passes	Tackles won (adj)	0.47	-0.32	0.26	0.78
Crosses	Tackles won (adj)	0.51	-0.25	0.48	0.76
xA	Shots on target	0.17	0.56	-0.19	0.75
Shots	Interceptions (adj)	0.22	-0.50	-0.04	0.72
xA	Tackles won (adj)	0.39	-0.30	0.15	0.69
xA	Interceptions (adj)	0.19	-0.50	0.16	0.69

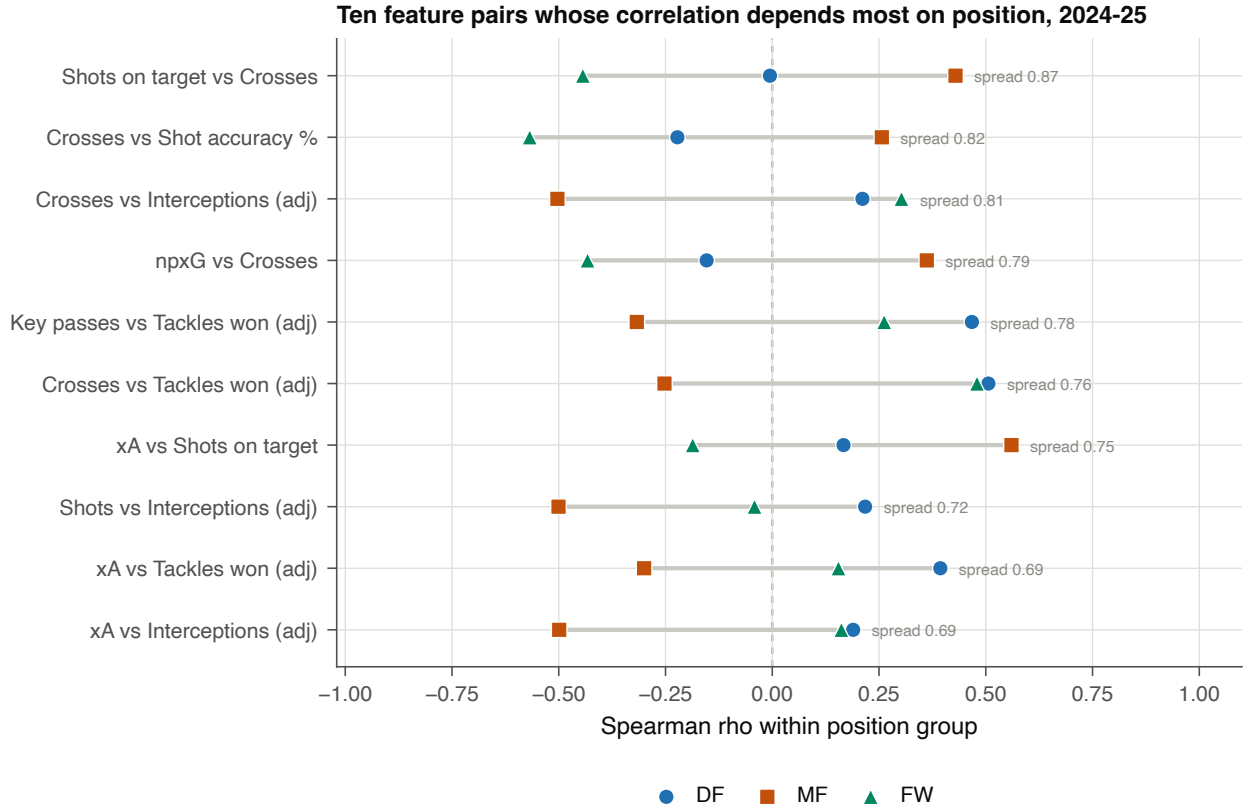


Figure 17: The ten feature pairs whose within-group correlation differs most across position groups in the earlier of the two reduced-sample seasons. Every pair changes sign between at least two groups.

6 Discussion

6.1 What two modes mean, and what they do not

The result is easy to overstate in either direction. It does not say that football has two kinds of player, and it does not say that role is a single number. It says that the distribution of players over

the 33 measured statistics has two regions of concentrated density joined by a populated middle, that the axis running between them is the dominant direction in the space, and that no division finer than that one is distinguishable from what a partitioning algorithm produces on data containing no divisions.

The axis itself is territorial, not technical. Its strongest loadings are touches by pitch third and progressive receptions, not passing accuracy or duel success, and the same ordering appears in the supervised attributions, where Touches def third carries more weight than any technical statistic for all three position groups. Football’s own vocabulary for role mixes territory with technique freely, describing a player as a ball playing centre back or a pressing forward, but the season aggregate record is far better at recording where a player was than what he did there. Any role structure recovered from it will be mostly a statement about territory, and reading it as a statement about ability is the most common way to misuse this kind of analysis.

Within position groups the two-way splits are ordered, not categorical, which the construction makes unavoidable: standardising within group before a two-way split forces the centroids to be mirror images, so each split reports a single direction of variation and its two ends. The three are not equally supported and the paper does not claim they are. The defender split is strongest on every measure applied to it. The midfield split is well supported with a more populated boundary. The forward split is the weakest and should be read with the most caution: its dip test does not reject unimodality at 0.9936, so although its silhouette separates from the clusterless null, what it separates is plausibly a cut through a continuum of forward involvement rather than two modes.

6.2 Why finer taxonomies keep being published

The methodological point generalises past football. A published role taxonomy is typically supported by three things: a selection criterion that picks the reported number of clusters, a stability check that shows the partition survives resampling, and centroid profiles that read sensibly against domain knowledge. Every one of those is satisfied here by partitions this paper shows to be artefacts.

The reason is that none of the three tests existence. A selection criterion compares candidate partitions against each other and is never offered the option of returning none. A bootstrap measures whether an algorithm applied twice to similar data behaves similarly, which a deterministic procedure on a smooth unimodal distribution does very reliably, so a high bootstrap adjusted Rand index is evidence of algorithmic stability and not of group structure. Interpretability is the weakest of the three, because a cut through any continuous distribution produces two centroids that differ along the direction of the cut, and a domain expert can narrate that difference whether or not it corresponds to a real boundary. The evidence here makes the concern concrete and measured rather than hypothetical. On data built to contain no groups, the same pipeline returns a silhouette maximum at two clusters, two nameable centroids, and a bootstrap adjusted Rand index of at least 0.911 on every clusterless dataset in every scope, against 0.962 on the real data. The difference is as small as 0.003. A stability check that passes a continuum at 0.911 is not a test of existence, and a taxonomy that rests on one rests on evidence a continuum satisfies equally well.

Calibration against matched clusterless data fixes this at low cost. It requires no assumption the analysis does not already make, since the simulated data are drawn using the observed mean and covariance, it costs a few dozen refits, and it is the single step that separates the two-mode result reported here from the six-way taxonomy the same pipeline would have produced with the same apparent confidence. Hartigan’s dip test and a density based method permitted to answer that nothing is clustered are useful supplements, and the agreement of all three is what makes the negative half of the result credible.

6.3 Listed positions and data-driven roles

The clusters do not recover listed positions, and where they disagree is the interesting part. The division places nearly every defender on one side and nearly every forward on the other while cutting the midfield close to in half, so it leaves the two extreme groups intact and divides the group that sits between them. The players it reassigns are recognisable: holding midfielders sit with the defenders and attacking midfielders sit with the forwards.

That the supervised model makes its errors in the same place is corroboration from machinery that shares nothing with the clustering. A classifier that reaches 93.4 percent accuracy while sending all but 1.31 percent of its errors through midfield is describing an ordered space, since a three-category problem would scatter errors more evenly across the off-diagonal. The practical reading is that listed position is a reasonable but coarse summary at the extremes and close to arbitrary in the middle, which is what anyone who has watched a modern midfield would expect and which the data now supports quantitatively.

6.4 What the modes predict

A description of role structure that never touched a result would be open to the objection that it describes the statistics and not the sport. The test against real outcomes closes that gap in two directions at once. The two-mode composition of a club carries information about its points that possession does not, out of sample and under three different holdouts, by season, by league and by club, so the axis the partition cuts is an axis along which football is actually decided. The six archetypes carry no information about points that the two modes did not already carry, so the finer taxonomy fails against results exactly as it failed against the clusterless null. Those are independent tests, one internal and one external, and they agree.

Two cautions belong with that. The composition alone predicts slightly worse than possession alone, so the modes are not a substitute for a measure of dominance; they are a statement about how dominance is produced, and the increment is the right thing to cite. The within league correlation also varies, from 0.19 in La Liga to 0.61 in Premier League, which is a reminder that the same composition buys different amounts in different competitions and that a pooled coefficient is a pooled coefficient.

6.5 Possession adjustment and its asymmetry

The possession adjustment result is independent of the clustering and probably the most directly reusable finding here. Defensive counts are conventionally adjusted as though opportunity scales inversely with possession, so that a team with twice the ball concedes half the defensive events. The elasticities measured from the data run from 0.172 for tackles to 0.958 for clearances. For four of the five the bootstrap interval excludes one, so the conventional correction overcorrects them, and does so unequally. For clearances it does not: the interval reaches one, and the conventional exponent is, for that one statistic, not distinguishably wrong. A correction applied at unit elasticity to the other four converts a defensive-count comparison between a possession-dominant club and a counter-attacking one from biased in one direction to biased in the other, and the exponents are moreover averages over a possession range across which three of them measurably bend. The adjustment is also one sided, and that needs stating because the symmetry argument is superficially compelling. If defensive opportunity falls with possession then attacking opportunity should rise with it, and correcting only one side looks arbitrary. The counterfactual reported in Section 5.12 tests this and rejects it: applying the correction in both directions makes the description worse. Attacking output does not scale with team possession the way the symmetric account requires,

because a player’s attacking opportunity depends on where his team uses the ball rather than only on how much of it it has, and a single elasticity cannot represent that.

6.6 Goalkeepers

Goalkeepers are analysed separately because they share almost no features with outfielders, and the archive restores the full block that makes the analysis possible at all. The division that emerges runs along what a goalkeeper is asked to do rather than how well he does it. The situational axis carries an effect size of 0.386 against 0.109 for technique, so a partition of goalkeepers built from these statistics is mostly a partition of the teams in front of them and of the sweeping duties those teams impose. This is a caution for anyone building goalkeeper archetypes from public data, since the resulting labels will describe a role assignment made by a coach rather than a property of the goalkeeper, and it is easy to present the former as the latter.

6.7 Limitations

Season aggregates discard context. A tackle count does not say where the tackles happened, an expected goals total does not distinguish six poor chances from two good ones, and a touch map summarised into three thirds is a coarse instrument for a question about territory that turns out to be territorial. Everything here is therefore a claim about what season aggregate data can support, not about what football contains, and the two should not be conflated: it remains entirely possible that finer roles exist and that this description cannot see them. Event and tracking data would be the way to find out.

The confound that possession adjustment does not remove is team strength. Correcting the defensive counts removes the exposure effect from those five features, but the attacking statistics still scale with a team’s own share of the ball, and the leading component remains related to team quality. Correcting both directions would require an instrument this data cannot supply, so the residual is acknowledged, not solved. The unit of analysis is the player-season, and player-seasons are nested: the same player appears in up to 5 rows, the same club in up to 5 team-seasons, and clubs within leagues. The pooled fits and the bootstrap treat rows as exchangeable, which they are not, so the pooled intervals are narrower than a fully nested analysis would give. The paper’s defence against this is design, not adjustment: every central claim is also made on independent refits by season and by league, and the test against results holds out whole seasons, whole leagues and, in Section 5.7, whole clubs, so that no held-out unit shares a player or a club with the training data. Where a pooled interval is quoted, the independent refits are the check on it. Per ninety rates remain noisy above the eligibility threshold, which is why the outlier list of Section 5.10 carries a minutes caveat and why rankings computed on partial seasons should be shrunk before they are read.

The archive is a mirror, not the source, and although its coverage was verified against the surviving public tables where those overlap, it cannot be re-derived from the live site. One season was excluded outright because a whole feature block was absent from it. Season aggregation also assumes a player’s role is constant across a season, which the drift analysis shows holds for most players and fails for a minority. Finally, the two-mode result covers recent European football at the top level, and the five competitions tested resemble one another more than they resemble the rest of the sport.

6.8 Applications

For recruitment the honest use of this work is narrow and real. The involvement axis is stable enough to support questions about squad composition, such as whether a club’s minutes are concentrated at

one end of the distribution, and the club level signatures make that legible. What the work does not support is a scouting taxonomy: a shortlist built by matching a target’s archetype label will be matching a label that the data does not establish, and the similarity search over the continuous profile is the better instrument for the same question because it does not force a boundary that is not there.

For anyone doing quantitative football analysis the two transferable items are the possession elasticities, which can be applied directly to any defensive count comparison, and the null calibration, which costs little and should be run before any archetype claim is made.

6.9 Future work

The obvious extension is data. Event and tracking data would restore the spatial detail that season aggregates discard and would test the two-mode claim against a description rich enough to reveal finer structure if it exists, at the cost of the reproducibility a public source provides. The most interesting version of that test is the one this paper cannot run: whether the modes are a property of football or a property of the measurement, which requires a dataset where a finer taxonomy is known to be present.

Within public data, three directions follow. The role drift analysis covers one club and would be worth extending, which is a matter of scraping time, not method. Extending beyond the big five, to competitions with different possession profiles, would test whether the elasticities and the two-mode result hold where the league is less concentrated. Modelling attacking exposure properly, rather than through a single elasticity, would let the symmetric adjustment be revisited on a footing the current test cannot provide.

7 Conclusion

The question this paper set out to answer is whether football player roles, as season aggregate statistics describe them, form types or a continuum. The answer is that they form neither cleanly, and that the useful description sits between the two: an ordered space with one dominant axis, density concentrated at two ends, and a populated middle that no partition divides well.

Two modes are real. Against three clusterless references, the softest a matched Gaussian and the hardest a copula that keeps every real marginal and correlation, each simulated 25 times, the two-mode partition is better separated than every simulation under every null in every scope, by no fewer than 8.01 standard deviations against the hardest, and the leading component fails a test of unimodality outside the forward group by a dip statistic that clears the hardest null by 13.67 on the pool. Nothing finer survives the same test. Beyond two modes the observed separation converges on what a clusterless continuum produces, density based clustering assigns every player to noise within each position group, and the mixture criterion prefers ever more components without settling, which is the signature of a distribution being approximated, not a set of groups being found.

The two modes are not the listed positions under another name. They agree with position group at an adjusted Rand index of 0.293, which is far above chance and far below recovery, and the shape of the disagreement is the finding. Defenders and forwards are assigned to opposite sides almost deterministically while midfielders divide close to evenly, so the partition cuts across the listed categories at their busiest point. A supervised model trained on the same features reaches 93.4 percent accuracy and yet sends all but 1.31 percent of its errors through midfield, which is the same geometry recovered by machinery that shares nothing with the clustering.

The result is not local. It reproduces in every one of 5 seasons and every one of 5 competitions, refitted independently in each, with a worst case agreement of 0.916 against the pooled partition

and a separating axis that correlates with the pooled axis above 0.9983 everywhere. Centroids fitted in one league assign another league’s players at a mean adjusted Rand index of 0.944. Where season against season agreement is lower, it decays with the gap between seasons while the transfer index over the same pairs stays flat, so the decay measures players changing, not the solution failing.

The result also survives the test a description of statistics can fail and a description of football cannot. Over 490 team-seasons with real results, the two-mode composition of a club predicts its points beyond possession out of sample, while the six archetypes add nothing beyond the two modes. The structure the null calibration found is the structure that results respond to, and the structure it rejected is the structure results ignore.

Three secondary results stand on their own. Possession adjustment of defensive counts is conventionally applied as though a team with twice the ball concedes half the defensive opportunities. The elasticities measured here run from 0.172 to 0.958, and with bootstrap intervals the conventional exponent is rejected for 4 of the five statistics and not for clearances, which is the one statistic where a purely reactive exposure story should hold and where it does. The adjustment is also one sided: attacking output does not scale with possession the way the symmetric argument would require, and applying the correction in both directions makes the description worse. Goalkeeping, analysed separately, divides along what a goalkeeper is asked to do rather than how well he does it, with the situational axis carrying an effect size of 0.386 against 0.109 for technique.

What follows for practice is a caution about method, not about football. Published role taxonomies are typically supported by a selection criterion, a stability check and a plausible reading of the resulting profiles, and every one of those is satisfied here by partitions this paper shows to be artefacts. A silhouette maximum, a high bootstrap adjusted Rand index and an interpretable centroid are all obtainable from data containing no groups whatsoever, because the algorithm supplies the groups, and this paper measured it: on clusterless data the bootstrap adjusted Rand index never fell below 0.911. The calibration against matched clusterless data is cheap, it needs no assumption the analysis does not already make, and it is the step that separates the two-mode result reported here from the finer taxonomy that the same pipeline would otherwise have produced with equal confidence.

Author contributions

C.C. conceived the research question. Z.C. designed the study, carried out the analysis, and wrote the paper. Both authors approved the final version.

Data and code availability

The analysis code, the retrieval step that fetches the archive from its stated upstream and records a digest per file, the cluster assignments for both samples and the sources of this document are available at <https://github.com/ZacharyTChung/pl-player-profiling>. The primary sample is the public worldfootballR mirror of FBref described in Section 3, and the reduced sample was retrieved from FBref and Understat as described there.

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References

- Bart Aalbers and Jan Van Haaren. Distinguishing between roles of football players in play-by-play match event data, 2018. URL <https://arxiv.org/abs/1809.05173>.
- Margareta Ackerman and Shai Ben-David. Clusterability: A theoretical study. In *Proceedings of the Twelfth International Conference on Artificial Intelligence and Statistics (AISTATS 2009)*, volume 5 of *Proceedings of Machine Learning Research*, pages 1–8, 2009. URL <https://proceedings.mlr.press/v5/ackerman09a.html>.
- Andreas Adolfosson, Margareta Ackerman, and Naomi C. Brownstein. To cluster, or not to cluster: An analysis of clusterability methods. *Pattern Recognition*, 88:13–26, 2019. doi: 10.1016/j.patcog.2018.10.026.
- Serhat Emre Akhanli and Christian Hennig. Comparing clusterings and numbers of clusters by aggregation of calibrated clustering validity indexes. *Statistics and Computing*, 30(5):1523–1544, 2020. doi: 10.1007/s11222-020-09958-2.
- Serhat Emre Akhanli and Christian Hennig. Clustering of football players based on performance data and aggregated clustering validity indexes. *Journal of Quantitative Analysis in Sports*, 19(2): 103–123, 2023. doi: 10.1515/jqas-2022-0037.
- Muthu Alagappan. From 5 to 13: Redefining the positions in basketball. In *MIT Sloan Sports Analytics Conference*, 2012. URL <https://www.sloansportsconference.com/research-papers/from-5-to-13-redefining-the-positions-in-basketball>.
- Asa Ben-Hur, André Elisseeff, and Isabelle Guyon. A stability based method for discovering structure in clustered data. In *Pacific Symposium on Biocomputing*, volume 7, pages 6–17, 2002. URL <https://psb.stanford.edu/psb-online/proceedings/psb02/benhur.pdf>.
- Alina Bialkowski, Patrick Lucey, Peter Carr, Yisong Yue, Sridha Sridharan, and Iain Matthews. Large-scale analysis of soccer matches using spatiotemporal tracking data. In *Proceedings of the 2014 IEEE International Conference on Data Mining (ICDM)*, pages 725–730, 2014. doi: 10.1109/ICDM.2014.133. URL <https://doi.org/10.1109/ICDM.2014.133>.
- Ricardo J. G. B. Campello, Davoud Moulavi, and Jörg Sander. Density-based clustering based on hierarchical density estimates. In *Advances in Knowledge Discovery and Data Mining (PAKDD 2013)*, volume 7819 of *Lecture Notes in Computer Science*, pages 160–172. Springer, 2013. doi: 10.1007/978-3-642-37456-2_14. URL https://doi.org/10.1007/978-3-642-37456-2_14.
- Tom Decroos and Jesse Davis. Player vectors: Characterizing soccer players’ playing style from match event streams. In *Machine Learning and Knowledge Discovery in Databases: ECML PKDD 2019*, Lecture Notes in Computer Science, pages 569–584. Springer, 2020. doi: 10.1007/978-3-030-46133-1_34. URL https://doi.org/10.1007/978-3-030-46133-1_34.
- Bradley Efron and Carl Morris. Stein’s paradox in statistics. *Scientific American*, 236(5):119–127, 1977.
- Harm Eggels, Ruud van Elk, and Mykola Pechenizkiy. Explaining soccer match outcomes with goal scoring opportunities predictive analytics. In *Proceedings of the 3rd Workshop on Machine Learning and Data Mining for Sports Analytics (MLSA 2016)*, co-located with *ECML/PKDD 2016*, volume 1842 of *CEUR Workshop Proceedings*, 2016. URL https://ceur-ws.org/Vol-1842/paper_07.pdf.

- Soumaya El Foukahi, Kaoutar El Handri, Mohamed El Ghmary, and Hassan Echoukairi. Unveiling midfielder roles in elite football: A data-driven clustering approach. In *Artificial Intelligence and Mathematics*, Springer Proceedings in Mathematics & Statistics, pages 465–477. Springer, 2026. doi: 10.1007/978-3-032-18042-1_37.
- Manuel J. A. Eugster. Performance profiles based on archetypal athletes. *International Journal of Performance Analysis in Sport*, 12(1):166–187, 2012. doi: 10.1080/24748668.2012.11868592.
- Daniel Fernández, Martí Casals, Martí Oliver, Montse Plensa, and Marica Manisera. Reporting of clustering techniques in sports sciences: a scoping review. *Electronic Journal of Applied Statistical Analysis*, 17(3):653–675, 2024. doi: 10.1285/i20705948v17n3p653.
- Linda Gu, Wenlong Wang, Spyridon Plakias, and Shaoliang Zhang. Playing style identification in team sports: A systematic review from statistical dimensionality reduction to unsupervised machine learning. *International Journal of Sports Science & Coaching*, 20(6):2742–2761, 2025. doi: 10.1177/17479541251372586.
- J. A. Hartigan and P. M. Hartigan. The dip test of unimodality. *The Annals of Statistics*, 13(1): 70–84, 1985. doi: 10.1214/aos/1176346577.
- Christian Hennig. What are the true clusters? *Pattern Recognition Letters*, 64:53–62, 2015. doi: 10.1016/j.patrec.2015.04.009.
- Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2:193–218, 1985. doi: 10.1007/BF01908075. URL <https://doi.org/10.1007/BF01908075>.
- Samuel Kalman and Jonathan Bosch. NBA lineup analysis on clustered player tendencies: A new approach to the positions of basketball and modeling lineup efficiency of soft lineup aggregates. In *MIT Sloan Sports Analytics Conference*, 2020. URL <https://www.sloansportsconference.com/research-papers/nba-lineup-analysis-on-clustered-player-tendencies-a-new-approach-to-the-positions-of-basketball-modeling-lineup-efficiency>.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>.
- Ted Knutson. Introducing possession-adjusted player stats. StatsBomb, June 2014. URL <https://blogarchive.statsbomb.com/articles/soccer/introducing-possession-adjusted-player-stats/>. Published 23 June 2014.
- Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation forest. In *Proceedings of the 2008 Eighth IEEE International Conference on Data Mining (ICDM)*, pages 413–422. IEEE, 2008. doi: 10.1109/ICDM.2008.17. URL <https://doi.org/10.1109/ICDM.2008.17>.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html. arXiv:1705.07874.
- Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction, 2018. URL <https://arxiv.org/abs/1802.03426>.

- Glenn W. Milligan and Martha C. Cooper. An examination of procedures for determining the number of clusters in a data set. *Psychometrika*, 50(2):159–179, 1985. doi: 10.1007/BF02294245.
- Luca Pappalardo, Paolo Cintia, Paolo Ferragina, Emanuele Massucco, Dino Pedreschi, and Fosca Giannotti. PlayeRank: Data-driven performance evaluation and player ranking in soccer via a machine learning approach. *ACM Transactions on Intelligent Systems and Technology*, 10(5), 2019. doi: 10.1145/3343172. URL <https://doi.org/10.1145/3343172>. arXiv:1802.04987.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. URL <https://jmlr.org/papers/volume12/pedregosa11a/pedregosa11a.pdf>.
- Spyridon Plakias, Serafeim Moustakidis, Christos Kokkotis, Marina Papalexi, Themistoklis Tsatalas, Giannis Giakas, and Dimitrios Tsaopoulos. Identifying soccer players’ playing styles: A systematic review. *Journal of Functional Morphology and Kinesiology*, 8(3):104, 2023. doi: 10.3390/jfmk8030104.
- Alex Rathke. An examination of expected goals and shot efficiency in soccer. *Journal of Human Sport and Exercise*, 12(Proc2), 2017. doi: 10.14198/jhse.2017.12.proc2.05. URL <https://doi.org/10.14198/jhse.2017.12.proc2.05>.
- Pieter Robberechts. soccerdata: A collection of scrapers for gathering soccer data. Python package, version 1.9.1, 2026. URL <https://github.com/probberechts/soccerdata>.
- Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987. doi: 10.1016/0377-0427(87)90125-7. URL [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7).
- Emiliano Seri, Roberto Rocci, and Thomas Brendan Murphy. Partial membership models for soft clustering of multivariate football player performance data, 2024. URL <https://arxiv.org/abs/2409.01874>.
- Sports Reference LLC. FBref.com: Football statistics and history. <https://fbref.com/en/>, 2026. Advanced statistics were licensed from a commercial event-data provider, StatsBomb until 2022 and Opta thereafter; the provider-derived columns were withdrawn on 20 January 2026.
- Robert Tibshirani, Guenther Walther, and Trevor Hastie. Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63(2):411–423, 2001. doi: 10.1111/1467-9868.00293. URL <https://doi.org/10.1111/1467-9868.00293>.
- Understat. Understat: Expected goals statistics. <https://understat.com>, 2026. Player and team expected goals for the seasons ending 2025 and 2026, retrieved 2026.
- Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008. URL <https://jmlr.org/papers/v9/vandermaaten08a.html>.
- Dinesh Vatvani. Upgrading Expected Goals. StatsBomb Blog, 2022. URL <https://blogarchive.statsbomb.com/articles/soccer/upgrading-expected-goals/>.

Ulrike von Luxburg. Clustering stability: An overview. *Foundations and Trends in Machine Learning*, 2(3):235–274, 2010. doi: 10.1561/22000000008.

Jason Zivkovic. worldfootballR: Extract and clean world football (soccer) data. R package and public data mirror, 2026. URL https://github.com/JaseZiv/worldfootballR_data. Archived FBref advanced season tables retrieved 7 September 2026.

A Feature dictionaries

The two samples are described by different feature sets drawn from different pulls, so each has its own dictionary. Table 6 lists the 33 features of the primary sample with the FBref table each column was read from. Table 7 lists the 18 features of the reduced sample, which combines what survived the withdrawal with an independent expected goals source. In both, counting statistics are expressed per ninety minutes played and rate features are used as published rather than divided a second time.

Every feature in the reduced set was confirmed to hold non-null values in both of its seasons before being admitted, which is the check that distinguishes a live column from one whose values were withdrawn. The primary set is audited differently, by per-column and per-season coverage on load, because its failure mode is a block missing from one season rather than a column emptied across all of them.

Table 6: The primary feature set, with the FBref table each column was read from. Counting statistics are per ninety minutes played; the seven rate features are used as published. Five defensive counts are possession adjusted, as described in Section 4.1.

Feature	FBref table	Definition
np xg p90	standard	Non-penalty expected goals (per 90)
shots p90	shooting	Shots taken, excluding penalties (per 90)
goals non penalty p90	standard	Goals excluding penalties (per 90)
xag p90	standard	Expected assisted goals, the expected goals of shots a pass created (per 90)
key passes p90	passing	Passes leading directly to a shot (per 90)
sca p90	gca	Shot creating actions, the two offensive actions before a shot (per 90)
gca p90	gca	Goal creating actions, the two offensive actions before a goal (per 90)
passes final third p90	passing	Completed passes into the final third (per 90)
passes penalty area p90	passing	Completed passes into the penalty area (per 90)
crosses p90	misc	Crosses attempted (per 90)
progressive passes p90	passing	Completed passes moving the ball substantially toward goal (per 90)
progressive carries p90	possession	Carries moving the ball substantially toward goal (per 90)
progressive receptions p90	possession	Passes received in a progressive position (per 90)
carries final third p90	possession	Carries into the final third (per 90)
touches def pen p90	possession	Touches in the defending team's own penalty area (per 90)
touches def third p90	possession	Touches in the defensive third (per 90)
touches mid third p90	possession	Touches in the middle third (per 90)
touches att third p90	possession	Touches in the attacking third (per 90)
touches att pen p90	possession	Touches in the opposition penalty area (per 90)
take ons p90	possession	Take-ons attempted (per 90)
tackles padj p90	defense	Tackles, possession adjusted (per 90)
interceptions padj p90	misc	Interceptions, possession adjusted (per 90)
blocks padj p90	defense	Blocks of a pass or shot, possession adjusted (per 90)
clearances padj p90	defense	Clearances, possession adjusted (per 90)
ball recoveries p90	misc	Loose balls recovered (per 90)
fouls committed padj p90	misc	Fouls committed, possession adjusted (per 90)
shot accuracy pct	shooting	Shots on target as a share of shots
take on success pct	possession	Take-ons completed as a share of those attempted
tackle win pct	defense	Duels won as a share of those contested
aerials won pct	misc	Aerial duels won as a share of those contested
pass cmp short pct	passing	Short passes completed as a share of those attempted
pass cmp medium pct	passing	Medium passes completed as a share of those attempted
pass cmp long pct	passing	Long passes completed as a share of those attempted

Table 7: The outfield feature set. Counting statistics are per ninety minutes. The two rate features are used as published.

Feature	Source	Definition
np_xg_p90	Understat	Non-penalty expected goals
xa_p90	Understat	Expected assists
key_passes_p90	Understat	Passes leading directly to a shot
xg_chain_p90	Understat	Expected goals of possessions the player was involved in
xg_buildup_p90	Understat	As xGChain, excluding shots and key passes
shots_p90	FBref	Shots taken, excluding penalties
shots_on_target_p90	FBref	Shots on target
goals_non_penalty_p90	FBref	Goals excluding penalties
assists_p90	FBref	Assists
crosses_p90	FBref	Crosses attempted
interceptions_padj_p90	FBref	Interceptions, possession adjusted
tackles_won_padj_p90	FBref	Tackles won, possession adjusted
fouls_committed_padj_p90	FBref	Fouls committed, possession adjusted
fouls_drawn_p90	FBref	Fouls drawn
offsides_p90	FBref	Times caught offside
cards_yellow_p90	FBref	Yellow cards
shot_accuracy_pct	FBref	Shots on target as a share of shots, a rate
goals_per_shot	FBref	Goals per shot, a rate

B Minutes threshold sensitivity

Both samples exclude outfield players below 450 minutes, and this appendix tests the threshold on the reduced sample, where refitting the whole procedure three times is cheap. That threshold balances two errors: a low threshold admits players whose per-90 rates are dominated by sampling noise, while a high threshold discards genuine squad roles such as the impact substitute. Table 8 repeats the entire selection and stability procedure at 270, 450 and 900 minutes, treating each threshold as an independent pool in which imputation and standardisation are recomputed and the declared selection rule is reapplied from scratch. Figure 18 plots the same quantities.

The threshold turns out to matter very little. The rule selects two clusters at every threshold, the silhouette moves only from 0.286 to 0.302, and bootstrap stability is between 0.924 and 0.947 throughout. More directly, the partition itself barely moves: measured on the players common to both pools, the adjusted Rand index against the partition at the threshold actually used is 0.967 at 270 minutes and 0.986 at 900. A reader who preferred a different threshold would reach the same conclusions.

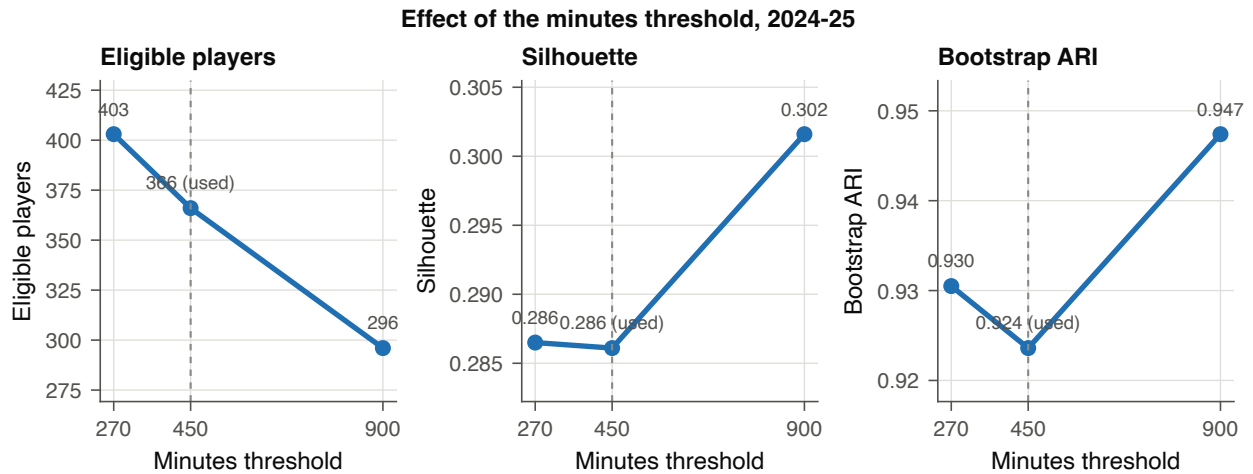


Figure 18: Eligible sample, silhouette and bootstrap stability across the candidate minutes thresholds. The dashed line marks the threshold used in the main analysis.

Table 8: Sensitivity of the analysis to the minutes threshold in the earlier of the two reduced-sample seasons. Agreement is the adjusted Rand index against the partition at the threshold actually used, computed on the players common to both pools.

Minutes	Eligible	k	Silhouette	Bootstrap ARI	Agreement
270	403	2	0.286	0.930	0.967
450	366	2	0.286	0.924	baseline
900	296	2	0.302	0.947	0.986

C UMAP hyperparameter grid

Table 9 reports trustworthiness for all twelve configurations of the neighbour count and minimum distance, on the reduced sample, which is the only one for which a manifold embedding is reported. Selection was by trustworthiness alone, decided before any embedding was inspected visually.

Table 9: UMAP hyperparameter grid, ranked by trustworthiness. Selection was by trustworthiness alone and was decided before any embedding was inspected.

Neighbours	Minimum distance	Trustworthiness
10	0	0.9322
10	0.10	0.9316
15	0	0.9306
15	0.10	0.9303
30	0.10	0.9276
30	0	0.9257
10	0.50	0.9186
15	0.50	0.9175
50	0.10	0.9169
50	0	0.9156
30	0.50	0.9083
50	0.50	0.9045

D Cluster selection criteria in full

Table 10 gives the silhouette coefficient, Davies-Bouldin index, Calinski-Harabasz index and gap statistic for every candidate number of clusters on the reduced sample, for the global analysis and within each position group. The consensus rule described in Section 4 is applied to these values, and the row the rule selects is marked in each scope.

The corresponding evidence for the primary sample is Figure 1, which plots the observed silhouette across the same candidate range in every scope against the clusterless null. That figure carries more information than a criteria table would, because the question the primary sample is used to answer is not which candidate scores best but whether any of them beats what a continuum produces.

Table 10: Cluster selection criteria for every candidate number of clusters, by scope. The consensus rule is applied to these values.

Scope	k	Silhouette	Davies-Bouldin	Calinski-Harabasz	Gap
global*	2	0.287	1.451	145.6	1.216
global	3	0.215	1.695	101.9	1.283
global	4	0.154	1.954	86.1	1.339
global	5	0.145	1.861	75.4	1.381
global	6	0.144	1.797	68.3	1.415
global	7	0.135	1.905	61.3	1.438
global	8	0.118	1.881	56.1	1.458
global	9	0.123	1.840	53.5	1.477
global	10	0.123	1.805	49.8	1.484
global	11	0.128	1.657	47.6	1.506
global	12	0.121	1.911	44.6	1.517
DF*	2	0.204	2.037	25.8	0.943
DF	3	0.151	1.919	23.5	1.011
DF	4	0.142	1.983	19.7	1.036
DF	5	0.120	1.961	18.0	1.059

Scope	k	Silhouette	Davies-Bouldin	Calinski-Harabasz	Gap
DF	6	0.117	1.795	16.9	1.083
DF	7	0.106	1.898	15.6	1.109
DF	8	0.120	1.792	14.7	1.136
DF	9	0.120	1.754	14.1	1.154
DF	10	0.112	1.636	13.5	1.159
DF	11	0.098	1.636	12.8	1.162
DF	12	0.105	1.627	12.3	1.196
MF*	2	0.243	1.543	69.4	1.035
MF	3	0.147	1.969	49.0	1.091
MF	4	0.145	1.895	40.4	1.130
MF	5	0.145	1.873	34.7	1.153
MF	6	0.132	1.939	30.3	1.167
MF	7	0.122	2.054	27.4	1.199
MF	8	0.107	2.030	24.9	1.202
MF	9	0.103	1.923	23.8	1.212
MF	10	0.106	1.983	21.9	1.220
MF	11	0.094	1.944	20.4	1.227
MF	12	0.097	1.835	19.5	1.226
FW*	2	0.187	1.803	12.4	0.764
FW	3	0.175	1.576	10.8	0.837
FW	4	0.176	1.599	10.2	0.908
FW	5	0.196	1.181	9.3	0.909
FW	6	0.124	1.490	8.5	0.944
FW	7	0.132	1.350	7.8	1.017
FW	8	0.101	1.389	7.7	1.021
FW	9	0.127	1.229	7.7	1.070
FW	10	0.125	1.350	7.8	1.051
FW	11	0.128	1.199	7.1	1.064
FW	12	0.116	1.126	6.8	1.096

E Cluster membership

Every player’s archetype assignment is published as CSV in the accompanying repository rather than reproduced here, for both samples: `results/membership_primary.csv` carries the 9,263 eligible player-seasons of the primary sample with club, league, season, listed position, minutes and assigned archetype, and `results/membership_reduced.csv` carries the reduced sample with the same fields plus each player’s distance from his cluster centroid. Both are regenerated by the pipeline, so they cannot drift from the numbers reported above.

F Supplementary figures

This section collects material from both samples, and each figure states which one it describes. Figure 19 gives the full hierarchically clustered Spearman correlation matrix over the reduced outfield feature set, from which the divergence analysis in Section 5.15 is derived. Figure 20a reports the scree and cumulative variance curves: 9 components are required to reach the ninety percent threshold, so

the feature set is not close to low rank, and the first three components together account for 0.634 of the variance. Figure 20b shows the loadings biplot. Figure 21a gives the correlation structure of the surviving goalkeeper features and Figure 21b the distribution of the two goalkeeper composites. Figure 22 gives the same two views for the archive goalkeeper block, whose composites the main text uses.

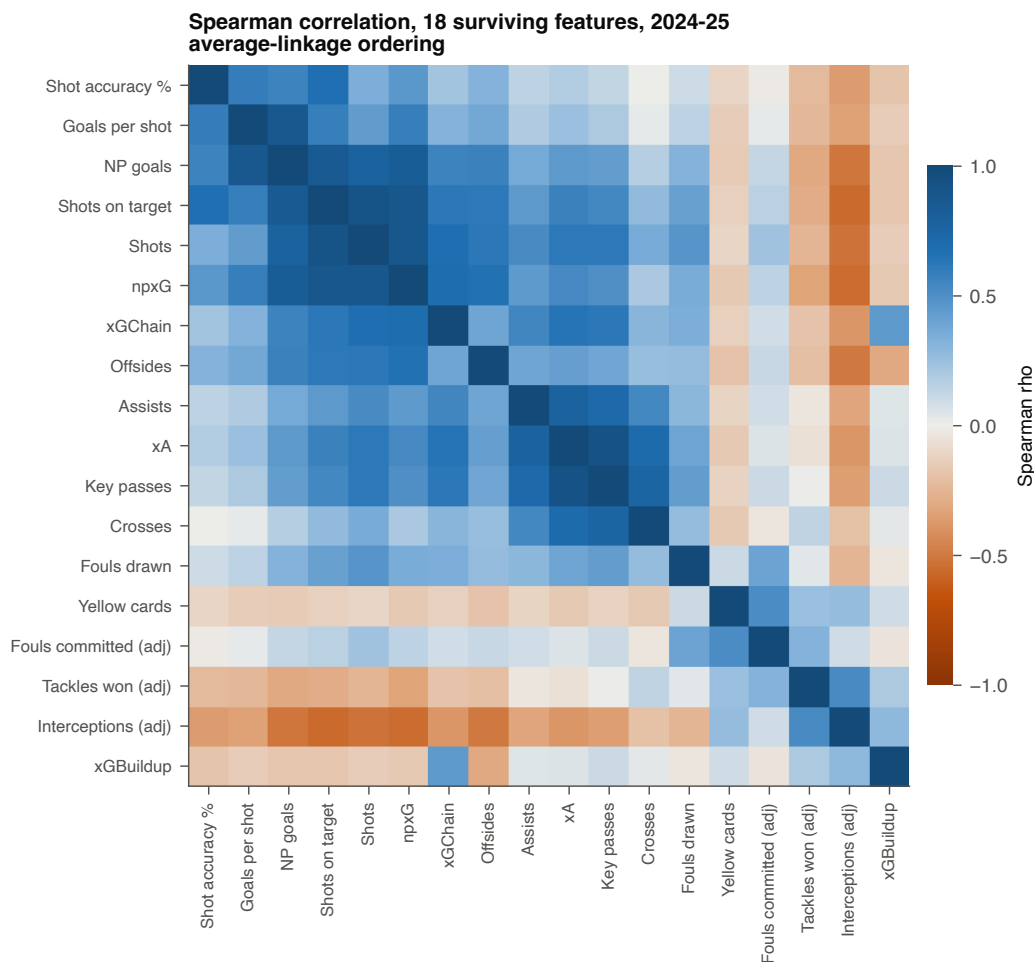
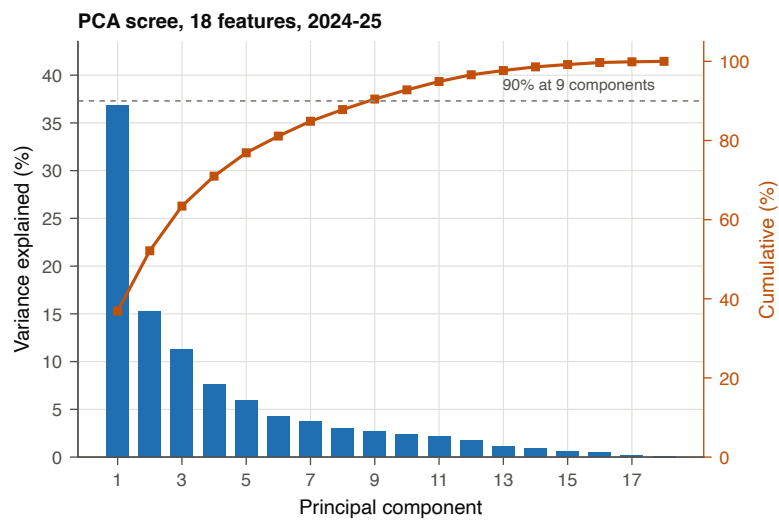
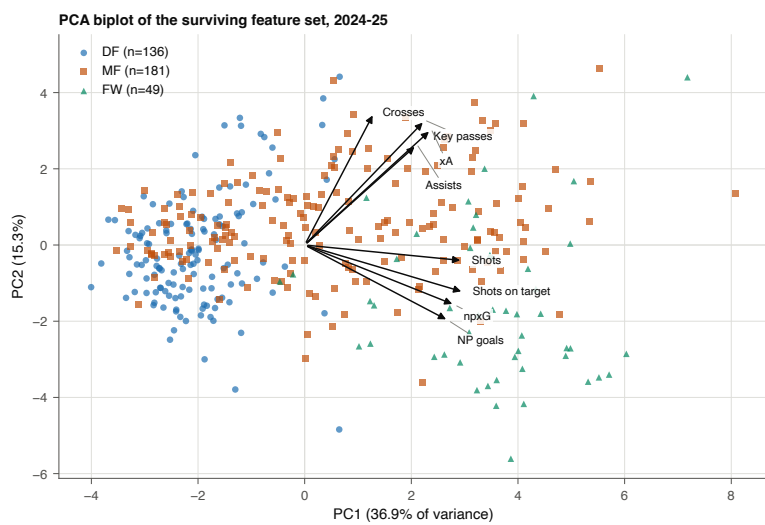


Figure 19: Hierarchically clustered Spearman correlation matrix over the outfield feature set in the earlier of the two reduced-sample seasons.

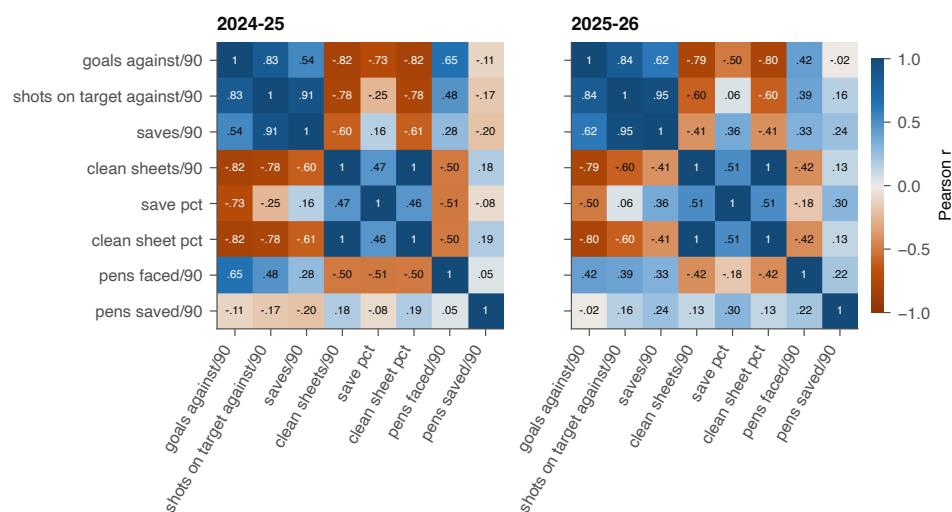


(a) Scree and cumulative variance.

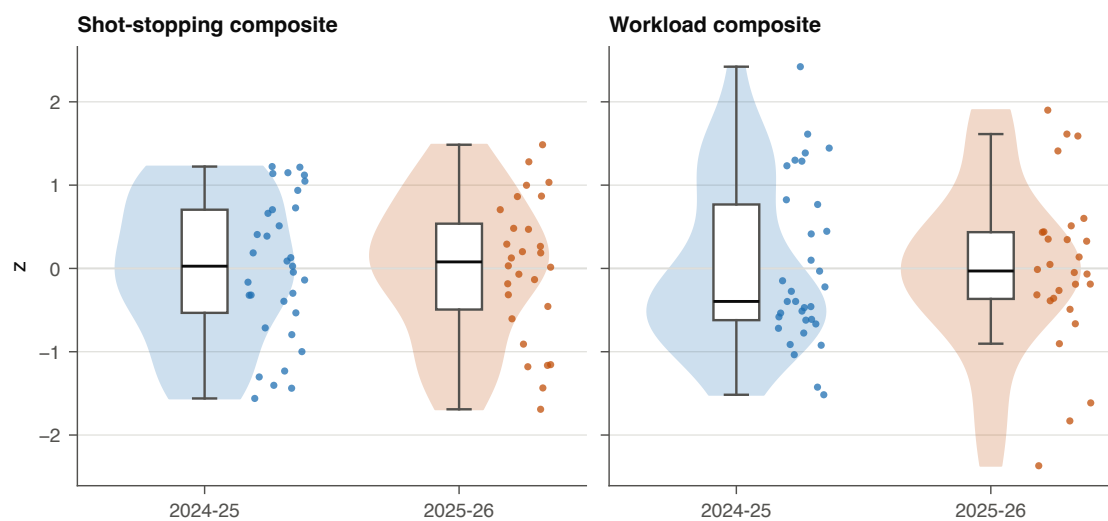


(b) Loadings biplot.

Figure 20: Principal component structure of the outfield feature set.

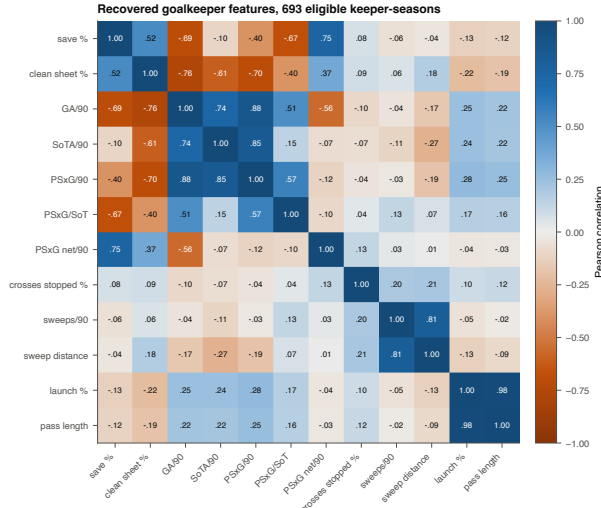


(a) Surviving goalkeeper features.

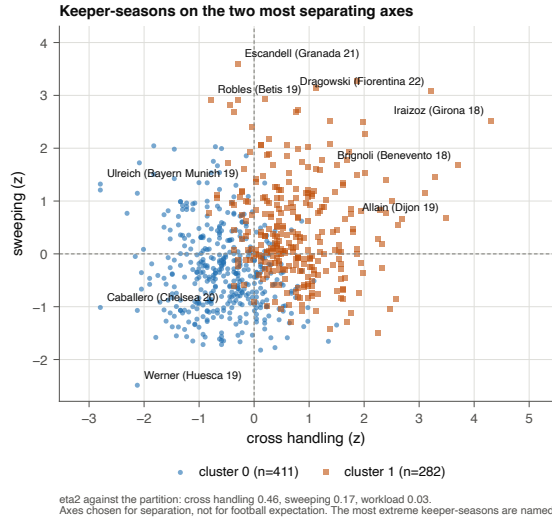


(b) Goalkeeper composite distributions.

Figure 21: Supplementary goalkeeper material, reduced sample.



(a) Correlation structure of the archive goalkeeper features.



(b) Distribution of the archive goalkeeper composites.

Figure 22: Supplementary goalkeeper material, archive sample. The full 13 feature block supports the composite axes that the reduced sample cannot, which is why the archive keeper analysis in Section 5.13 reports separations the reduced sample collapses.

G Individual archetype radars

Figures 23 to 25 give the archive archetype radars for each position group at full size, on the fixed axes described in Section 4, and Figures 26 to 28 give the corresponding radars on the reduced sample. Within each position group the two profiles are mirror images by construction, since the features are standardised within group before clustering. The two sets are drawn on different feature counts, so their axes are not directly comparable, and the useful comparison is of shape, not extent.

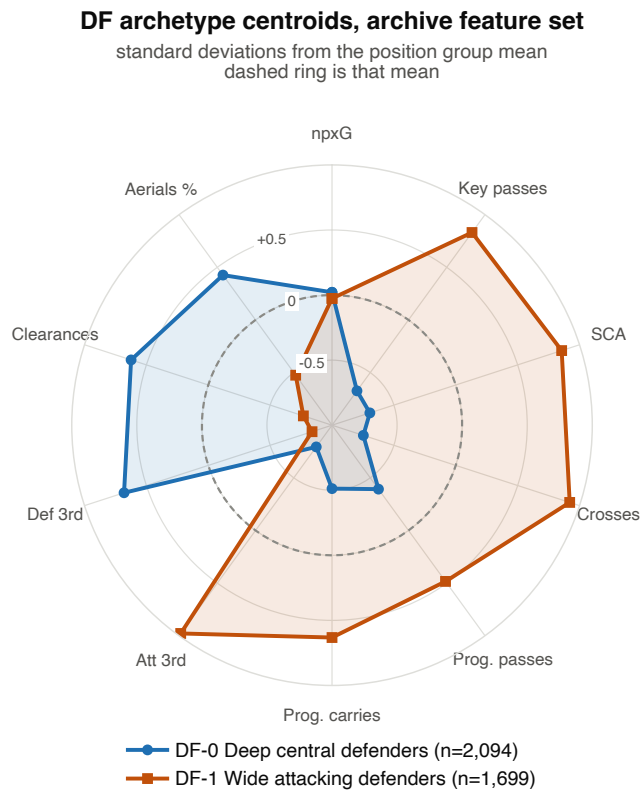


Figure 23: Defender archetypes, archive sample.

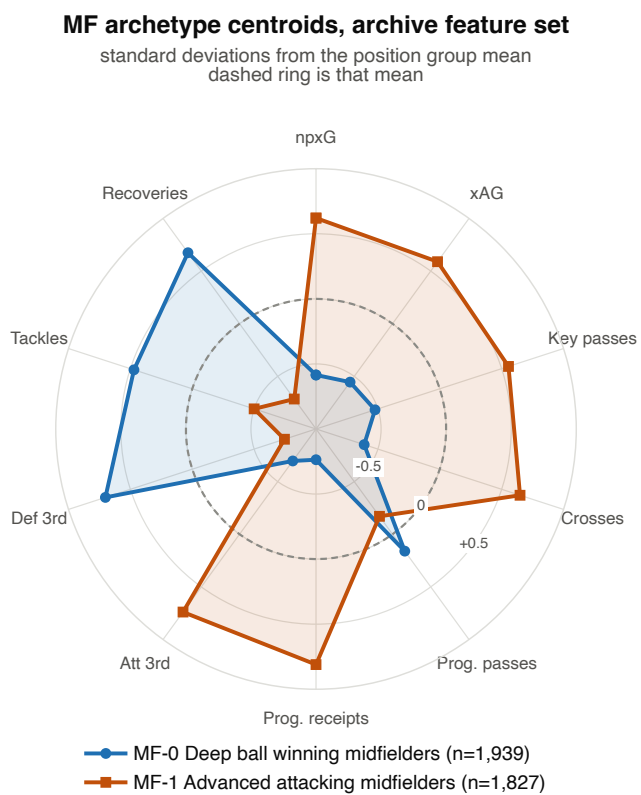


Figure 24: Midfielder archetypes, archive sample.

FW archetype centroids, archive feature set

standard deviations from the position group mean
dashed ring is that mean

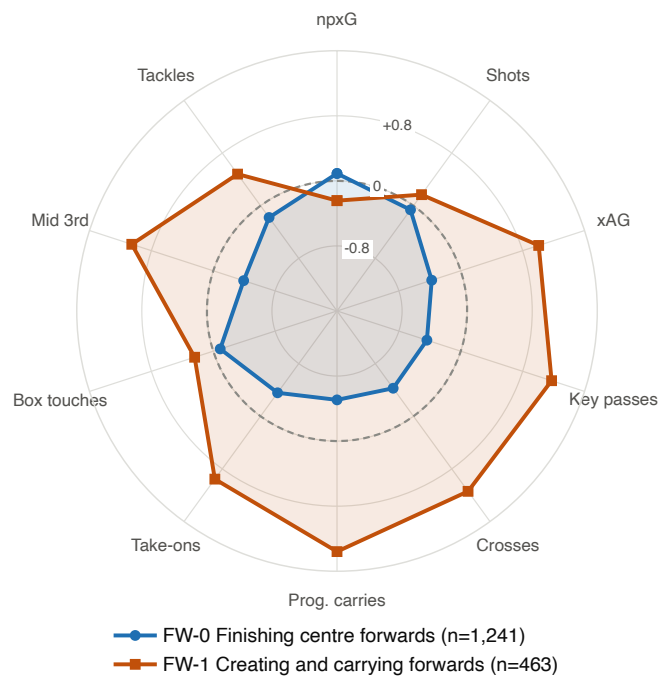


Figure 25: Forward archetypes, archive sample.

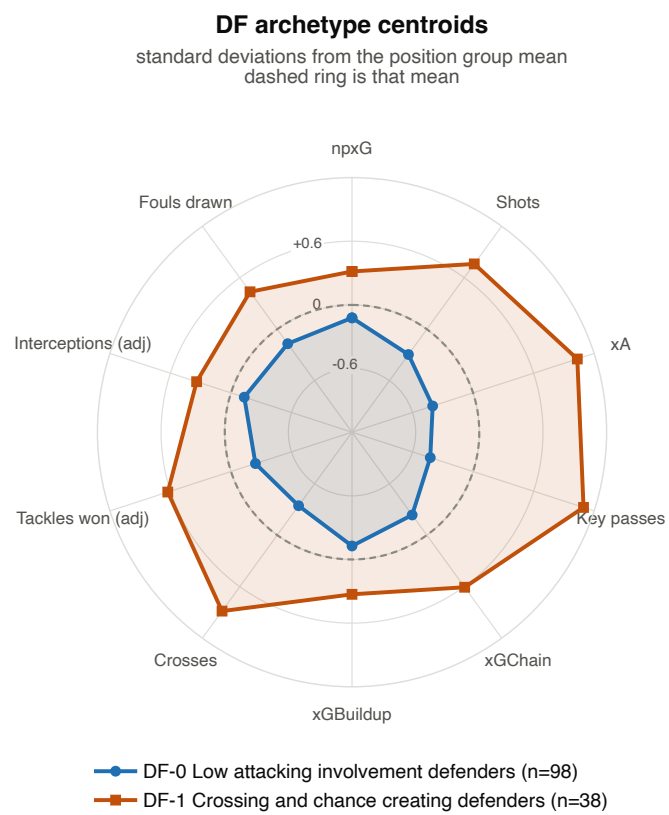


Figure 26: Defender archetypes.

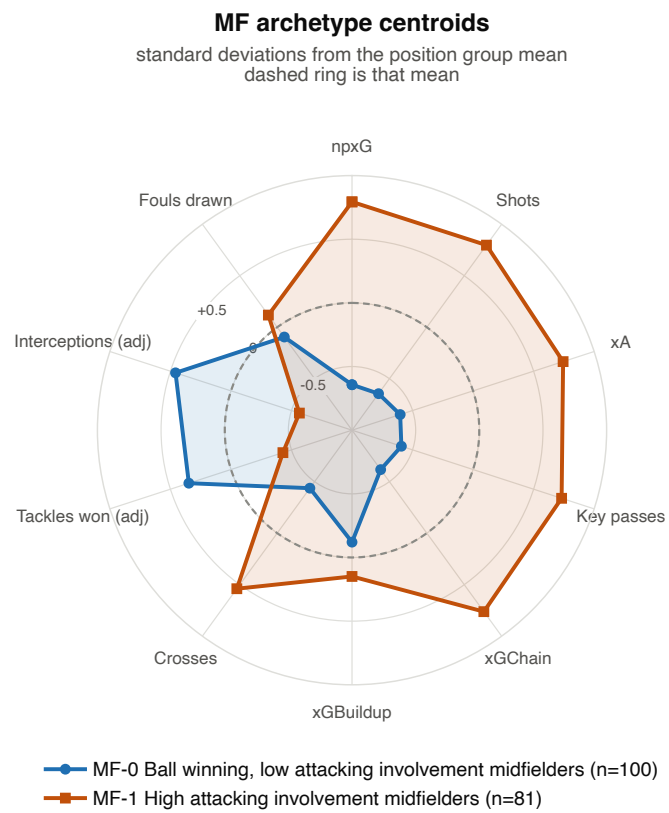


Figure 27: Midfielder archetypes.

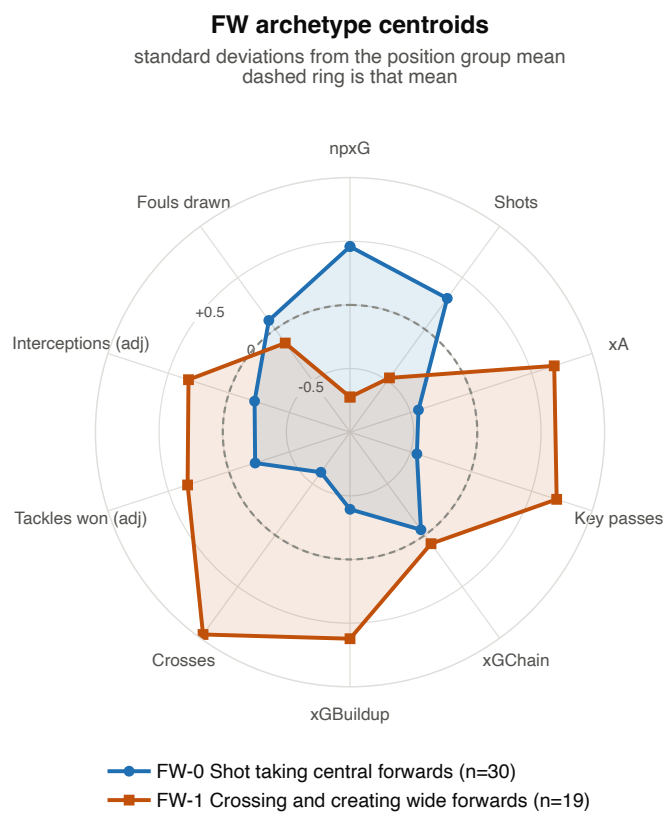


Figure 28: Forward archetypes.

H Further analyses on the reduced sample

The analyses collected here were run on the reduced sample of Section 3.4. They are reported in full because several of them depend on data that exists only for that window, and because the reduced sample is the robustness case for the main result rather than a separate study. None of them carries the paper’s argument, which is why they sit here rather than in Section 5.

H.1 Cluster selection and stability on the reduced sample

This subsection reports the clustering detail for the reduced sample of Section 3.4, which is the robustness case rather than the main analysis. Applying the same rule to the four selection criteria gives $k = 2$ globally and within each position group, matching the primary sample. The full criterion curves appear in Figure 29 and Appendix D.

Stability separates the position groups sharply. Across 200 bootstrap resamples the mean adjusted Rand index against the reference partition is 0.914 globally and 0.815 for midfielders, both of which describe a partition that reappears almost unchanged under resampling. For defenders the mean falls to 0.607 with a standard deviation of 0.216, and for forwards to 0.602 with a standard deviation of 0.281. The minimum adjusted Rand index observed across forward resamples is -0.065, which is to say that in the least favourable resample the partition recovered is no better than chance. Forwards number only 49 players above the minutes threshold, so this is partly a sample size problem, but it is a real limit on what can be claimed. The consensus co-assignment structure is shown in Figure 30a. Figure 31 shows the selected partition in the principal component space.

The relationship between discovered clusters and listed positions is weak. The global partition scores an adjusted Rand index of 0.226 against the position group with a cluster purity of 0.579, so the clusters are not a rediscovery of the positions that the source already records. Figure 30b shows where the two disagree. Read together with the embedding geometry, the interpretation is that the dominant axis in the surviving data is not defender against forward but attacking involvement against defensive involvement, which cuts across the listed groups and places a ball-playing full back nearer an ordinary midfielder than a defensive counterpart at his own listed position.

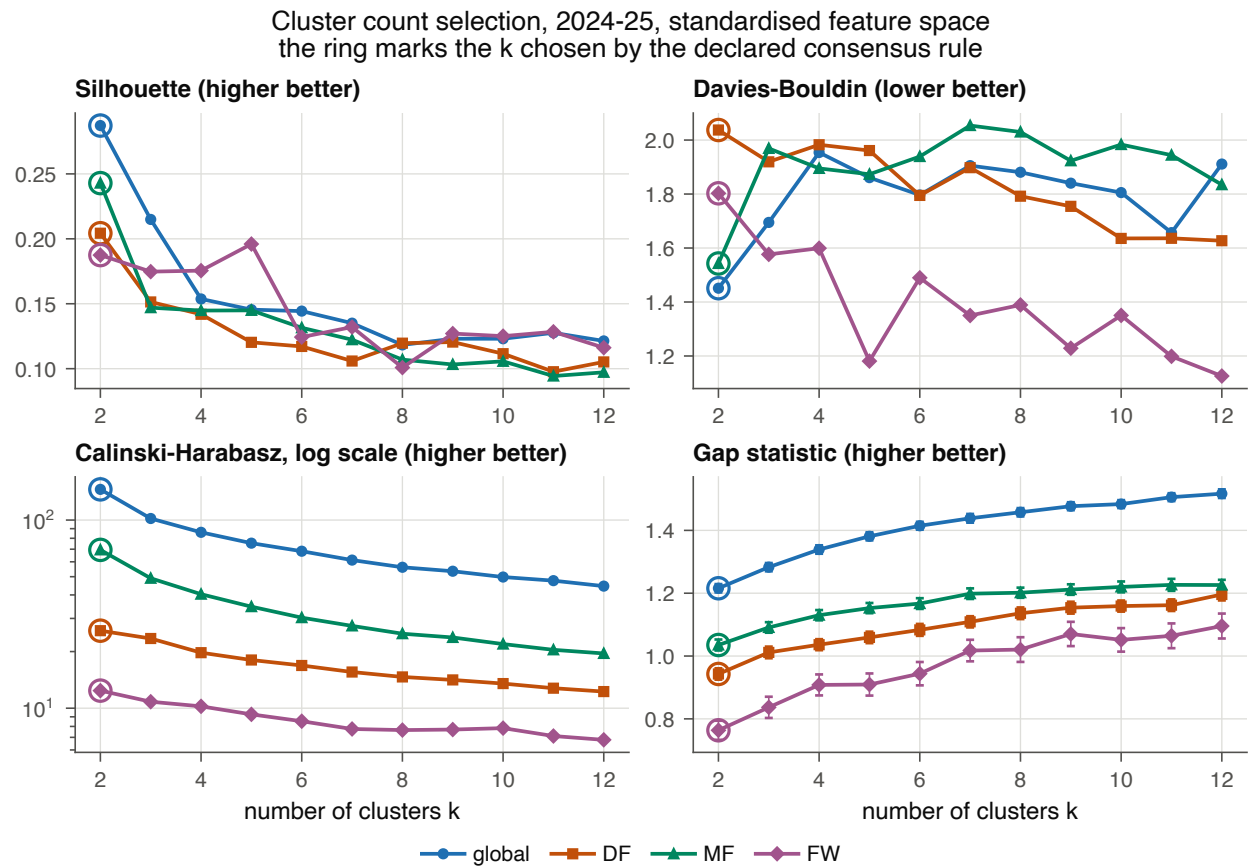
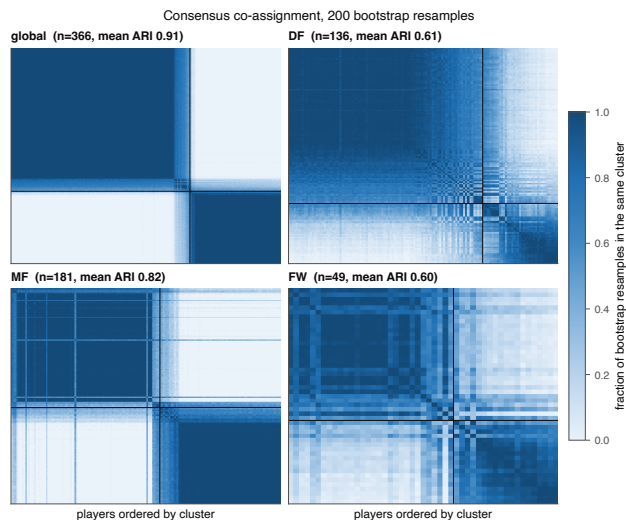
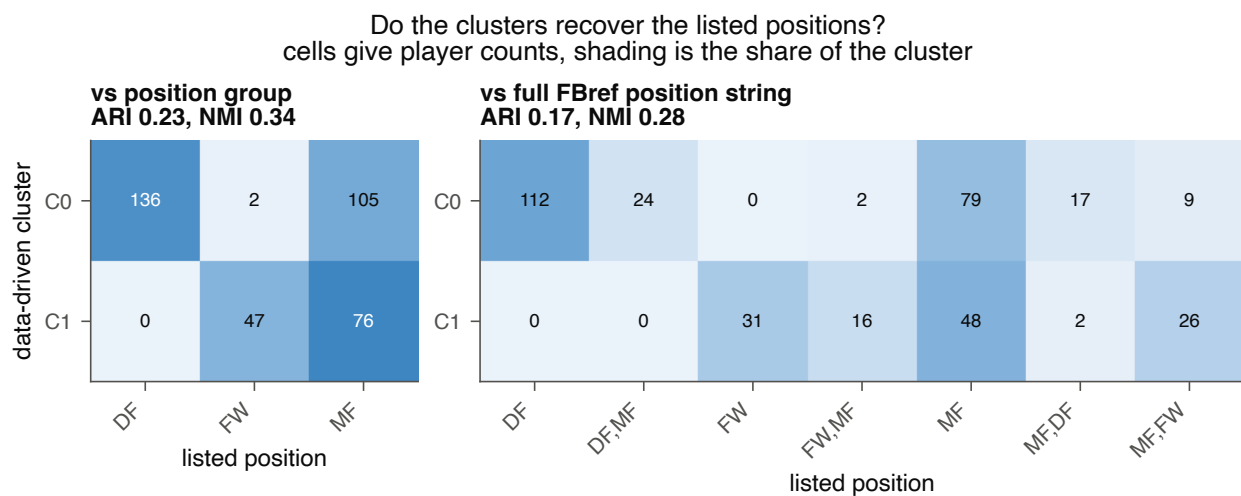


Figure 29: Cluster selection criteria across candidate cluster counts. The consensus rule was fixed before fitting and selects the same value in every scope.



(a) Bootstrap co-assignment.



(b) Clusters against listed positions.

Figure 30: Cluster stability and the correspondence with listed positions.

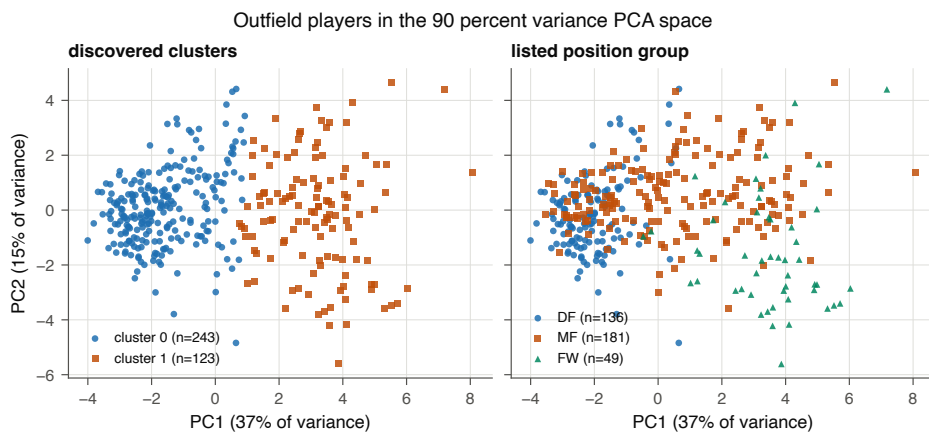


Figure 31: The selected partition shown in the principal component space.

Distribution by position group, 2024-25 (white tick marks the median)

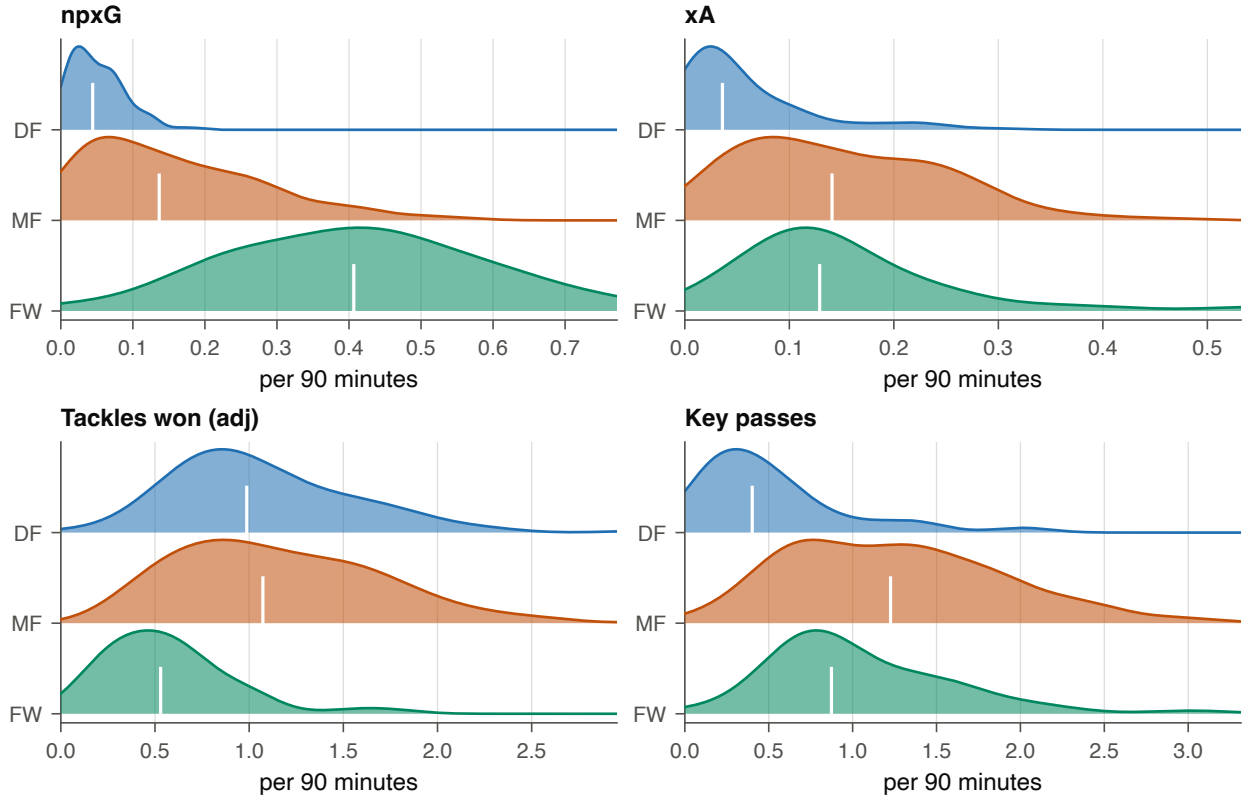


Figure 32: Distributions of four representative per-90 statistics across position groups in the earlier of the two reduced-sample seasons. The separation between groups is clear for shot volume and for defensive actions, and much weaker for creation.

H.2 Descriptive structure

After filtering, 366 outfield players are eligible in the earlier of the two reduced-sample seasons, comprising 136 defenders, 181 midfielders and 49 forwards. Summary statistics by position group appear in Table 11, and the distributions of four representative statistics are shown in Figure 32.

The feature set is strongly non-normal. All 18 of the 18 features reject the Shapiro-Wilk null at the five percent level, mean absolute skewness is 1.13, and the most skewed feature is goals per shot at 2.89. Mardia’s test rejects multivariate normality decisively on both the skewness and kurtosis components. Principal component analysis requires no distributional assumption to be computed, so this does not invalidate the decomposition, but it does mean the components should be read as directions of variance rather than as latent factors with any distributional interpretation, and it justifies the use of Spearman rather than Pearson correlation throughout.

Table 11: Headline per-90 statistics by position group in the earlier of the two reduced-sample seasons, reported as mean with standard deviation in parentheses.

Statistic	DF	MF	FW
np_xg_p90	0.05 (0.04)	0.16 (0.12)	0.41 (0.17)
xa_p90	0.06 (0.06)	0.15 (0.10)	0.16 (0.11)
key_passes_p90	0.51 (0.44)	1.26 (0.63)	1.05 (0.54)
xg_chain_p90	0.35 (0.16)	0.50 (0.20)	0.63 (0.20)
xg_buildup_p90	0.30 (0.14)	0.28 (0.13)	0.18 (0.09)
shots_p90	0.54 (0.33)	1.54 (0.85)	2.53 (0.69)
shot_accuracy_pct	29.22 (20.60)	31.91 (13.77)	40.30 (9.65)
crosses_p90	1.22 (1.62)	2.26 (1.75)	1.40 (1.62)
interceptions_padj_p90	0.96 (0.37)	0.74 (0.44)	0.24 (0.17)
tackles_won_padj_p90	1.07 (0.48)	1.16 (0.54)	0.56 (0.34)
fouls_committed_padj_p90	0.87 (0.39)	1.31 (0.55)	1.28 (0.61)
offsides_p90	0.06 (0.07)	0.16 (0.17)	0.53 (0.33)

H.3 Comparing embeddings

Principal component analysis, UMAP and t-SNE were fitted to the same standardised features and compared in Figure 33. The selected UMAP configuration used 10 neighbours at a minimum distance of 0, chosen by trustworthiness over the twelve configuration grid reported in Appendix C.

The three methods agree on the dominant structure and differ in how they present it. Trustworthiness is 0.855 for the linear projection, 0.932 for UMAP and 0.944 for t-SNE, while continuity is close to identical across the three. Nearest neighbour position purity at ten neighbours is 0.649 for the linear projection and 0.699 for UMAP, so roughly two thirds to three quarters of a player’s closest statistical neighbours share his listed position group. That figure is informative in both directions: it confirms that the surviving features carry real positional signal, and it establishes that a substantial minority of every player’s nearest neighbours are listed at a different position, which is the space in which data-driven roles can differ from listed ones.

All three embeddings display a continuous gradient from defenders through midfielders to forwards rather than separated groups. That gradient, and not a set of islands, is the honest description of the geometry, and it conditions how the clustering that follows should be read.

Two-dimensional embeddings of the same 18 features, 2024-25
T trustworthiness, C continuity, purity at k=10

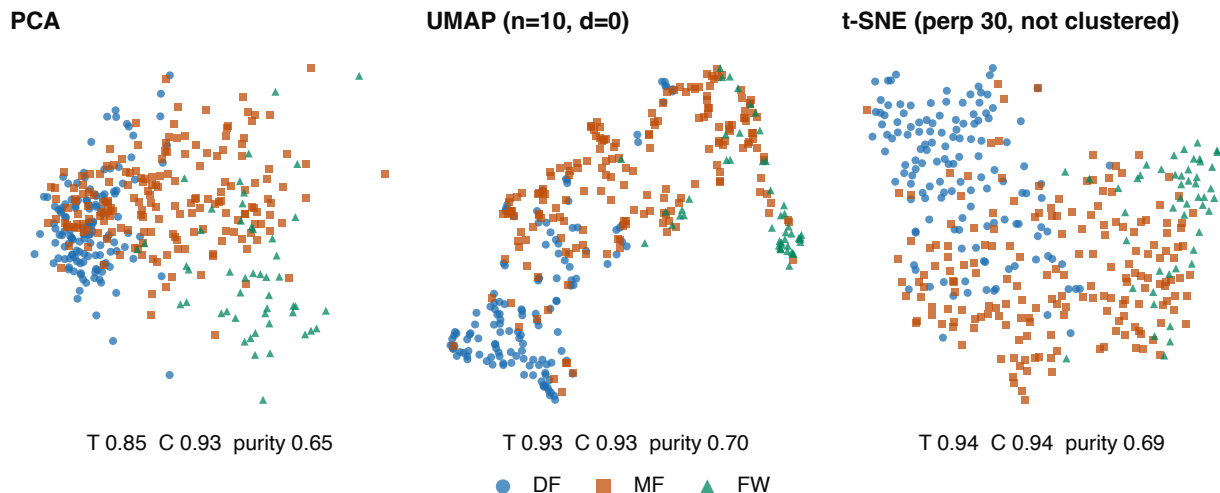


Figure 33: Two-dimensional embeddings of the same feature set, coloured by listed position group. The defender to forward gradient is continuous in all three, and t-SNE is shown for visual contrast only and is not used for clustering.

H.4 The third dimension

Two components are a convenient way to draw the space and a poor way to describe it. The first two carry shooting volume and wide creation and together account for 0.521 of the variance. The third carries duels and discipline, loading on fouls committed, cards and tackles won, and it is orthogonal to both of the others, so a planar projection does not compress it, it deletes it. Retaining it raises captured variance to 0.634. Figure 34 shows the outfield space in all three, drawn from two viewpoints so that structure can be told apart from the accident of a single projection.

Adding the dimension does not change the headline geometry. The defender to forward gradient remains continuous rather than resolving into islands, which is the same conclusion the planar embeddings reached and is now established against a projection that had the opportunity to falsify it. Figure 35 places the selected partition in the same space: the boundary cuts across the shooting axis and is close to indifferent to duels and discipline, which is consistent with a split that is substantially about whether a player shoots.

The third dimension does change the club picture, and this is the one place where it produces a finding the planar analysis could not. Club centroids correlate with final league position at -0.861 on the first component and -0.695 on the second, but only 0.209 on the third. Shooting volume and wide creation both track league position; the physical dimension of duels, fouls and cards essentially does not. A club does not finish higher by fouling more or less. That is unsurprising once stated, and it needs stating because the duelling axis is a full component of player variation that carries almost no information about team success. Figure 36 shows the clubs, labelled at the extremes of the table.

Finally, the three-dimensional view settles a structural question raised in Section 5.3 by measurement rather than assertion. Drawing each archetype centroid as a direction from the origin, the two centroids within a position group have a cosine of -1 for defenders, -1 for midfielders and -1 for forwards. They are exactly anti-parallel, which is forced by standardising within group before a two-way split. Figure 37 shows the consequence: six archetypes occupy three lines through the origin, not six independent directions.

Outfield players in three principal components, 2024-25

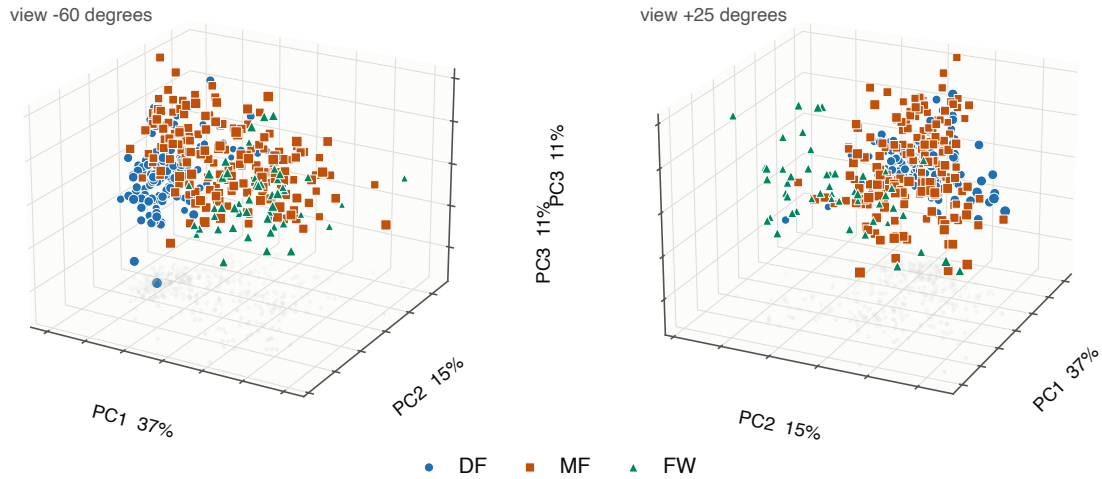


Figure 34: Outfield players in the first three principal components, from two viewpoints. Grey points on the floor are the vertical projection of each player, supplying a depth cue that a static rendering otherwise lacks.

The global partition in three components, crosses mark centroids, 2024-25

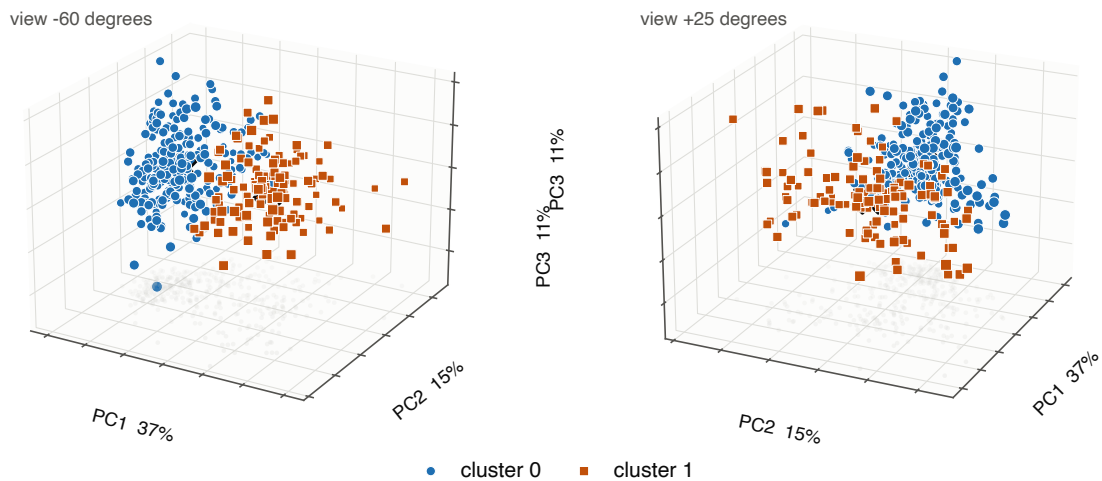


Figure 35: The selected global partition in three components, with cluster centroids marked by crosses. The split runs along the shooting axis and is close to indifferent to the third component.

**Clubs as minutes-weighted centroids, darker finished higher, 2024-25.
Only the top and bottom 5 are labelled**

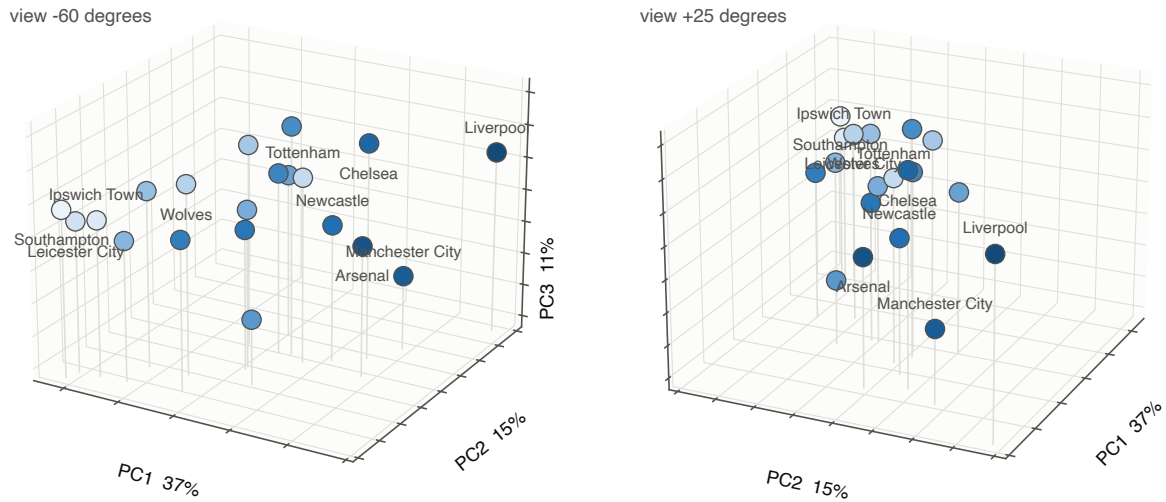


Figure 36: Clubs as minutes-weighted centroids in three components, shaded by final league position with darker shading for a higher finish. Vertical lines to the floor give the depth cue, and only the top and bottom of the table are labelled.

Archetype centroid directions: each position group is one axis, not two

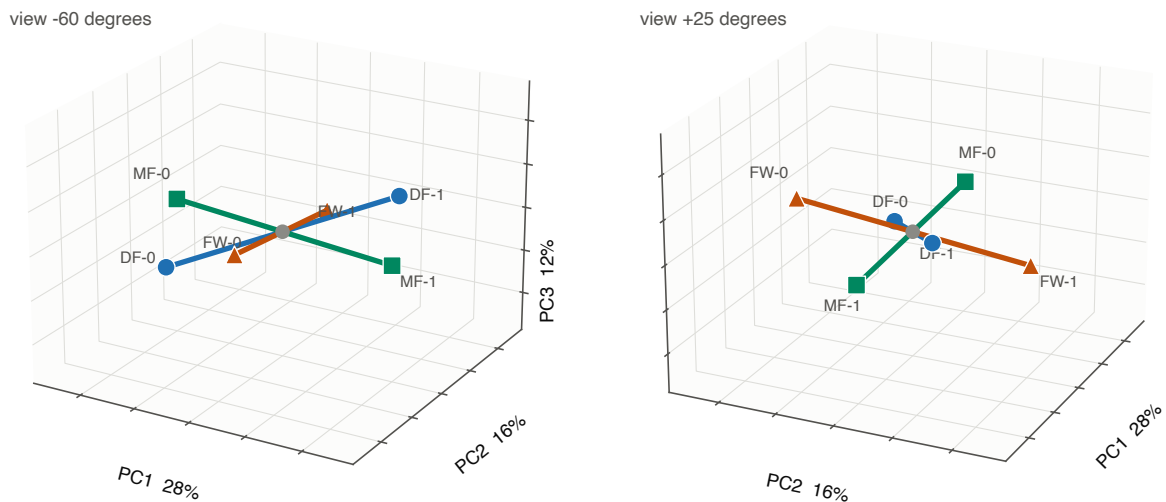


Figure 37: Archetype centroid directions, drawn as unit vectors so the figure shows orientation rather than magnitude. Each position group forms a single line through the origin, so the six archetypes describe three axes.

The same pipeline across the big five, 2024-25. England k=2, Spain k=2, Germany k=2, Italy k=2, France k=2

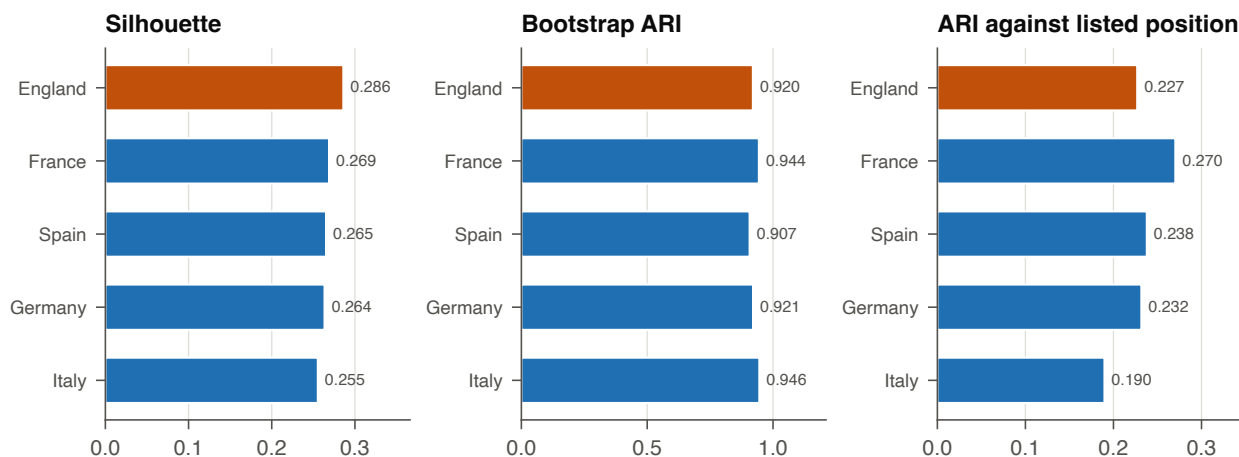


Figure 38: The pipeline applied unchanged to each of the big five leagues in the primary season. The selected number of clusters, the silhouette, bootstrap stability and agreement with listed positions are all close across competitions.

H.5 The same pipeline across the big five

Every result reported so far comes from one competition, which leaves an alternative explanation standing: perhaps English football happens to divide in two and another league would divide differently. The withdrawal applied to the whole site rather than to one competition, so the other four of the big five are in the same reduced state and the independent expected goals source covers all of them, which makes the alternative testable rather than merely acknowledged.

Repeating the pipeline unchanged on Spain, Germany, Italy and France for the earlier of the two reduced-sample seasons returns the same answer everywhere. The rule declared in advance selects 2 clusters in all 5 leagues. The silhouette at that choice spans 0.255 to 0.286 across the five, bootstrap stability spans 0.907 to 0.946, and agreement with listed position groups spans 0.190 to 0.270. Every one of those three ranges is narrow, and the third is uniformly weak: in no league does the partition recover the positions the source already records. Figure 38 shows the comparison.

The elasticity result generalises as well. Fitted independently inside each league, all 15 defensive elasticities lie between 0.081 and 0.692, so every one is below the value of one that the conventional possession adjustment assumes. The overcorrection identified in Section 5.11 is therefore a property of how defending responds to territory rather than a quirk of a single competition, and the conventional correction overcorrects across the big five.

This changes the standing of the main result. Two clusters is not a fact about England. It is what the surviving public feature set can support, and the same feature set supports the same coarse division wherever it is applied.

H.6 What a supervised model sees

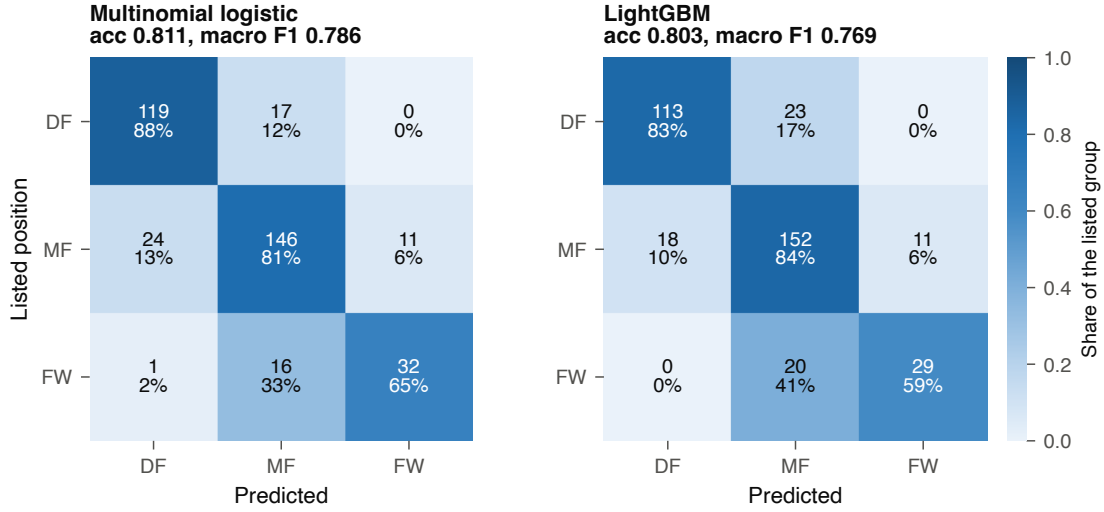
The clustering results leave one question open: are the surviving features too weak to carry positional information at all, or do they carry it while the clustering chooses to partition along a different axis? A supervised model settles it. Predicting the listed position group from the same 18 features, gradient boosting reaches an accuracy of 0.803 and a macro F1 of 0.769, and logistic regression reaches 0.811 and 0.786, against a majority class baseline of 0.494 and 0.221. The features clearly do

carry position, so the weak agreement between clusters and positions reflects the clustering finding a different and equally real axis, not an empty feature set. The confusion is almost entirely between adjacent groups, as Figure 39a shows, and forwards are the weakest class.

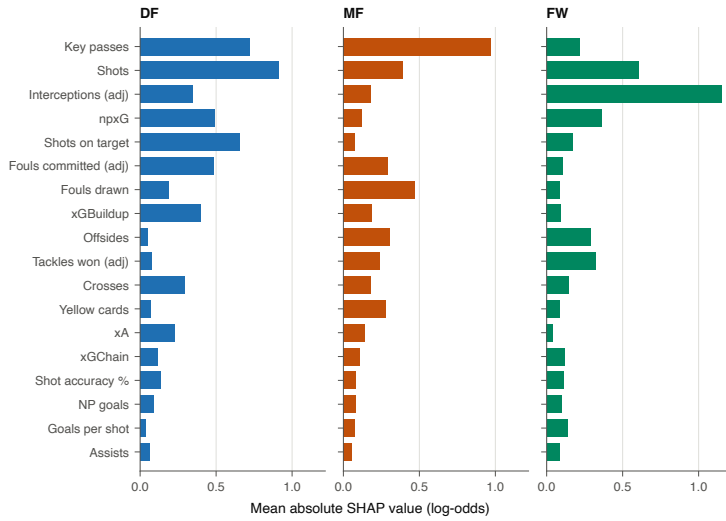
The SHAP decomposition in Figures 47 and 39b carries a finding that follows directly from the withdrawal. With possession and passing volume gone, positions are identified largely by absence. Defenders are recognised by low shooting and low key passes, forwards by low interceptions, and midfielders by an intermediate profile on both. A classifier of this kind, trained before January 2026, would have identified a defender by what he did with the ball. It now identifies him by what he does not do.

Players the classifier places in the wrong group overlap significantly with the players the clustering flagged as hybrids. The overlap rate is 0.623 for logistic regression at an enrichment of 1.48 over independence, and 0.556 at an enrichment of 1.32 for gradient boosting. The enrichment is modest because the hybrid flag is itself broad, covering 154 of 366 players, so this is corroboration rather than a strong result.

The finishing above role regression illustrates a hazard. Predicting non-penalty expected goals per ninety from non-shooting behaviour returns a cross validated coefficient of determination of 0.531. An earlier specification that also included xGChain returned 0.892, which looks far better and is in fact circular: Understat credits xGChain for possessions the player finished himself, so the difference between xGChain and xGBuildup contains the target and correlates with it at 0.904. The apparently strong model was reconstructing the outcome rather than predicting it. Even at the honest value, roughly half the variance is unexplained, so the residual leaderboard in Figure 46b is presented as an illustration and not as a measurement of finishing ability. Figure 46a shows the two outlier detectors against each other.



(a) Confusion between listed positions.



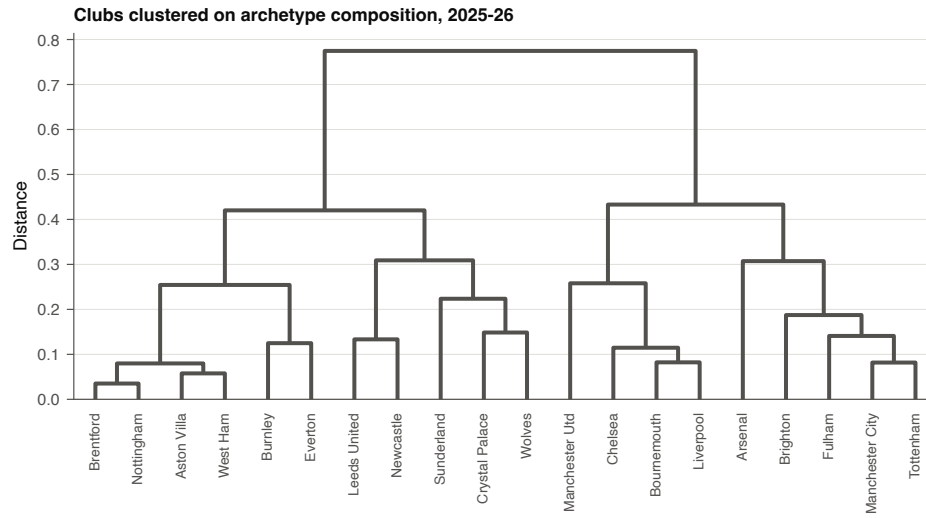
(b) Feature importance per position.

Figure 39: Supervised validation of the surviving feature set.

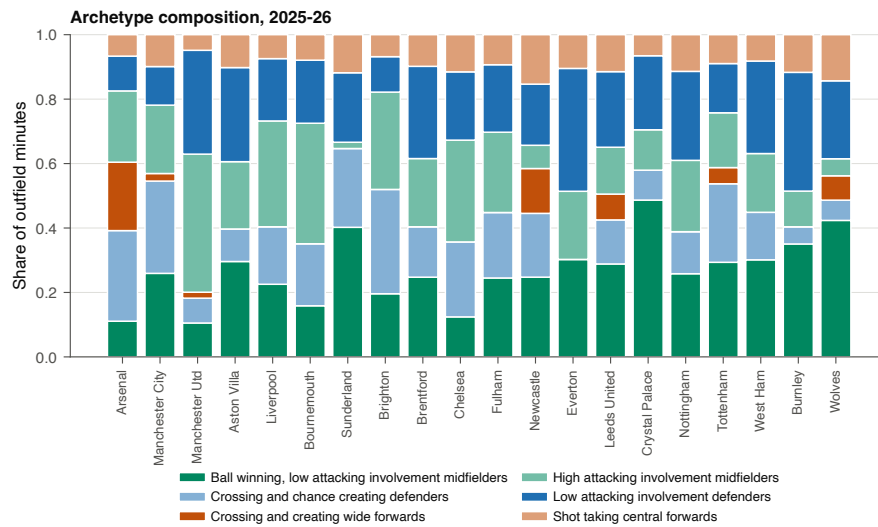
H.7 Team signatures

Aggregating players into clubs produces the one result in this study that is both strong and stable. The minutes-weighted position of a club's outfield players on the first principal component correlates with final league position at a Spearman coefficient of -0.771 in the earlier of the two reduced-sample seasons and -0.783 in the replication season, which is close to an exact reproduction. Since the first component is essentially attacking involvement, this says that clubs whose players collectively shoot and create more finish higher, which is unsurprising in itself, but the near identical coefficient across two independent seasons shows the team level summary is measuring something stable even where the player level partitions are not.

Hierarchical clustering on archetype composition groups the 20 clubs into 3 style families, shown in Figure 40a, with composition by club in Figure 40b. Figure 41 places each club's players in the shared embedding against the rest of the league, which makes squad shape legible in a way that a



(a) Style families.



(b) Archetype composition by club.

Figure 40: Club stylistic signatures, ordered by final league position.

composition bar cannot: clubs differ visibly in how far their defenders sit toward the midfield region.



Figure 41: Each club's outfield players highlighted in the shared embedding, with the rest of the league in grey. Axes are shared across all twenty panels.

H.8 Similarity, replication and noise

H.8.1 Player similarity

Nearest neighbour search within position group recovers football sense for players whose roles are defined by the surviving statistics, and fails informatively for players whose roles are not. Full backs return full backs, penalty box forwards return penalty box forwards, and centre backs return centre backs. The instructive failures are the midfielders. A holding midfielder's nearest neighbours are creative attacking midfielders, because with no tackles by pitch third, no blocks, no clearances, no pressures and no passing volume, the only things left that distinguish him are his set piece deliveries and key passes. The metric finds the players who share that signature and misses the entire defensive half of his job. An inverted full back returns three centre backs for the same reason, since the role is defined by possession behaviour that no longer exists in the data. Similarity scores among centre backs are systematically the lowest in the table, which is the same feature poverty appearing as an inability to tell defenders apart at all. Figure 42a shows the structure.

H.8.2 Cross-season replication

242 players are eligible in both seasons. The declared rule independently selects the same number of clusters in every scope in both seasons, so the structure replicates. Membership replicates only partly. The adjusted Rand index between seasons is 0.598 globally, with 27 players changing cluster, and within position group it falls to 0.526 for defenders, 0.381 for midfielders and 0.453 for forwards. The within group figures are worse than the bootstrap stability of Section H.1, which is what would be expected if the within group splits are finer than the data can reliably support. Figure 42b shows the transitions.

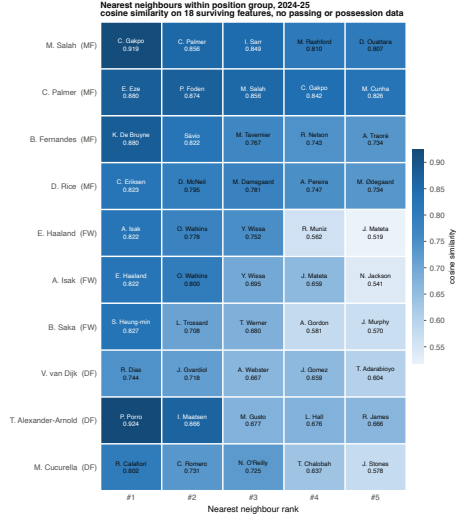
H.8.3 Sensitivity to sampling noise

Empirical Bayes shrinkage toward the position group mean, with the degree of shrinkage set by minutes played, provides the sharpest test of whether the partitions describe playing roles or sampling noise. Shrinkage is substantial rather than cosmetic, and the results divide cleanly.

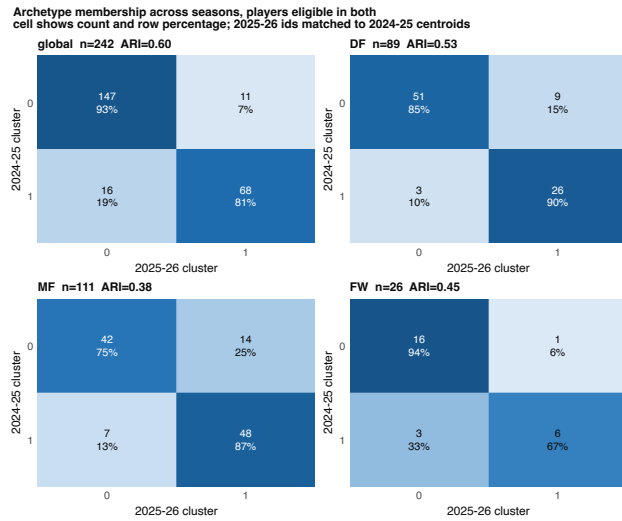
The global partition survives. Only 5 of 366 players change cluster, the adjusted Rand index between the original and shrunk partitions is 0.945, the selected number of clusters is unchanged, and the rate at which players move falls monotonically with minutes played, exactly as the noise account predicts.

The within group partitions fare worse, and they do not fare equally. For defenders the adjusted Rand index falls to 0.453, with 21 of 136 players moving, which is a partition largely dissolved by removing sampling noise. Midfielders retain 0.770 and forwards 0.626, both of which survive in part rather than collapsing. The forward figure is notably higher than it was before the possession correction of Section 5.11 was applied, which is itself informative: some of what previously looked like noise in the forward partition was exposure to opponent possession, and removing it left a more reproducible split. The mechanism is visible in the estimated variance components: several features are assigned an estimated between-player variance of zero within a position group, meaning the model attributes their entire observed spread to sampling noise and collapses them to the group mean, removing them from the clustering altogether. Shot accuracy retains very little reliability, because its true spread between players is small relative to the noise in estimating it from a season of shots.

Read alongside the bootstrap and replication results, the three tests do not deliver a single uniform verdict, and reporting them as though they did would overstate the case. Defenders fail



(a) Nearest neighbour similarity.



(b) Between-season transitions.

Figure 42: Similarity structure and cross-season replication.

all three: bootstrap stability of 0.607, cross-season agreement of 0.526, and near dissolution under shrinkage. Forwards fail two of the three, with a bootstrap minimum of -0.065 across resamples and cross-season agreement of 0.453, while partly surviving shrinkage. Midfielders pass all three. The conclusion that survives every test is therefore narrower than a single summary sentence would suggest: the global division between attacking and defensive involvement is a real property of players, the midfield split is trustworthy, and the defender split should not be used to characterise an individual at all. Figures 45a and 45b show the effect.

H.8.4 Role drift within a season

Every result so far treats a season as one observation per player, which assumes a player's profile is homogeneous across it. Match reports make that assumption testable, and because the withdrawal removed the affected columns from the match schema rather than emptying them, what remains

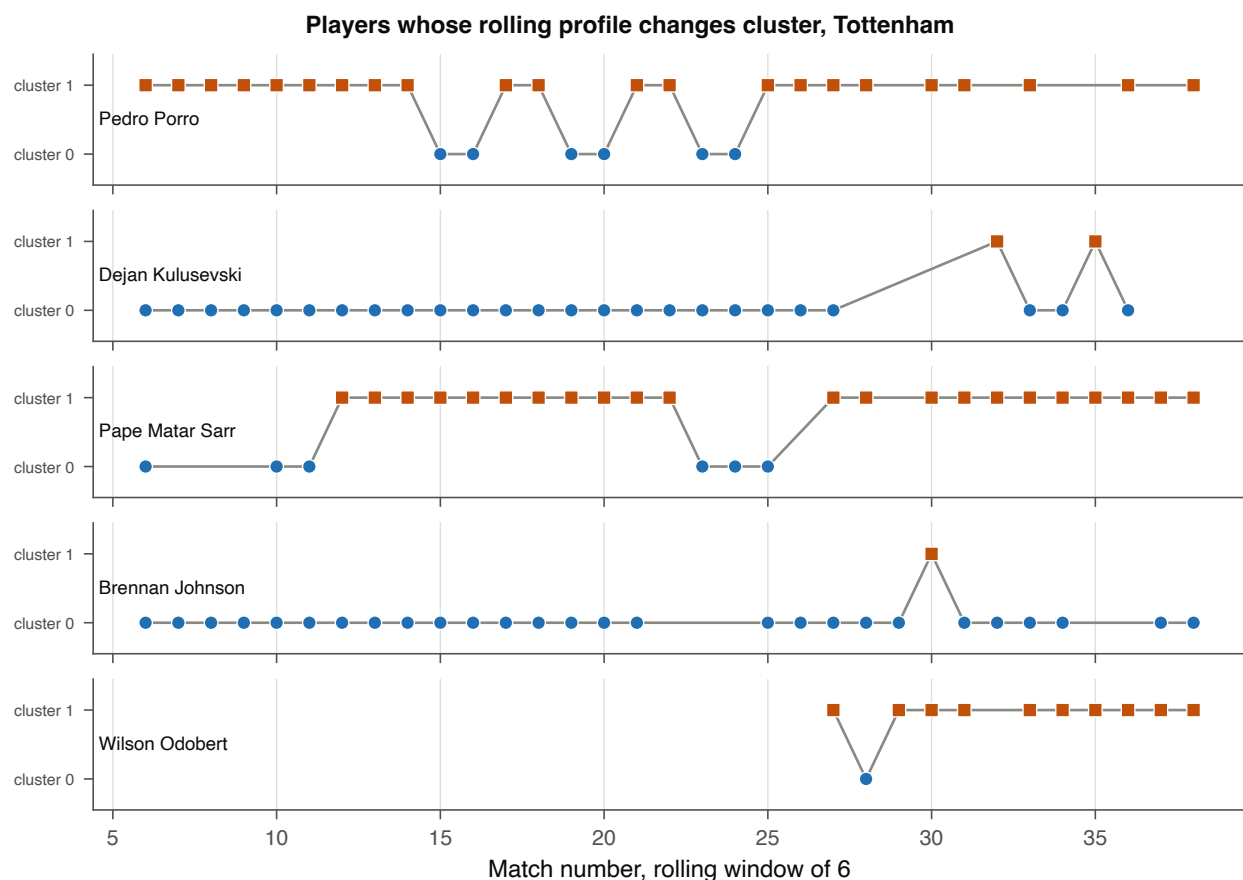


Figure 43: Rolling cluster assignment across the season for the Tottenham players who change cluster at least once, on a 6 match window. Players not shown occupied one cluster for the whole season.

there is fully populated. A full season is 380 reports at seven seconds each, so the scrape is restricted to one club, Tottenham, chosen because its eligible players span all six archetypes and therefore give drift the widest range to move across. 38 matches were retrieved and 26 players accumulated at least one complete rolling window of 6 matches.

The match feature set has 10 of the eighteen features used elsewhere, since the Understat expected goals block has no match equivalent. Rather than assume that the coarser space is a fair stand-in, we refit the season clustering using only those 10 features and compare it to the partition reported throughout. The two agree at an adjusted Rand index of 0.903, so the reduced space reproduces almost all of the main division and drift measured within it is interpretable.

Most players do not drift. 5 of 26, or 19 percent, change cluster at least once during the season, and the remainder occupy a single cluster throughout. The players who stay put are the ones whose roles are least ambiguous, holding midfielders on one side and the recognised attackers on the other, while the movers are predominantly full backs and wide players whose attacking involvement varies with the fixture. Figure 43 shows the trajectories.

The finding cuts both ways and should be read that way. For four fifths of players a season aggregate describes a single role and the homogeneity assumption holds. For the remaining fifth it averages over two, which means their season-level archetype is a summary of something that changed rather than a description of something stable. That is a limitation of season aggregates in general rather than of this feature set in particular, and it is one that only match level data can expose.

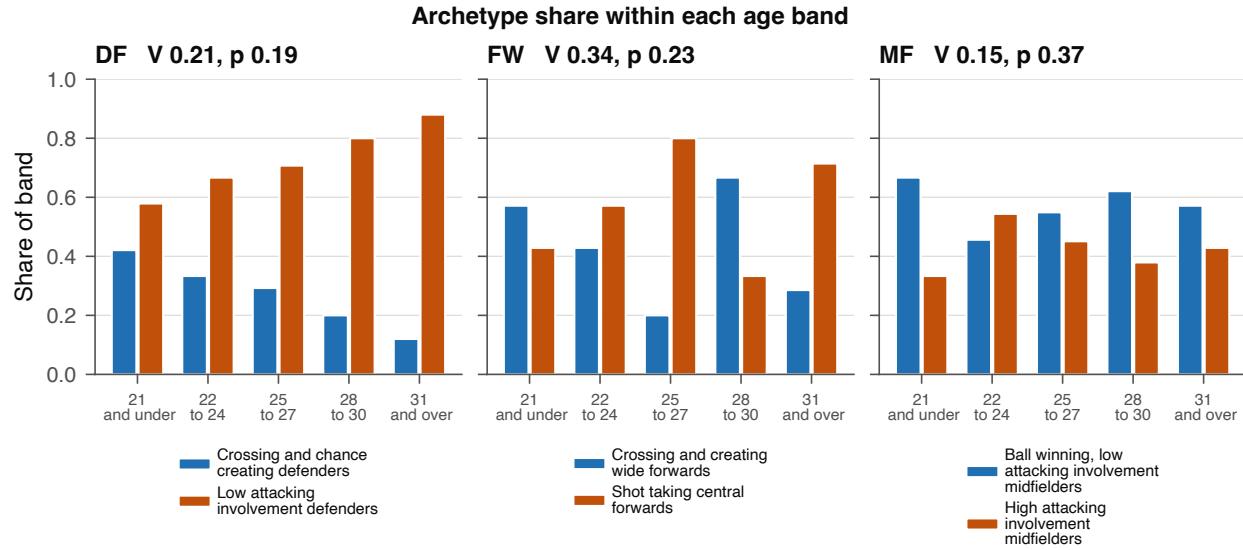


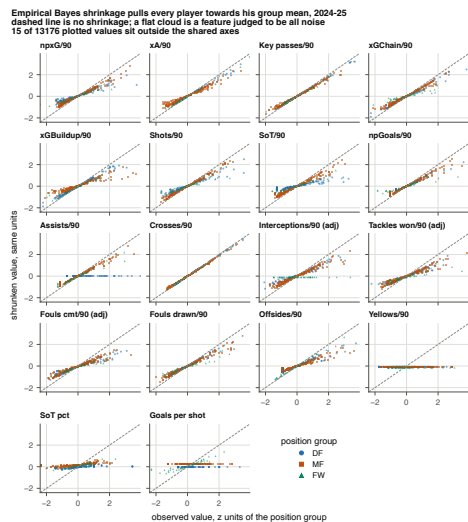
Figure 44: Share of each age band taken by each archetype, within position group. The defender panel falls monotonically with age, which a chi-square on bands does not detect and a rank correlation does.

H.8.5 Age and archetype

The two archetypes within a position group differ mainly in attacking involvement, and involvement is plausibly something a career passes through rather than a fixed trait, so the association with age is a natural test. It is tested within position group, because archetypes are nested inside positions and a pooled table would largely recover the fact that forwards and defenders have different age profiles.

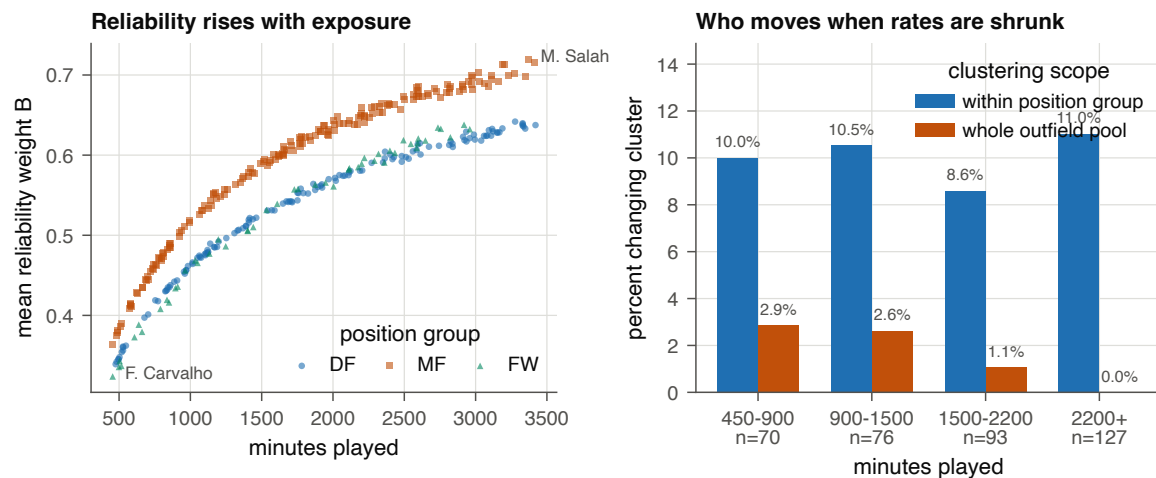
Only defenders show a relationship. The crossing and chance creating defenders average -1.65 years younger than the low involvement group, a difference that the rank test places at 0.015, and the share of the crossing archetype falls monotonically from the youngest band to the oldest. Midfielders show nothing at all, with a gap of 0.12 years at 0.698, and the global partition is unrelated to age at 0.415. Forwards point the same way as defenders but cannot support the claim, since six of the ten cells in their table have expected counts below five.

The defender result also illustrates a point about test choice that is easy to get wrong. A chi-square on the banded table returns 0.194 and would ordinarily be read as no association. That test spends its degrees of freedom on any departure from independence, while the pattern here is monotonic, and a rank correlation against age recovers it at 0.209 with a p-value of 0.015. The banded test is not wrong, it is underpowered against the alternative that actually holds, and reporting only the chi-square would have converted a real finding into a null one. Figure 44 shows the shares.



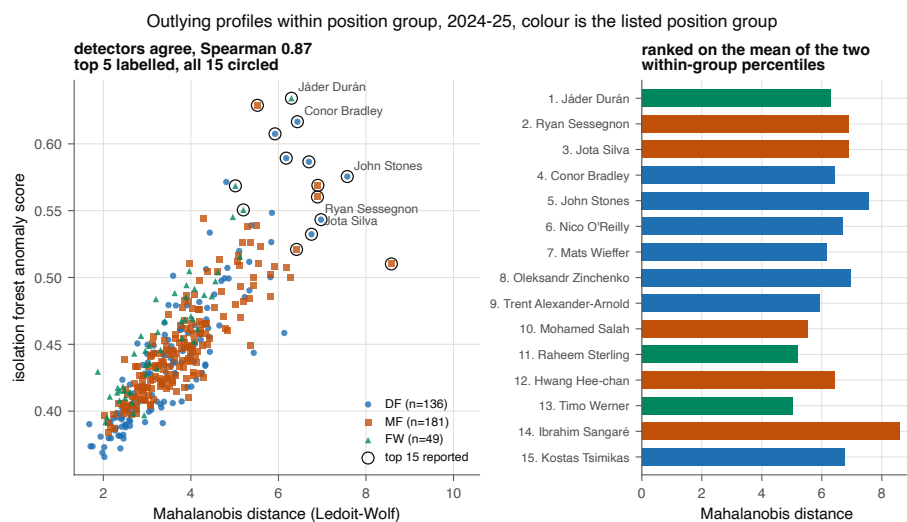
(a) Effect of shrinkage on assignments.

Reliability and cluster movement against minutes played, 2024-25

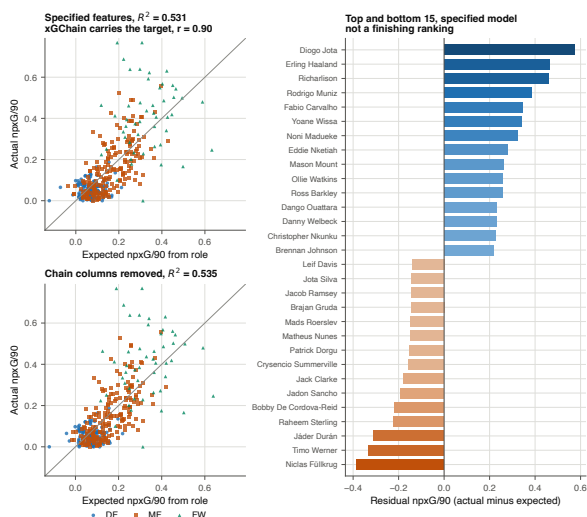


(b) Assignment changes by minutes played.

Figure 45: Empirical Bayes shrinkage as a test of whether the partitions describe roles or noise.



(a) Outliers within position group.



(b) Residuals from the role model.

Figure 46: Unusual players, and the residual leaderboard that should not be read as a measurement of finishing ability.

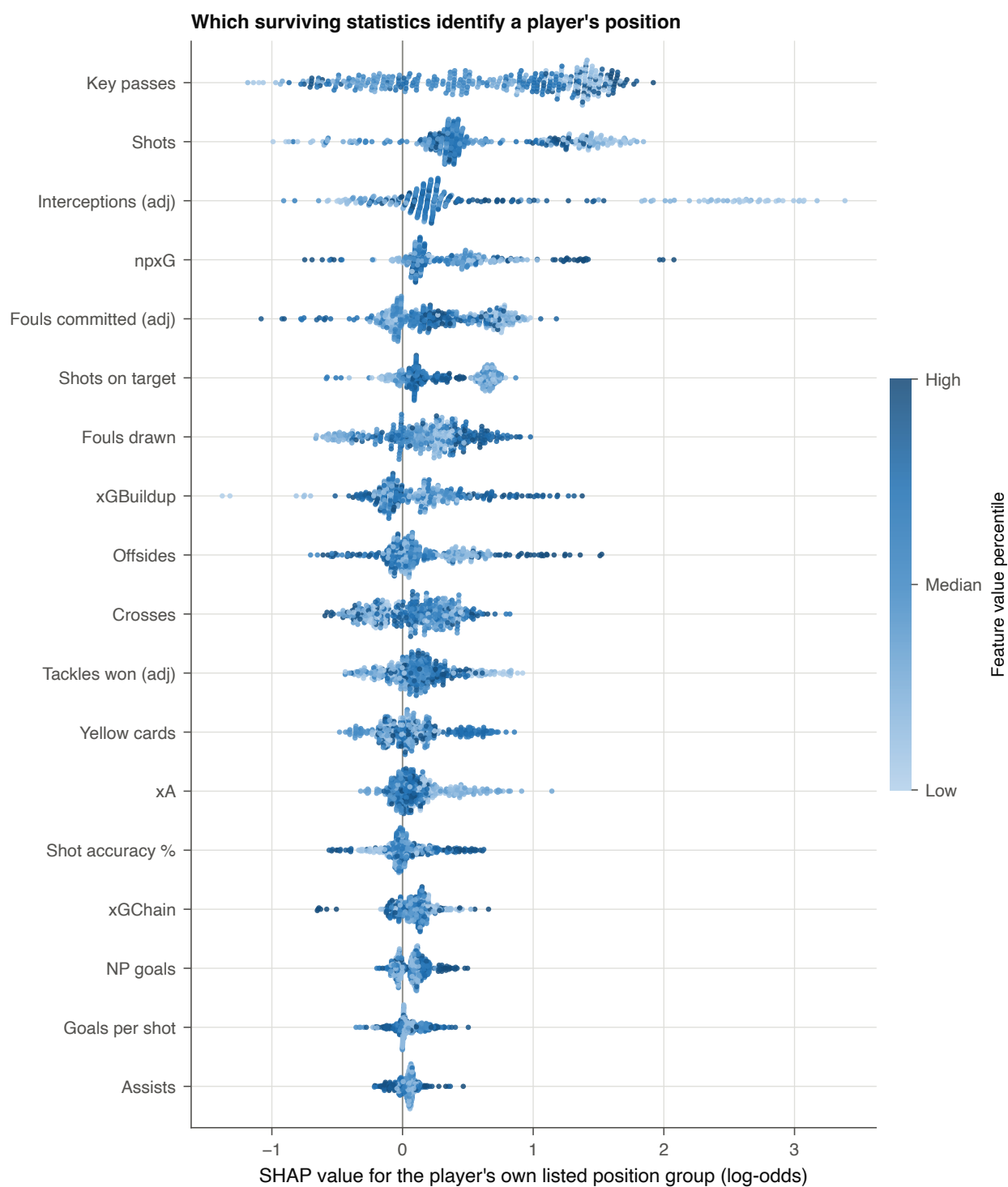


Figure 47: Global SHAP values for the gradient boosted position classifier. Shooting volume and interceptions carry most of the signal, so a player's position is now identified largely by the actions he does not perform.

H.9 Supplementary figures

The figures collected here belong to the same reduced sample and are reported for completeness. Figure 48 gives the centroid deviations and Figure 49 the archetype radars. Figures 50a to 52 give the goalkeeper material that survives on that sample, which is limited to shot stopping and workload for the reasons set out in Section 3.4.

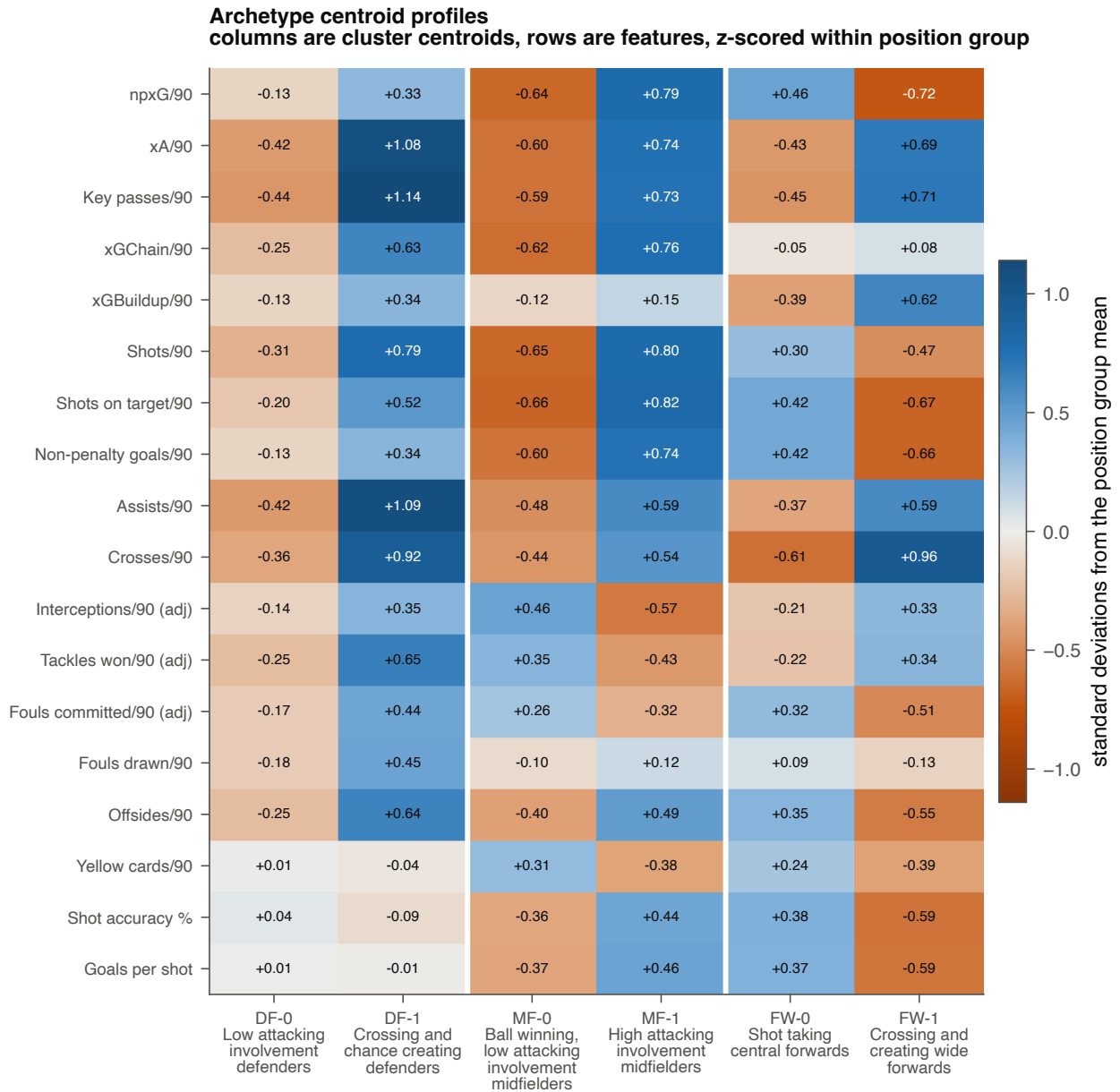
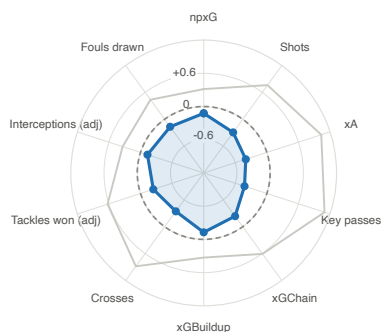


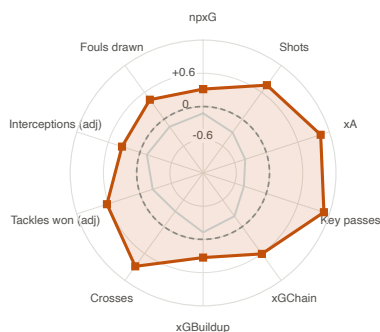
Figure 48: Centroid deviation of each archetype from its position group mean, reduced sample.

DF archetypes on the fixed radar axes, the other archetype shown in grey

**DF-0 Low attacking involvement
defenders (n=98)**

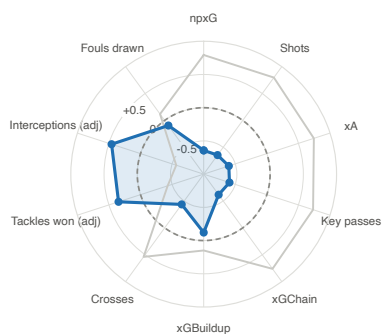


**DF-1 Crossing and chance
creating defenders (n=38)**

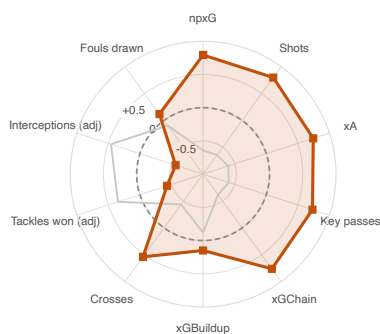


MF archetypes on the fixed radar axes, the other archetype shown in grey

**MF-0 Ball winning, low
attacking involvement
midfielders (n=100)**

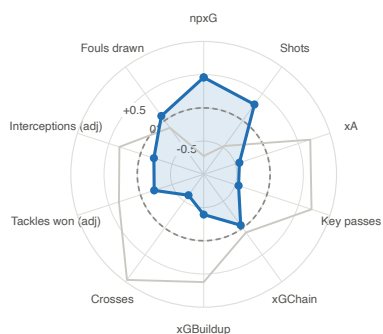


**MF-1 High attacking involvement
midfielders (n=81)**



FW archetypes on the fixed radar axes, the other archetype shown in grey

**FW-0 Shot taking central
forwards (n=30)**



**FW-1 Crossing and creating wide
forwards (n=19)**

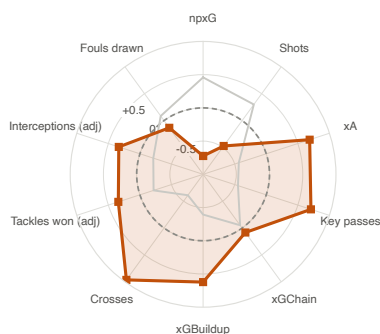
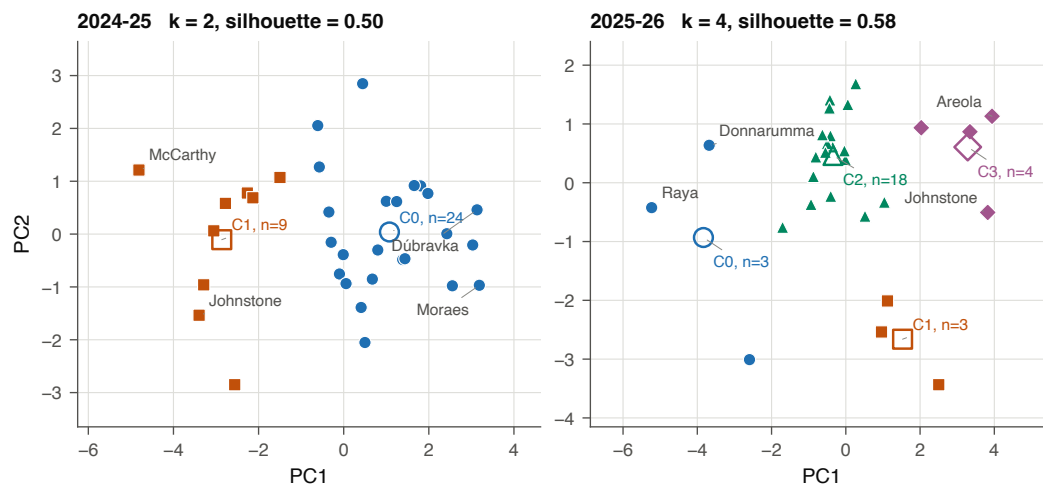


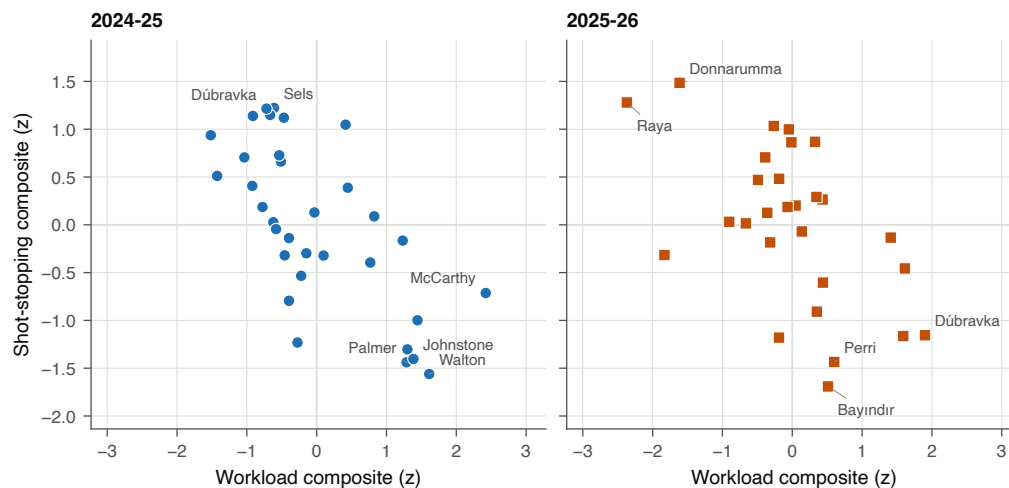
Figure 49: Archetype radars on the reduced sample.

Goalkeeper PCA, cluster means ringed. Silhouette bands, not separated groups



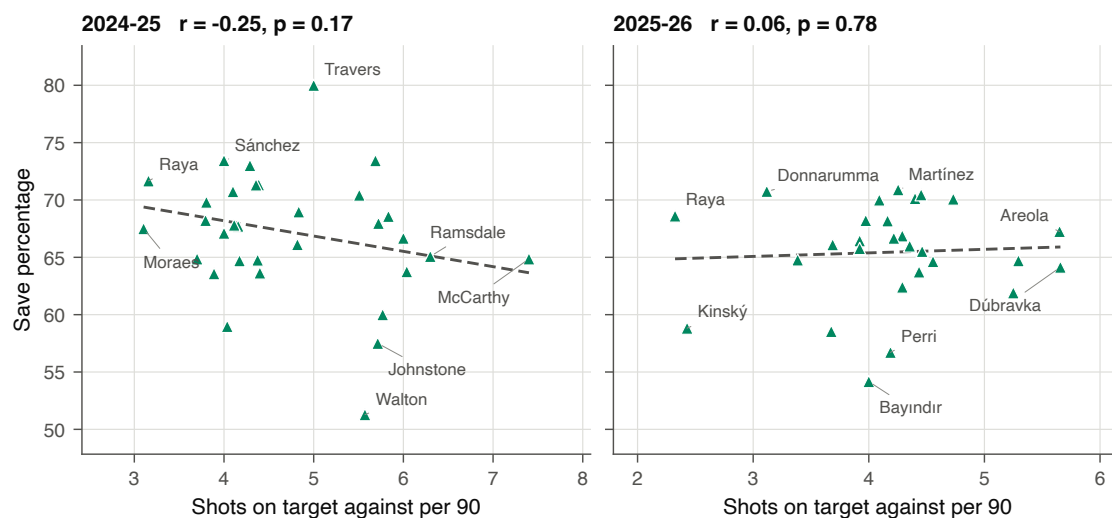
(a) Goalkeeper clusters in principal component space.

The only two composite axes the surviving data supports



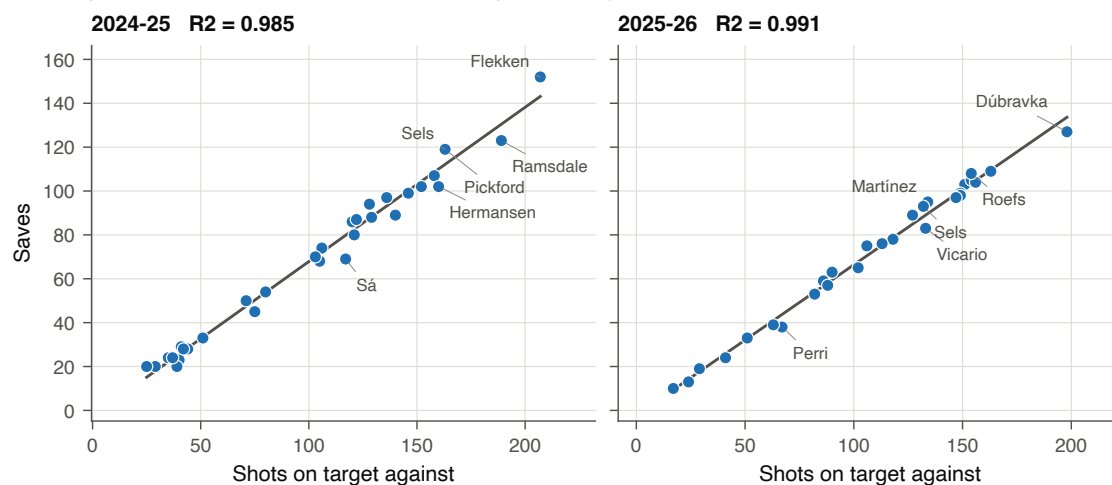
(b) The two surviving composite axes.

Figure 50: Goalkeeper structure on the reduced sample.



(a) Save percentage against workload.

Saves against shot volume. Residual is the shot-stopping index, unadjusted for shot quality



(b) Saves against shots on target.

Figure 51: Goalkeeper shot stopping on the reduced sample. With expected goals on target withdrawn, save percentage against volume is the only shot stopping view the reduced sample supports, and it cannot separate difficulty from quantity.

Shot-stopping index. Volume adjusted only, not shot-quality adjusted

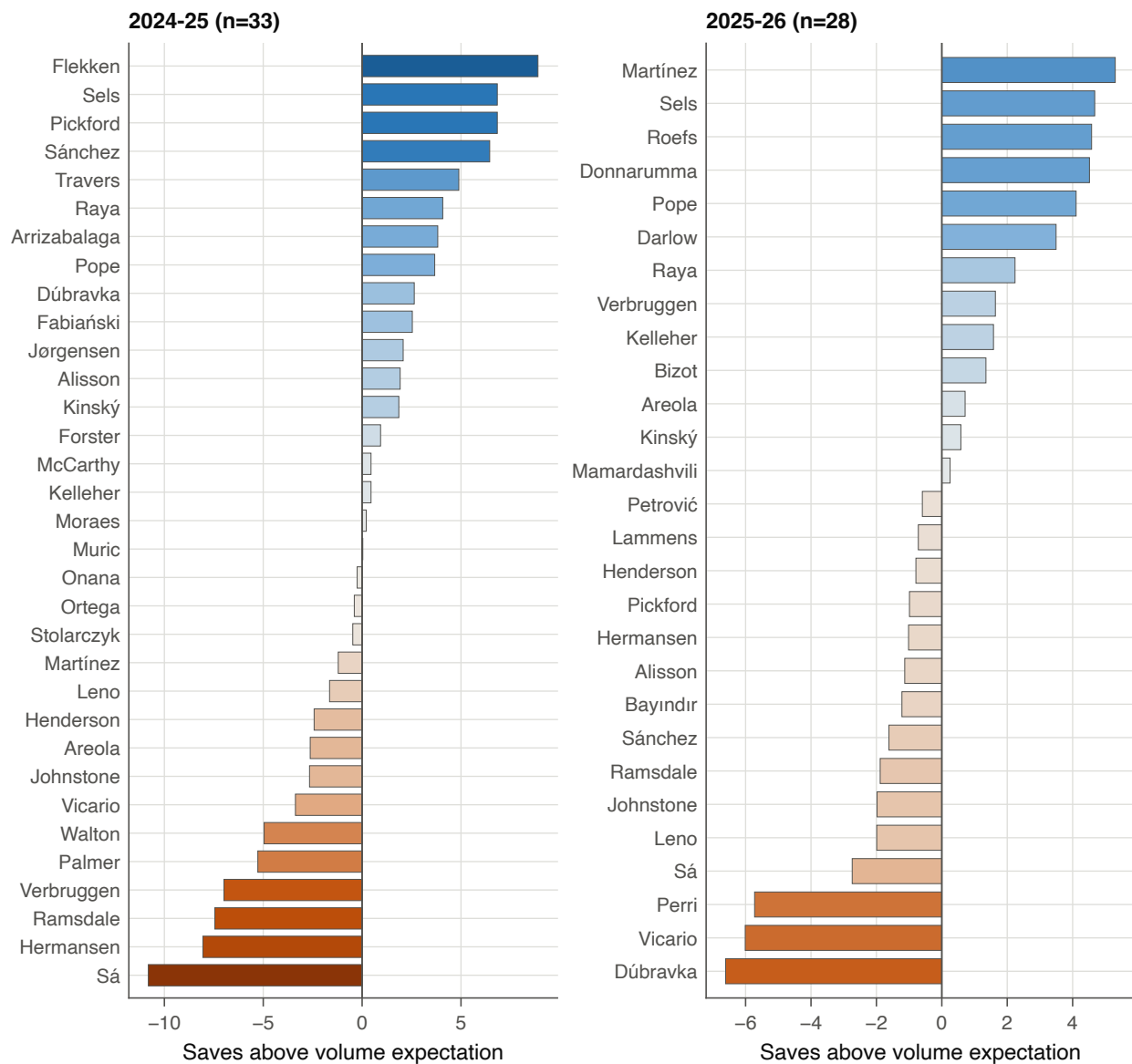


Figure 52: Goalkeeper shot stopping ranking on the reduced sample.